Solutions to Dummit & Foote's Abstract Algebra 3rd Edition

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Contents

Preface

This collection contains solutions to the exercises in David S. Dummit and Richard M. Foote's *Abstract Algebra*, Third Edition—a comprehensive graduate-level text that has become a standard reference in the field. I have worked through these problems over time to deepen my understanding of the material and to create a resource for others studying abstract algebra. While these solutions are written with care and rigor, they are not guaranteed to be error-free; if you discover mistakes or have suggestions for improvement, I welcome corrections.

Solutions are organized by chapter and section, mirroring the structure of the textbook. Within each section, exercises are presented in the order they appear in the book. Each solution aims to be self-contained but frequently references prior results; when a proof depends heavily on another exercise or theorem, I note this explicitly.

Exercise 0.0.1

Most problems in the book are presented in the standard exercise environment. These exercises are typically routine and don't require advanced mathematical insight beyond the techniques and concepts introduced in the text.

Exercise 0.0.2

Problems in the "special exercise" environment are included because they are particularly important, instructive, or deep. Some are challenging in their proof techniques or conceptual demands; others are simple but illustrate key ideas fundamental to the overall text. These exercises deepen understanding and often provide insight into material developed later in the book.

This is an ongoing project. As I work through the text, I refine earlier solutions and add new ones. While I strive for mathematical correctness, errors may remain–particularly in computational problems or long proofs. I encourage you to read critically, verify claims, and develop your own solutions. The goal is not to provide answers to be memorized, but to illustrate good problem-solving techniques and deepen your understanding of abstract algebra.

I am completing this project independently at my own pace. That said, I genuinely welcome corrections, suggestions, and alternative solutions from the community. If you find errors, have ideas for improvement, or wish to contribute, please open an issue or submit a pull request on GitHub. You may also reach out directly via Reddit or Discord with feedback or discussion. I appreciate the interest in helping this project grow and remain open to community involvement.

My deepest thanks to David S. Dummit and Richard M. Foote for writing *Abstract Algebra*. Their exposition is remarkably clear, their choice of exercises is thoughtful and comprehensive, and their book has been instrumental in my mathematical development. Many of these solutions exist because of the beauty and depth of their presentation.

Finally, I'd like to take a moment and thank the broader mathematics community for maintaining a culture of shared knowledge. The practice of working through textbooks and sharing solutions has a long and valuable tradition, and I am grateful to everyone who contributes to this collective effort. I hope that this project, in its small way, adds to that tradition and proves useful to others in their study of abstract algebra. Please enjoy!

Contributors

As of now, I am independently forming solutions to each exercise myself. However, I welcome any corrections, suggestions, or slicker solutions from others. A (growing) list of contributors will be maintained here. If you'd like to contact me, please reach out on Reddit at my profile or via Discord, where my username is blanketism. Thank you to everyone who decided to take time out of their busy lives to contribute to this project. I appreciate your help in making it better for everyone.

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0 Preliminaries

0.1 Basics

For Exercise 0.1.1 through Exercise 0.1.4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathcal{B} = \{X \in \mathcal{A} \mid MX = XM\}$, where $\mathcal{A}$ denotes the set of 2×2 matrices with real entries.

Exercise 0.1.1

Determine which of the following elements of $\mathcal{A}$ lie in $\mathcal{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Let M_i denote the i-th matrix in the list. M_1, M_3, M_5, M_6 is M, the zero matrix, I_2, and the exchange matrix respectively. Among these, M_1, M_3, M_5 belong to $\mathcal{B}$, while $M_6 \notin \mathcal{B}$. Direct calculation shows M_2 and M_4 are not in $\mathcal{B}$. ■

Exercise 0.1.2

Prove that if $P, Q \in \mathcal{B}$, then $P + Q \in \mathcal{B}$ where $+$ denotes the usual sum of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(P+Q) = MP + MQ = PM + QM = (P+Q)M$, so $P + Q \in \mathcal{B}$. ■

Exercise 0.1.3

Prove that if $P, Q \in \mathcal{B}$, then $P \cdot Q \in \mathcal{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(PQ) = (MP)Q = (PM)Q = P(MQ) = P(QM) = (PQ)M$, so $PQ \in \mathcal{B}$. ■

Exercise 0.1.4

Find conditions on p, q, r, s which determine precisely when

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathcal{B}.$$

Solution. Let X be the matrix above. Computing $MX = XM$ gives the following system of equations:

$$\begin{cases} p + r = p \\ q + s = p + q \\ r = r \\ s = r + s \end{cases}$$

Simplifying results in $r = 0$ and $p = s$. Then

$$\mathcal{B} = \left\{ \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} \,\middle|\, p, q \in \mathbb{R} \right\}.$$ ■

Exercise 0.1.5

Determine whether the following functions f are well-defined:
(a) $f : \mathbb{Q} \to \mathbb{Z}$ defined by $f(a/b) = a$.
(b) $f : \mathbb{Q} \to \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.
(a) No: $1/2 = 2/4$, but $f(1/2) = 1 \neq 2 = f(2/4)$.
(b) Yes: $a/b = c/d$ implies $a^2/b^2 = c^2/d^2$, so $f(a/b) = f(c/d)$. ■

Exercise 0.1.6

Determine whether the function $f : \mathbb{R}^+ \to \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well-defined.

Solution. No. Note that $1 = 1.000\ldots = 0.999\ldots$, but $f(1.000\ldots) = 1$ and $f(0.999\ldots) = 9$. ■

Exercise 0.1.7

Let $f : A \to B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f.

Solution. That $\sim$ is an equivalence relation follows from $=$ being an equivalence relation.

The equivalence class of $a \in A$ is $\{x \in A \mid x \sim a\}$. By definition, this is $\{x \in A \mid f(x) = f(a)\}$, which is precisely the fiber of f over $f(a)$. ■

0.2 Properties of the Integers

Exercise 0.2.1

For each of the following pairs of integers a and b, determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y.

(a) $a = 20, b = 13$
(b) $a = 69, b = 372$
(c) $a = 792, b = 275$
(d) $a = 11391, b = 5673$
(e) $a = 1761, b = 1567$
(f) $a = 507885, b = 60808$

Solution.

	(a, b)	$\operatorname{lcm}(a, b)$	$ax + by$
(a)	1	260	$2(20) - 3(13)$
(b)	3	8556	$27(69) - 5(372)$
(c)	11	19800	$8(792) - 23(275)$
(d)	3	21540381	$253(5673) - 126(11391)$
(e)	1	2759487	$-105(1761) + 118(1567)$
(f)	691	44693880	$142(60808) - 17(507885)$

Here, the lcm was computed using $ab = (a, b)\operatorname{lcm}(a, b)$. ■

Exercise 0.2.2

Prove that if the integer k divides the integers a and b, then k divides $as + bt$ for every pair of integers s and t.

Solution. Since $k \mid a$ and $k \mid b$, there exist $x, y \in \mathbb{Z}$ such that $a = kx$ and $b = ky$. Then for any $s, t \in \mathbb{Z}$, we have

$$as + bt = (kx)s + (ky)t = k(xs + yt),$$

which shows that $k \mid (as + bt)$. ■

Exercise 0.2.3

Prove that if n is composite, then there are integers a and b such that n divides ab, but n does not divide either a or b.

Solution. Since n is composite, put $n = ab$ with $1 < a, b < n$. Then $n \mid ab$, but $n \nmid a$ and $n \nmid b$. ■

Exercise 0.2.4

Let a, b and N be fixed integers with a and b nonzero, and let $d = (a, b)$ be the greatest common divisor of a and b. Suppose x_0 and y_0 are particular solutions to $ax + by = N$. Prove, for any integer t, that the integers

$$x = x_0 + \frac{b}{d}t \quad \text{and} \quad y = y_0 - \frac{a}{d}t$$

are also solutions to $ax + by = N$.
REMARK. *This is, in fact, the general solution.*

Solution.

$$\begin{aligned} ax + by &= a\left(x_0 + \frac{b}{d}t\right) + b\left(y_0 - \frac{a}{d}t\right) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N \end{aligned}$$

■

Exercise 0.2.5

Determine the value $\varphi(n)$ for each integer $n \leq 30$, where φ denotes the Euler φ-function.

Solution.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	8	16	6	18	8	12	10	22	8	20	12	18	12	28	8

■

Exercise 0.2.6

Prove the Well Ordering Property of $\mathbb{Z}$ by induction, and prove the minimal element is unique.

Solution. Suppose S has no minimal element, and let T be $\mathbb{Z}^+ \setminus S$, the set of nonnegative integers not in S. Since 0 is minimal in $\mathbb{Z}^+$, then $0 \notin S$, hence $0 \in T$. Fix $k \in \mathbb{Z}^+$, and suppose every j such that $0 \leq j \leq k$ is in T. Observe $k + 1 \notin S$, for otherwise it would be minimal in S, since every integer less than or equal to k is in T. Then $k + 1 \in T$. By induction, every nonnegative integer is in T, contradicting that S is nonempty. Therefore, S must have a minimal element.

If $s_1, s_2 \in S$ are both minimal, then $s_1 \leq s_2$ and $s_2 \leq s_1$ imply $s_1 = s_2$. ■

Exercise 0.2.7

If p is a prime, prove that there do not exist nonzero integers a and b such that $a^2 = pb^2$.

Solution. Suppose there existed a, b such that $a^2 = pb^2$. If $(a, b) \neq 1$, divide both a and b by (a, b) to get relatively prime integers. Then $p \mid a^2$, which implies $p \mid a$. Put $a = pk$ for $k \in \mathbb{Z}$. Then

$$a^2 = (pk)^2 = p^2k^2 = pb^2,$$

hence $b^2 = pk^2$. But then $p \mid b$, contradicting that $(a, b) = 1$. ■

Exercise 0.2.8

Let p be a prime, $n \in \mathbb{Z}^+$. Find a formula for the largest power of p which divides $n! = n(n-1)(n-2)\dots 2\cdot 1$.
HINT. *It involves the greatest integer function.*

Solution. Let $k_1 \in \mathbb{Z}^+$ such that $n \geq k_1 p$. Then there are exactly k_1 factors of p, namely $p, 2p, \dots, k_1 p$. Let $k_2 \in \mathbb{Z}^+$ such that $n \geq k_2 p^2$, so that $p^2, 2p^2, \dots, k_2 p^2$ contribute exactly k_2 factors of p^2, and hence k_2 factors of p. Note that there exists some m such that $n \geq k_m p^m$, but $n < k_{m+1} p^{m+1}$. Then the largest power of p which divides $n!$ is given by

$$\sum_{i=1}^{m} k_i = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor. \qquad \blacksquare$$

Exercise 0.2.9

Write a computer program to determine the greatest common divisor (a, b) of two integers a and b and to express (a, b) in the form $ax + by$ for some integers x and y.

Solution. Please see `euclidean_algorithm.py` for the solution. $\blacksquare$

Exercise 0.2.10

Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ-function. Conclude in particular that φ tends to infinity as n tends to infinity.

Solution. Fix $N \in \mathbb{Z}^+$, and consider $n \in \mathbb{Z}^+$ such that $\varphi(n) = N$. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ be the prime factorization of n. Then

$$\varphi(n) = \varphi(p_1^{\alpha_1})\varphi(p_2^{\alpha_2})\cdots\varphi(p_k^{\alpha_k}) = \prod_{i=1}^{k} p_i^{\alpha_i - 1}(p_i - 1).$$

Since $\varphi(n) = N$, there are only finitely many possibilities for the primes p_i and their exponents α_i that satisfy this equation. Hence, there are only finitely many integers n such that $\varphi(n) = N$.

If φ was finite as n tends to infinity, then there would exist some $M \in \mathbb{Z}^+$ such that $\varphi(n) \leq M$ for infinitely many n. However, this contradicts the fact that for each fixed $N \in \mathbb{Z}^+$, there are only finitely many n such that $\varphi(n) = N$. Hence, φ must tend to infinity as n tends to infinity. $\blacksquare$

Exercise 0.2.11

Prove that if d divides n then $\varphi(d)$ divides $\varphi(n)$ where φ denotes Euler's φ-function.

Solution. Let $n \in \mathbb{Z}^+$ with $d \mid n$. Put their prime factorizations as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \quad d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_j^{\beta_j},$$

where $j \leq k$ and $1 \leq \beta_i \leq \alpha_i$ for all $1 \leq i \leq j$. Then $\varphi(d) \mid \varphi(n)$, since

$$\frac{\varphi(n)}{\varphi(d)} = \prod_{i=1}^{j} \frac{p_i^{\alpha_i - 1}(p_i - 1)}{p_i^{\beta_i - 1}(p_i - 1)} \cdot \prod_{i=j+1}^{k} p_i^{\alpha_i - 1}(p_i - 1) = \prod_{i=1}^{j} p_i^{\alpha_i - \beta_i} \cdot \prod_{i=j+1}^{k} p_i^{\alpha_i - 1}(p_i - 1) \in \mathbb{Z}. \qquad \blacksquare$$

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n

Exercise 0.3.1

Write down explicitly all the elements in the residue classes of $\mathbb{Z}/18\mathbb{Z}$.

Solution. Note that $\bar{x} = \{x + 18k \mid k \in \mathbb{Z}\}$. Explicitly, we have:

$$\begin{aligned}
\bar{0} &= \{0, 18, -18, 36, -36, \ldots\} & \bar{1} &= \{1, 19, -17, 37, -35, \ldots\} & \bar{2} &= \{2, 20, -16, 38, -34, \ldots\} \\
\bar{3} &= \{3, 21, -15, 39, -33, \ldots\} & \bar{4} &= \{4, 22, -14, 40, -32, \ldots\} & \bar{5} &= \{5, 23, -13, 41, -31, \ldots\} \\
\bar{6} &= \{6, 24, -12, 42, -30, \ldots\} & \bar{7} &= \{7, 25, -11, 43, -29, \ldots\} & \bar{8} &= \{8, 26, -10, 44, -28, \ldots\} \\
\bar{9} &= \{9, 27, -9, 45, -27, \ldots\} & \overline{10} &= \{10, 28, -8, 46, -26, \ldots\} & \overline{11} &= \{11, 29, -7, 47, -25, \ldots\} \\
\overline{12} &= \{12, 30, -6, 48, -24, \ldots\} & \overline{13} &= \{13, 31, -5, 49, -23, \ldots\} & \overline{14} &= \{14, 32, -4, 50, -22, \ldots\} \\
\overline{15} &= \{15, 33, -3, 51, -21, \ldots\} & \overline{16} &= \{16, 34, -2, 52, -20, \ldots\} & \overline{17} &= \{17, 35, -1, 53, -19, \ldots\}
\end{aligned}$$ ■

Exercise 0.3.2

Prove that the distinct equivalence classes in $\mathbb{Z}/n\mathbb{Z}$ are precisely $\bar{0}, \bar{1}, \bar{2}, \ldots, \overline{n-1}$.
HINT. *Use the Division Algorithm.*

Solution. Let $m \in \mathbb{Z}$. By the Division Algorithm, there exists unique $q, r \in \mathbb{Z}$ such that

$$m = nq + r, \quad 0 \le r < n.$$

Then $m \equiv r \bmod n$, so $m \in \bar{r}$. By definition, $\bar{r}$ is one of $\bar{0}, \bar{1}, \ldots, \overline{n-1}$. Thus, every equivalence class in $\mathbb{Z}/n\mathbb{Z}$ is one of $\bar{0}, \bar{1}, \ldots, \overline{n-1}$. Moreover, they are all distinct. Otherwise, if $\bar{x} = \bar{y}$ for $0 \le x, y < n$, then $x \equiv y \bmod n$, or $n \mid (x - y)$. However, since $-n < x - y < n$, we must have $x - y = 0$, or $x = y$. ■

Exercise 0.3.3

Prove that if

$$a = a_n 10^n + a_{n-1} 10^{n-1} + \cdots + a_1 10 + a_0$$

is any positive integer, then

$$a \equiv (a_n + a_{n-1} + \cdots + a_1 + a_0) \bmod 9.$$

HINT. *Use the fact that* $10 \equiv 1 \bmod 9$.
REMARK. *Note that this is the usual arithmetic rule that the remainder after division by 9 is the same as the sum of the decimal digits mod 9—in particular an integer is divisible by 9 if and only if the sum of its digits is divisible by 9.*

Solution. Using the hint, then

$$\begin{aligned}
a &\equiv (a_n 10^n + a_{n-1} 10^{n-1} + \cdots + a_1 10 + a_0) \bmod 9 \\
&\equiv (a_n 1^n + a_{n-1} 1^{n-1} + \cdots + a_1 1 + a_0) \bmod 9 \\
&= (a_n + a_{n-1} + \cdots + a_1 + a_0) \bmod 9
\end{aligned}$$ ■

Exercise 0.3.4

Compute the remainder when 37^{100} is divided by 29.

Solution. Modding 37^{100} by 29 results in 23. ■

Exercise 0.3.5

Compute the last two digits of 9^{1500}.

Solution. Modding 9^{1500} by 100 results in 1. ■

Exercise 0.3.6

Prove that the square of the elements in $\mathbb{Z}/4\mathbb{Z}$ are just $\bar{0}$ and $\bar{1}$.

Solution.

$$
\begin{aligned}
0^2 &= 0 \equiv 0 \bmod 4 \\
1^2 &= 1 \equiv 1 \bmod 4 \\
2^2 &= 4 \equiv 0 \bmod 4 \\
3^2 &= 9 \equiv 1 \bmod 4
\end{aligned}
$$

■

Exercise 0.3.7

Prove that for any integers a and b that $a^2 + b^2$ never leaves a remainder of 3 when divided by 4.
HINT. *Use Exercise 0.3.6.*

Solution. By Exercise 0.3.6, a^2 and b^2 are both either 0 mod 4 or 1 mod 4, so their sum can be:

$$
\begin{aligned}
0 + 0 &\equiv 0 \bmod 4 \\
0 + 1 &\equiv 1 \bmod 4 \\
1 + 0 &\equiv 1 \bmod 4 \\
1 + 1 &\equiv 2 \bmod 4
\end{aligned}
$$

In any of the sums, there is no remainder of 3 when dividing by 4. ■

Exercise 0.3.8

Prove that the equation $a^2 + b^2 = 3c^2$ has no solutions in nonzero integers a, b and c.
HINT. *Consider the equation mod 4 as in the previous two exercises and show that a, b and c would all have to be divisible by 2. Then each of a^2, b^2 and c^2 has a factor of 4 and by dividing through by 4 show that there would be a smaller set of solutions to the original equation. Iterate to reach a contradiction.*

Solution. We consider $a^2 + b^2 = 3c^2$ in mod 4. By Exercise 0.3.7, $a^2 + b^2$ is either 0, 1, or 2 mod 4. By Exercise 0.3.6, c^2 is either 0 or 1 mod 4, so that $3c^2$ is either 0 or 3 mod 4. Thus, both sides must be 0 mod 4.

It is clear that $4 \mid 3c^2$ implies $4 \mid c^2$, and hence $2 \mid c$. Observe that if a and b were both odd, then $a^2 + b^2 \equiv 2 \mod 4$. If one of a or b were odd and the other even, then $a^2 + b^2 \equiv 1 \mod 4$. Thus, both a and b must be even. Put $a = 2\alpha$, $b = 2\beta$, and $c = 2\gamma$ for nonzero $\alpha, \beta, \gamma \in \mathbb{Z}$. Substituting into the original equation gives

$$(2\alpha)^2 + (2\beta)^2 = 3(2\gamma)^2.$$

We may divide by 4 to obtain $\alpha^2 + \beta^2 = 3\gamma^2$, which is the same form as the original equation but with integers with smaller absolute values. Repeating this process leads to an infinite descent, which is impossible. Therefore, there are no solutions in nonzero integers a, b, and c. ■

Exercise 0.3.9

Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

Solution. Put $a = 2k+1$ for $k \in \mathbb{Z}$. Then $a^2 = 4k(k+1)+1 \equiv 1 \mod 8$ since $k(k+1)$ is always even. ■

Exercise 0.3.10

Prove that the number of elements of $(\mathbb{Z}/n\mathbb{Z})^\times$ is $\varphi(n)$, where φ denotes the Euler φ-function.

Solution. It suffices to show that $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ if and only if $(a, n) = 1$.

($\Rightarrow$) If $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then there exists $\bar{b} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{b} = \bar{1}$. Then $ab \equiv 1 \mod n$ implies the existence of $k \in \mathbb{Z}$ such that $ab - 1 = kn$, or $ab - kn = 1$. Thus, $(a, n) = 1$.

($\Leftarrow$) By $(a, n) = 1$, there exists $x, y \in \mathbb{Z}$ such that $ax + ny = 1$, or $ax = 1 - ny$. We then have $ax \equiv 1 \mod n$, so that $\bar{a} \cdot \bar{x} = \bar{1}$. Thus, $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.11

Prove that if $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Solution. Since $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, there exist $\bar{c}, \bar{d} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{c} = \bar{b} \cdot \bar{d} = \bar{1}$. Then

$$(\bar{a} \cdot \bar{b})(\bar{c} \cdot \bar{d}) = (\bar{a} \cdot \bar{c})(\bar{b} \cdot \bar{d}) = \bar{1} \cdot \bar{1} = \bar{1},$$

so that $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.12

Let $n \in \mathbb{Z}, n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove if a and n are not relatively prime, there exists an integer b with $1 \leq b < n$ such that $ab \equiv 0 \mod n$ and deduce that there cannot be an integer c such that $ac \equiv 1 \mod n$.

Solution. Let $(a, n) = d > 1$. Then $a/d, n/d \in \mathbb{Z}$ and $1 \leq a/d < n/d$. Put $b = n/d$, so that $1 \leq b < n$ and

$$ab = a(n/d) = (a/d)n \equiv 0 \mod n.$$

If there existed some $c \in \mathbb{Z}$ such that $ac \equiv 1 \mod n$, then $(ac)b \equiv b \mod n$. However, $ab \equiv 0 \mod n$, so that $0 \equiv b \mod n$, which is impossible since $1 \leq b < n$. ■

Exercise 0.3.13

Let $n \in \mathbb{Z}, n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove that if a and n are relatively prime then there is an integer c such that $ac \equiv 1 \bmod n$.
HINT. *Use the fact that the GCD of two integers is a $\mathbb{Z}$-linear combination of the integers.*

Solution. Since $(a, n) = 1$, there exists $c, x \in \mathbb{Z}$ such that $ac + nx = 1$. Then $ac \equiv 1 \bmod n$. ■

Exercise 0.3.14

Conclude from the previous two exercises that $(\mathbb{Z}/n\mathbb{Z})^\times$ is the set of elements $\overline{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $(a, n) = 1$ and hence prove Proposition 4. Verify this directly in the case $n = 12$.

Solution. The previous two exercises show that a and n are relatively prime if and only if there exists b such that $ab \equiv 1 \bmod n$, which is exactly the proposition. For $n = 12$, the elements $1, 5, 7, 11$ are relatively prime to 12, so that $(\mathbb{Z}/12\mathbb{Z})^\times = \{\overline{1}, \overline{5}, \overline{7}, \overline{11}\}$ whose inverses are $\overline{1}, \overline{7}, \overline{5}, \overline{11}$ respectively. ■

Exercise 0.3.15

For each of the following pairs of integers a and n, show that a is relatively prime to n and determine the multiplicative inverse of $\overline{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

(a) $a = 13, n = 20$
(b) $a = 69, n = 89$
(c) $a = 1891, n = 3797$
(d) $a = 6003722857, n = 77695236973$

Solution. Use the Euclidean Algorithm to find $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ for each pair (a, n). Then $ax \equiv 1 \bmod n$, and x is the multiplicative inverse of $\overline{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

(a) 17.
(b) 40.
(c) 253.
(d) 77695236753. ■

Exercise 0.3.16

Write a computer program to add and multiply mod n, for any n given as input. The output of these operations should be the least residues of the sums and products of two integers. Also include the feature that if $(a, n) = 1$, an integer c between 1 and $n - 1$ such that $\overline{a} \cdot \overline{c} = \overline{1}$ may be printed on request.
REMARK. *Your program should not, of course, simply quote "mod" functions already built into many systems.*

Solution. Please see `modular_arithmetic.py` for the solution. ■

1 Introduction to Groups

1.1 Basic Axioms and Examples

Let G be a group.

Exercise 1.1.1

Determine which of the following binary operations are associative.

(a) The operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$.
(b) The operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$.
(c) The operation $\star$ on $\mathbb{Q}$ defined by

$$a \star b = \frac{a+b}{5}.$$

(d) The operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \star (c, d) = (ad + bc,\ bd)$.
(e) The operation $\star$ on $\mathbb{Q} - \{0\}$ defined by

$$a \star b = \frac{a}{b}.$$

Solution.

(a) Not associative: $(1 \star 0) \star 1 = 0 \neq 2 = 1 \star (0 \star 1)$.
(b) Associative.
(c) Not associative: $(1 \star 0) \star 2 = 11/25 \neq 7/25 = 1 \star (0 \star 2)$.
(d) Associative.
(e) Not associative: $(1 \star 2) \star 3 = 1/6 \neq 3/2 = 1 \star (2 \star 3)$. ■

Exercise 1.1.2

Decide which of the binary operations in the preceding exercise are commutative.

(a) The operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$.
(b) The operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$.
(c) The operation $\star$ on $\mathbb{Q}$ defined by

$$a \star b = \frac{a+b}{5}.$$

(d) The operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \star (c, d) = (ad + bc,\ bd)$.
(e) The operation $\star$ on $\mathbb{Q} - \{0\}$ defined by

$$a \star b = \frac{a}{b}.$$

Solution.

(a) Not commutative: $1 - 0 \neq 0 - 1$.
(b) Commutative.
(c) Commutative.
(d) Commutative.
(e) Not commutative: $1/2 \neq 2/1$. ■

Exercise 1.1.3

Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well-defined).

Solution. Let $\bar{a}, \bar{b}, \bar{c} \in \mathbb{Z}/n\mathbb{Z}$. By associativity of addition in $\mathbb{Z}$, we have:

$$(\bar{a} + \bar{b}) + \bar{c} = \overline{(a+b)+c} = \overline{a+(b+c)} = \bar{a} + (\bar{b} + \bar{c}).$$ ■

Exercise 1.1.4

Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well-defined).

Solution. Use the same argument as in the previous exercise, replacing $+$ with $\cdot$. ■

Exercise 1.1.5

Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

Solution. Since $0 \cdot a = 0$ for all $a \in \mathbb{Z}$, so $\bar{a} \cdot \bar{0} = \bar{0}$. ■

Exercise 1.1.6

Determine which of the following sets are groups under addition:

(a) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd
(b) the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even
(c) the set of rational numbers of absolute value < 1
(d) the set of rational numbers of absolute value ≥ 1 together with 0
(e) the set of rational numbers with denominators equal to 1 or 2
(f) the set of rational numbers with denominators equal to 1, 2, or 3

Solution. Let S denote the set in each part.

(a) S has identity element 0, inverse element $-a/b$ for $a/b \in S$, and inherits associativity from $\mathbb{Q}$. Moreover, if $a/b, c/d \in S$ such that $(a,b) = (c,d) = 1$, then

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd},$$

where bd is odd. Since an odd number cannot have a factor of 2, the denominator remains odd when reduced to lowest terms. Hence, S is closed under addition, and it is a group.

(b) No: $1/2 \in S$, but $1/2 + 1/2 = 2/2 = 1/1 \notin S$.

(c) Same as (b).

(d) No: $3/2, 1 \in S$, but $3/2 - 1 = 1/2 \notin S$.

(e) Identity, inverses, and associativity are satisfied. Let $a, b \in \mathbb{Z}$. Then

$$\frac{a}{1} + \frac{b}{1} = \frac{a+b}{1}, \quad \frac{a}{2} + \frac{b}{2} = \frac{a+b}{2}, \quad \frac{a}{1} + \frac{b}{2} = \frac{2a+b}{2},$$

where the last case is chosen without loss of generality. Hence, S is closed.

(f) No: $1/3, 1/2 \in S$, but $1/3 + 1/2 = 5/6 \notin S$. ■

Exercise 1.1.7

Let $G = \{x \in \mathbb{R} \mid 0 \le x < 1\}$ and for $x, y \in G$ let $x \star y$ be the fractional part of $x + y$ (i.e., $x \star y = x + y - [x + y]$ where $[a]$ is the greatest integer less than or equal to a). Prove that $\star$ is a well-defined binary operation on G and that G is an Abelian group under $\star$.

Solution. Let $x, y, z \in G$.

- Note that $0 \le x + y < 2$. If $x + y < 1$, then $[x + y] = 0$, and $x \star y = x + y \in G$. If $1 \le x + y < 2$, then $[x + y] = 1$, and $x \star y = x + y - 1 \in G$. Hence, $x \star y \in G$, and $\star$ is well-defined.
- $x \star 0 = x - [x] = x$. Then 0 is the identity element of G under $\star$.
- $x \star (1 - x) = x + (1 - x) - [x + (1 - x)] = 0$. Hence, $1 - x$ is the inverse of x under $\star$.
- $x \star y = x + y - [x + y] = y + x - [y + x] = y \star x$. Hence, G is Abelian under $\star$.
- Note that $[\alpha + \beta] = [\alpha] + \beta$ for $\beta \in \mathbb{Z}$. Then

$$\begin{aligned}(x \star y) \star z &= (x + y - [x + y]) \star z \\ &= (x + y + z - [x + y]) - [x + y + z - [x + y]] \\ &= x + y + z - [x + y] - [x + y + z - [x + y]] \\ &= x + y + z - [x + y] - ([x + y + z] - [x + y]) \\ &= x + y + z - [x + y + z] \\ &= x \star (y + z - [y + z]) = x \star (y \star z).\end{aligned}$$

■

Exercise 1.1.8

Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.

(a) Prove that G is a group under multiplication (called the **group of roots of unity in** $\mathbb{C}$).
(b) Prove that G is not a group under addition.

Solution.

(a) Since $1^1 = 1$, then $1 \in G$. If $z \in G$ for some n, then $(z^{-1})^n = (z^n)^{-1} = 1^{-1} = 1$, so that $z^{-1} \in G$. Associativity from $\mathbb{C}$ is inherited by G. Finally, let $z, w \in G$ such that $z^n = w^m = 1$ for $n, m \in \mathbb{Z}^+$. Then

$$(zw)^{nm} = z^{nm}w^{nm} = (z^n)^m(w^m)^n = 1 \cdot 1 = 1,$$

so that $zw \in G$. Hence, G is a group under multiplication.

(b) No; note $1, -1 \in G$, but $1 + (-1) = 0 \notin G$ since $0^n = 0 \ne 1$ for any $n \in \mathbb{Z}^+$. ■

Exercise 1.1.9

Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.

(a) Prove that G is a group under addition.
(b) Prove that the nonzero elements of G are a group under multiplication.
HINT. *"Rationalize the denominators" to find multiplicative inverses.*

Solution.

(a) Note that addition of two elements in G is given by

$$(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2} \in G.$$

From this, we see that G is closed under addition. The identity is $0 + 0\sqrt{2} = 0$, and the inverse of $a + b\sqrt{2}$ is $-a - b\sqrt{2}$. Associativity is inherited from $\mathbb{R}$. Hence, G is a group under addition.

(b) Note that multiplication of two elements in G is given by

$$(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2} \in G.$$

From this, we see that G is closed under multiplication. The identity is $1 + 0\sqrt{2} = 1$, and the inverse of a nonzero element $a + b\sqrt{2}$ is

$$(a + b\sqrt{2})^{-1} = \frac{1}{a + b\sqrt{2}} = \frac{1}{a + b\sqrt{2}} \cdot \frac{a - b\sqrt{2}}{a - b\sqrt{2}} = \frac{a - b\sqrt{2}}{a^2 - 2b^2} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2}\sqrt{2}.$$

Observe that $a - b\sqrt{2} \neq 0$, since that would imply $a = b\sqrt{2}$, contradicting that $a \in \mathbb{Q}$. Moreover, $a + b\sqrt{2} \neq 0$ implies $a^2 - 2b^2 \neq 0$. Hence, the inverse is well-defined and in G. Associativity is inherited from $\mathbb{R}$. Hence, the nonzero elements of G are a group under multiplication. ■

Exercise 1.1.10

Prove that a finite group is Abelian if and only if its group table is a symmetric matrix.

Solution. Let $G = \{g_1, g_2, \ldots, g_n\}$ be a finite group.

($\Rightarrow$) If G is Abelian, then $g_i g_j = g_j g_i$ for all i, j. Hence, the (i, j)-th entry of the group table is equal to the (j, i)-th entry for all i, j, so that the group table is symmetric.

($\Leftarrow$) If the group table is symmetric, then $g_i g_j = g_j g_i$ for all i, j. Hence, G is Abelian. ■

Exercise 1.1.11

Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.

Solution. For each $\bar{x} \in \mathbb{Z}/12\mathbb{Z}$, add it to itself until we arrive at $\bar{0}$. $|\bar{0}| = 1$, and $\bar{1} = 12$ since $12 \cdot \bar{1} = \overline{12} = \bar{0}$. In particular, we have:

$\bar{x}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{6}$	$\bar{7}$	$\bar{8}$	$\bar{9}$	$\overline{10}$	$\overline{11}$
$\lvert\bar{x}\rvert$	1	12	6	4	3	12	2	12	3	4	6	12

■

Exercise 1.1.12

Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/12\mathbb{Z})^\times$: $\bar{1}, \overline{-1}, \bar{5}, \bar{7}, \overline{-7}, \overline{13}$.

Solution.

$\bar{x}$	$\bar{1}$	$\overline{-1}$	$\bar{5}$	$\bar{7}$	$\overline{-7}$	$\overline{13}$
$\lvert\bar{x}\rvert$	1	2	2	2	2	1

■

Exercise 1.1.13

Find the orders of the following elements of the additive group $\mathbb{Z}/36\mathbb{Z}$: $\bar{1}, \bar{2}, \bar{6}, \bar{9}, \overline{10}, \overline{12}, \overline{-1}, \overline{-10}, \overline{-18}$.

Solution.

$\bar{x}$	$\bar{1}$	$\bar{2}$	$\bar{6}$	$\bar{9}$	$\overline{10}$	$\overline{12}$	$\overline{-1}$	$\overline{-10}$	$\overline{-18}$
$\lvert\bar{x}\rvert$	36	18	6	4	18	3	36	18	2

■

Exercise 1.1.14

Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/36\mathbb{Z})^\times$: $\overline{1}, \overline{-1}, \overline{5}, \overline{13}, \overline{-13}, \overline{17}$.

Solution.

$\overline{x}$	$\overline{1}$	$\overline{-1}$	$\overline{5}$	$\overline{13}$	$\overline{-13}$	$\overline{17}$
$\lvert\overline{x}\rvert$	1	2	6	3	6	2

■

Exercise 1.1.15

Prove that $(a_1a_2\dots a_n)^{-1} = a_n^{-1}a_{n-1}^{-1}\dots a_1^{-1}$ for all $a_1, a_2, \dots, a_n \in G$.

Solution. We proceed by induction. This holds for $n = 2$ by Proposition 1.1 (iv). Suppose it holds for some $n \in \mathbb{Z}^+$. Then for $n+1$, we have

$$(a_1a_2\dots a_na_{n+1})^{-1} = a_{n+1}^{-1}(a_1a_2\dots a_n)^{-1} = a_{n+1}^{-1}a_n^{-1}a_{n-1}^{-1}\dots a_1^{-1}.$$

■

Exercise 1.1.16

Let x be an element of G. Prove that $x^2 = 1$ if and only if $|x|$ is either 1 or 2.

Solution.
($\Rightarrow$) Suppose $x^2 = 1$. Then $|x| \le 2$. Since $|x| \in \mathbb{Z}^+$, then $|x|$ is either 1 or 2.
($\Leftarrow$) If $|x| = 1$, then $x = 1$ and $x^2 = 1$. If $|x| = 2$, then $x^2 = 1$ by definition. ■

Exercise 1.1.17

Let x be an element of G. Prove that if $|x| = n$ for some positive integer n, then $x^{-1} = x^{n-1}$.

Solution. Note that $x^n = x^{n-1}x = 1$. Then $x^{n-1} = x^{-1}$. ■

Exercise 1.1.18

Let x, y be elements of G. Prove that the following are equivalent:
(i) $xy = yx$; (ii) $y^{-1}xy = x$; (iii) $x^{-1}y^{-1}xy = 1$.

Solution. $xy = yx \iff y^{-1}xy = y^{-1}yx = x \iff x^{-1}y^{-1}xy = x^{-1}x = 1$. ■

Exercise 1.1.19

Let $x \in G$ and let $a, b \in \mathbb{Z}^+$.
(a) Prove that $x^{a+b} = x^ax^b$ and $(x^a)^b = x^{ab}$.
(b) Prove that $(x^a)^{-1} = x^{-a}$.
(c) Establish part (a) for arbitrary integers a and b.

Solution.
(a) There are a instances of x in x^a and b instances of x in x^b. Hence, there are $a+b$ instances of x in x^ax^b, so that $x^{a+b} = x^ax^b$. There are a instances of x in x^a, and there are b instances of x^a in $(x^a)^b$. Hence, there are ab instances of x in $(x^a)^b$, so that $(x^a)^b = x^{ab}$.
(b) By Exercise 1.1.15, then $(x^a)^{-1}$ consists of a instances of x^{-1}, so that $(x^a)^{-1} = x^{-a}$.

(c) We consider the following cases:

- If $a, b \in \mathbb{Z}^+$, then the result follows from part (a).
- If $a \in \mathbb{Z}^+$ and $b < 0$ (the case of $a < 0$ and $b \in \mathbb{Z}^+$ is analogous), suppose $a + b \geq 0$. Then

$$x^a = x^{a+b-b} = x^{a+b}x^{-b},$$

so that $x^{a+b} = x^a x^b$. If $a + b < 0$, then

$$(x^{a+b})^{-1} = x^{-a-b} = x^{-b}x^{-a} = (x^a x^b)^{-1},$$

implying that $x^{a+b} = x^a x^b$, since inverses are unique. Moreover,

$$(x^a)^b = (x^a)^{-(-b)} = ((x^a)^{-b})^{-1} = (x^{-ab})^{-1} = x^{ab}.$$

- If $a < 0$ and $b < 0$, then $a + b < 0$ and

$$x^{a+b} = x^{b+a} = (x^{-b-a})^{-1} = (x^{-b}x^{-a})^{-1} = (x^{-a})^{-1}(x^{-b})^{-1} = x^a x^b.$$

Moreover,

$$(x^a)^b = ((x^{-a})^{-1})^b = ((x^{-a})^b)^{-1} = (x^{(-a)b})^{-1} = x^{ab}.$$ ■

Exercise 1.1.20

For x an element in G, show that x and x^{-1} have the same order.

Solution. Suppose $|x| = n \in \mathbb{Z}^+$. Then

$$(x^{-1})^n = (x^n)^{-1} = 1^{-1} = 1$$

so that $|x^{-1}| \leq n$ and has finite order. Similarly, suppose $|x^{-1}| = m \in \mathbb{Z}^+$. Then

$$x^m = ((x^{-1})^m)^{-1} = 1^{-1} = 1$$

so that $|x| \leq m$ and has finite order. Then $n = m$, and $|x| = |x^{-1}|$. Specifically, x has finite order if and only if x^{-1} has finite order, hence x has infinite order if and only if x^{-1} has infinite order. ■

Exercise 1.1.21

Let G be a finite group and let x be an element of order n. Prove that if n is odd, then $x = (x^2)^k$ for some $k \geq 1$.

Solution. Put $n = 2k - 1$ for $k \in \mathbb{Z}^+$. Then $x^n = x^{2k-1} = 1$, so that $x^{2k} = x$. Hence, $x = (x^2)^k$. ■

Exercise 1.1.22

If x and g are elements of the group G, prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.

Solution. We first prove the following lemma.

> ***Lemma.*** $(g^{-1}xg)^n = g^{-1}x^n g$ for all $n \in \mathbb{Z}^+$.
>
> *Proof.* We proceed by induction, where $n = 1$ is clear. If it holds for $n \in \mathbb{Z}^+$, then for $n+1$, we have
>
> $$(g^{-1}xg)^{n+1} = (g^{-1}xg)^n(g^{-1}xg) = (g^{-1}x^n g)(g^{-1}xg) = g^{-1}x^{n+1}g.$$
>
> Hence, the result follows by induction for all $n \in \mathbb{Z}^+$. ■

Let $|x| = m \in \mathbb{Z}^+$. Then

$$(g^{-1}xg)^m = g^{-1}x^m g = g^{-1}1g = 1,$$

so that $|g^{-1}xg| \le m$. Similarly, let $|g^{-1}xg| = n \in \mathbb{Z}^+$. Then

$$x^n = (gg^{-1})x^n(gg^{-1}) = g(g^{-1}xg)^n g^{-1} = g1g^{-1} = 1,$$

so that $|x| \le n$. Hence, $m = n$, and $|x| = |g^{-1}xg|$.

Let $a, b \in G$. Then $|ab| = |b^{-1}(ab)b| = |ba|$ by the above result. ■

Exercise 1.1.23

Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$.

Solution. Let $|x^s| = r \in \mathbb{Z}^+$. Then

$$(x^s)^t = x^{st} = x^n = 1,$$

so that $r \le t$. Moreover,

$$x^{sr} = (x^s)^r = 1$$

implies that $|x| = st \le sr$, hence $t \le r$. Thus, $|x^s| = r = t$. ■

Exercise 1.1.24

If a and b are commuting elements of G, prove that $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$.
HINT. *Do this by induction for positive n first.*

Solution. The case for $n = 0$ and $n = 1$ is clear. If it holds for some $n \in \mathbb{Z}^+$, then for $n+1$, we have

$$(ab)^{n+1} = (ab)^n(ab) = (a^n b^n)(ab) = a^n(b^n a)b = a^n(ab^n)b = a^{n+1}b^{n+1}.$$

For $n < 0$, then $(ab)^n = ((ab)^{-n})^{-1} = (b^{-n}a^{-n})^{-1} = a^n b^n$. ■

Exercise 1.1.25

Prove that if $x^2 = 1$ for all $x \in G$, then G is Abelian.

Solution. Since $x^2 = 1$, then $x = x^{-1}$ for all $x \in G$. Let $a, b \in G$. Then

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba. \quad ■$$

Exercise 1.1.26

Assume H is a nonempty subset of $(G, \star)$ which is closed under the binary operation on G and is closed under inverses, i.e., for all $h, k \in H$, hk, $h^{-1} \in H$. Prove that H is a group under the operation $\star$ restricted to H (such a subset H is called a **subgroup** of G).

Solution. Let $H \subseteq (G, \star)$ be nonempty and closed under the binary operation on G and inverses. Let $h \in H$. Then $h^{-1} \in H$, so that $hh^{-1} = 1 \in H$. Hence, H has an identity. Associativity is inherited from G. Hence, $(H, \star|_H)$ is a group. ∎

Exercise 1.1.27

Prove that if x is an element of the group G, then $\{x^n \mid n \in \mathbb{Z}\}$ is a subgroup of G (called the **cyclic subgroup** of G generated by x).

Solution. Let H be the set in question. Then $1 = x^0 \in H$, so that H is nonempty. Let $x^m, x^n \in H$. Then $x^m x^n = x^{m+n} \in H$ and $(x^m)^{-1} = x^{-m} \in H$. Hence, H is closed under the binary operation on G and inverses. By Exercise 1.1.26, then H is a subgroup of G. ∎

Exercise 1.1.28

Let $(A, \star)$ and $(B, \diamond)$ be groups and let $A \times B$ be their direct product (as defined in Example 6). Verify all the group axioms for $A \times B$:

(a) Prove that the associative law holds, i.e., for all $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$:

$$((a_1, b_1)(a_2, b_2))(a_3, b_3) = (a_1, b_1)((a_2, b_2)(a_3, b_3))$$

(b) Prove that $(1, 1)$ is the identity of $A \times B$, and

(c) Prove that the inverse of (a, b) is (a^{-1}, b^{-1}).

Solution.

(a) Let $(a_1, b_1), (a_2, b_2), (a_3, b_3) \in A \times B$. Then

$$\begin{aligned}
[(a_1, b_1)(a_2, b_2)](a_3, b_3) &= (a_1a_2, b_1b_2)(a_3, b_3) \\
&= ((a_1a_2)a_3, (b_1b_2)b_3) \\
&= (a_1(a_2a_3), b_1(b_2b_3)) \\
&= (a_1, b_1)(a_2a_3, b_2b_3) \\
&= (a_1, b_1)[(a_2, b_2)(a_3, b_3)]
\end{aligned}$$

(b) For $a \in A, b \in B$, we have $(a, b)(1, 1) = (a \star 1, b \diamond 1) = (a, b)$.

(c) $(a, b)(a^{-1}, b^{-1}) = (a \star a^{-1}, b \diamond b^{-1}) = (1, 1)$. ∎

Exercise 1.1.29

Prove that $A \times B$ is an Abelian group if and only if both A and B are Abelian.

Solution. Let $a, a' \in A$ and $b, b' \in B$. Suppose $A \times B$ is Abelian. Then

$$(aa', bb') = (a, b)(a', b') = (a', b')(a, b) = (a'a, b'b)$$

so that A and B are Abelian. If A and B are both Abelian, then

$$(a, b)(a', b) = (aa', bb') = (a'a, b'b) = (a', b')(a, b)$$

hence $A \times B$ is Abelian. ■

Exercise 1.1.30

Prove that the elements $(a, 1)$ and $(1, b)$ of $A \times B$ commute and deduce that the order of (a, b) is the least common multiple of $|a|$ and $|b|$.

Solution. Let $a \in A$ and $b \in B$. Then

$$(a, 1)(1, b) = (a \cdot 1, 1 \cdot b) = (a, b) = (1 \cdot a, b \cdot 1) = (1, b)(a, 1)$$

so that $(a, 1)$ and $(1, b)$ commute. Let $\ell = \operatorname{lcm}(|a|, |b|)$. Then

$$(a, b)^\ell = (a^\ell, b^\ell) = (1, 1)$$

so that $|(a, b)| \leq \ell$. Moreover, if $(a, b)^k = (1, 1)$ for some $k \in \mathbb{Z}^+$, then $a^k = 1$ and $b^k = 1$ so that both $|a|$ and $|b|$ divide k. Hence, ℓ divides k so that $\ell \leq k$. Thus, $|(a, b)| = \ell$. ■

Exercise 1.1.31

Prove that any finite group G of even order contains an element of order 2.
HINT. *Let $t(G)$ be the set $\{g \in G \mid g \neq g^{-1}\}$. Show that $t(G)$ has an even number of elements and every non-identity element of $G - t(G)$ has order 2.*

Solution. Let $g \in G$ with $g \neq g^{-1}$, so $g \in t(G)$. Since $(g^{-1})^{-1} = g$, then $g^{-1} \in t(G)$. Then elements of $t(G)$ come in pairs, so that $t(G)$ has even order. Since G has even order, then $G - t(G)$ has even order. Moreover, $1 \notin t(G)$ since $1^{-1} = 1$, hence G contains a non-identity element h. Then $h = h^{-1}$, so that $h^2 = 1$ and $|h| = 2$. ■

Exercise 1.1.32

If x is an element of finite order n in G, prove that the elements $1, x, x^2, \ldots, x^{n-1}$ are all distinct. Deduce that $|x| \leq |G|$.

Solution. Let $a, b \in \mathbb{Z}^+$ be such that $0 \leq a < b < n$. Suppose $x^a = x^b$. Then

$$1 = x^b x^{-a} = x^{b-a}$$

so that $|x| \leq b - a < n$, which is a contradiction. Hence, the elements $1, x, x^2, \ldots, x^{n-1}$ are all distinct. Since these elements are all in G, then $|x| = n \leq |G|$. ■

Exercise 1.1.33

Let x be an element of finite order n in G.

(a) Prove that if n is odd then $x^i \neq x^{-i}$ for all $i = 1, 2, \ldots, n - 1$.
(b) Prove that if $n = 2k$ and $1 \leq i < n$ then $x^i = x^{-i}$ if and only if $i = k$.

Solution.

(a) Note that $x^{n-i} = x^{-i}$. If $i \neq n-i$, then $x^i \neq x^{n-i}$ by Exercise 1.1.32, so $x^i \neq x^{-i}$. If $i = n-i$, then $2i = n$, which is impossible since n is odd. Hence, $x^i \neq x^{-i}$ for all $i = 1, 2, \ldots, n-1$.

(b) Part (a) shows that $x^i \neq x^{-i}$ for $i \neq k$. If $i = k$, the result follows. ■

Exercise 1.1.34

If x is an element of infinite order in G, prove that the elements $x^n, n \in \mathbb{Z}$ are all distinct.

Solution. If $x^a = x^b$ for $a, b \in \mathbb{Z}$ such that $a < b$, then $x^{b-a} = 1$, contradicting that x has infinite order. Hence, the elements $x^n, n \in \mathbb{Z}$ are all distinct. ■

Exercise 1.1.35

If x is an element of finite order n in G, use the Division Algorithm to show that any integral power of x equals one of the elements in the set $\{1, x, x^2, \ldots, x^{n-1}\}$.
REMARK. *These are the distinct elements of the cyclic subgroup of G generated by x.*

Solution. Let $|x| = n \in \mathbb{Z}^+$ and let $m \in \mathbb{Z}$. By the Division Algorithm, there exist unique integers q and r such that

$$m = qn + r$$

where $0 \leq r < n$. Then

$$x^m = x^{qn+r} = x^{qn}x^r = (x^n)^q x^r = 1^q x^r = x^r$$

so that any integral power of x equals one of the elements in the set $\{1, x, x^2, \ldots, x^{n-1}\}$. ■

Exercise 1.1.36

Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that G has no elements of order 4 (so by Exercise 1.1.32, every element has order ≤ 3). Use the cancellation laws to show that there is a unique group table for G. Deduce that G is Abelian.

Solution. Since G has even order, it has an element of order 2 by Exercise 1.1.31. Without loss of generality, let a be that element so that $a^2 = 1$. Since there are no elements of order 4, then the remaining elements b and c may have order 2 or 3.

We now observe that $ab \neq 1$ since that would imply that $b = a^{-1} = a$ contradicting that a and b are distinct elements. Similarly, $ab \neq a$ since that would imply that $b = 1$, and $ab \neq b$ since that would imply that $a = 1$. Thus, $ab = ba = c$. Similarly, $ac \neq 1, a, c$ so that $ac = ca = b$, hence $b^2 = (ca)(ac) = c^2$.

We now consider the possible orders of b and c. If $b^2 \neq 1$, then $|b| = 3$. But then $b^3 = bb^2 = bc = a$, contradicting that $|b| = 3$. Thus, $b^2 = 1$. Similarly, $c^2 = 1$. The group table for G is therefore as follows:

$\star$	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

from which we may deduce that G is Abelian. ■

1.2 Dihedral Groups

In these exercises, D_{2n} has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

Exercise 1.2.1

Compute the order of each of the elements in the following groups:
(a) D_6 (b) D_8 (c) D_{10}

Solution. For any reflection $sr^k \in D_{2n}$, it is clear that $sr^k \neq 1$. Moreover,

$$(sr^k)^2 = s(r^k s)r^k = s(sr^{-k})r^k = r^{-k}r^k = 1$$

so that every reflection has order 2. We now compute the orders of the rotations in each group.
(a) $D_6 = \{1, r, r^2, s, sr, sr^2\}$. Then $|1| = 1$ and $|r| = |r^2| = 3$.
(b) $D_8 = \{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$. Then $|1| = 1$, $|r^2| = 2$ and $|r| = |r^3| = 4$.
(c) $D_{10} = \{1, r, r^2, r^3, r^4, s, sr, sr^2, sr^3, sr^4\}$. Then $|1| = 1$ and $|r| = |r^2| = |r^3| = |r^4| = 5$. ■

Exercise 1.2.2

Use the generators and relations above to show that if x is any element of D_{2n} which is not a power of r, then $rx = xr^{-1}$.

Solution. If x is not a power of r, then x is a reflection, i.e., $x = sr^k$ for some $0 \leq k < n$. Then

$$rx = r(sr^k) = (rs)r^k = (sr^{-1})r^k = s(r^{-1}r^k) = sr^{k-1} = (sr^k)r^{-1} = xr^{-1}$$ ■

Exercise 1.2.3

Use the generators and relations above to show that every element of D_{2n} which is not a power of r has order 2. Deduce that D_{2n} is generated by the two elements s and sr, both of which have order 2.
HINT. *cf.* *Exercise 1.1.33*.

Solution. If an element is not a power of r, then it is a reflection. See Exercise 1.2.1 as to why every reflection has order 2. Moreover, for any $r^k, sr^k \in D_{2n}$ for $0 \leq k < n$, we may rewrite them as

$$r^k = (s(sr))^k \quad \text{and} \quad sr^k = s(s(sr))^k,$$

showing that every element of D_{2n} can be expressed in terms of s and sr. ■

Exercise 1.2.4

If $n = 2k$ is even and $n \geq 4$, show that $z = r^k$ is an element of order 2 which commutes with all elements of D_{2n}. Show also that z is the only non-identity element of D_{2n} which commutes with all elements of D_{2n}.
HINT. *cf.* *Exercise 1.1.33*.

Solution. Since $k \neq 1$, then $r^k \neq 1$. Moreover, $(r^k)^2 = r^n = 1$, hence $|r^k| = 2$. Since r^k commutes with other powers of r, consider $sr^j \in D_{2n}$ for $0 \leq j < n$. Then

$$r^k(sr^j) = (r^k s)r^j = (sr^{-k})r^j = s(r^{-k}r^j) = sr^{j-k} = (sr^j)r^k,$$

Now let $w \in D_{2n}$ such that $wx = xw$ for all $x \in D_{2n}$. If $w = r^m$ for $0 \le m < n$, then commuting with s gives $w = w^{-1}$, which implies $m = k$ by Exercise 1.1.33. If $w = sr^m$ instead, then commuting with r implies $r = r^{-1}$, which cannot be true since $n \ge 4$. ■

Exercise 1.2.5

If n is odd and $n \ge 3$, show that the identity is the only element of D_{2n} which commutes with all elements of D_{2n}.
HINT. *cf. Exercise 1.1.33.*

Solution. Using the same argument as the previous exercise and the odd case in Exercise 1.1.33, then w cannot be a rotation since $r^k \neq r^{-k}$. The reflection case is also the same as the previous exercise. ■

Exercise 1.2.6

Let x and y be elements of order 2 in any group G. Prove that if $t = xy$, then $tx = xt^{-1}$ (so that if $n = |xy| < \infty$ then x, y satisfy the same relations in G as s, r do in D_{2n}).

Solution. Since $|x| = |y| = 2$, then $x = x^{-1}$ and $y = y^{-1}$. Then

$$tx = (xy)x = x(yx) = x(xy)^{-1} = xt^{-1}$$ ■

Exercise 1.2.7

Show that $\langle a, b \mid a^2 = b^2 = (ab)^n = 1\rangle$ gives a presentation for D_{2n} in terms of the two generators a and b of order 2 computed in Exercise 1.2.3 above.
HINT. *Show that the relations for r and s follow from the relations for a and b and conversely, the relations for a and b follow from those for r and s.*

Solution. If $a^2 = b^2 = (ab)^n = 1$, put $r = ab$ and $s = a$. Then

$$r^n = (ab)^n = 1, \quad s^2 = a^2 = 1, \quad rs = (ab)a = a(ba) = a(b^{-1}a^{-1}) = a(ab)^{-1} = sr^{-1}.$$

Conversely, put $a = s$ and $b = sr$. The calculations are similar. ■

Exercise 1.2.8

Find the order of the cyclic subgroup of D_{2n} generated by r.

Solution. Since $|r| = n$, the cyclic subgroup generated by r has order n by Exercise 1.1.35. ■

Exercise 1.2.9

Let G be the group of rigid motions in $\mathbb{R}^3$ of a tetrahedron. Show that $|G| = 12$.

Solution. Label the vertices of a tetrahedron 1 through 4. Vertex 1 has 4 choices, vertex 2 has 3 remaining choices, and vertices 3 and 4 are determined after. It follows that $4(3) = 12$ symmetries. ■

Exercise 1.2.10

Let G be the group of rigid motions in $\mathbb{R}^3$ of a cube. Show that $|G| = 24$.

Solution. 8 choices for vertex 1 and 3 adjacent vertices for vertex 2; $8(3) = 24$ symmetries. ∎

Exercise 1.2.11

Let G be the group of rigid motions in $\mathbb{R}^3$ of an octahedron. Show that $|G| = 24$.

Solution. 6 choices for vertex 1 and 4 adjacent vertices for vertex 2; $6(4) = 24$ symmetries. ∎

Exercise 1.2.12

Let G be the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron. Show that $|G| = 60$.

Solution. 20 choices for vertex 1 and 3 adjacent vertices for vertex 2; $20(3) = 60$ symmetries. ∎

Exercise 1.2.13

Let G be the group of rigid motions in $\mathbb{R}^3$ of an icosahedron. Show that $|G| = 60$.

Solution. 12 choices for vertex 1 and 5 adjacent vertices for vertex 2; $12(5) = 60$ symmetries. ∎

Exercise 1.2.14

Find a set of generators for $\mathbb{Z}$.

Solution. Any integer is of the form $n1$, so $\mathbb{Z} = \langle 1 \rangle$. ∎

Exercise 1.2.15

Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

Solution. Each element of $\mathbb{Z}/n\mathbb{Z}$ is of the form $k\bar{1}$ for $0 \le k < n$. Then $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \mid n\bar{1} = \bar{0} \rangle$. ∎

Exercise 1.2.16

Show that the group $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1y_1)^2 = 1 \rangle$ is the dihedral group D_4 (where x_1 may be replaced by the letter r and y_1 by s).
HINT. *Show that the last relation is the same as $x_1y_1 = y_1x_1^{-1}$.*

Solution. Note that $y_1^2 = 1$, then $y_1^{-1} = y_1$. Then

$$(x_1y_1)^2 = 1 \iff x_1y_1x_1y_1 = 1 \iff x_1y_1x_1 = y_1^{-1} = y_1 \iff x_1y_1 = y_1x_1^{-1}$$

Then the relations of D_4 are satisfied by x_1 and y_1 if we set $r = x_1$ and $s = y_1$. ∎

Exercise 1.2.17

Let X_{2n} be the group whose presentation is displayed in (1.2).

$$X_{2n} = \langle x, y \mid x^n = y^2 = 1, xy = yx^2\rangle$$

(a) Show that if $n = 3k$, then X_{2n} has order 6, and it has the same generators and relations as D_6 when x is replaced by r and y by s.
(b) Show that if $(3, n) = 1$, then x satisfies the additional relation: $x = 1$. In this case deduce that X_{2n} has order 2.
HINT. *Use the facts that $x^n = 1$ and $x^3 = 1$.*

Solution.

(a) If $n = 3k$, then

$$X_{2n} = X_{6k} = \langle x, y \mid x^{3k} = y^2 = 1, xy = yx^2\rangle$$

As shown in the text, $x^3 = 1$. Assume the relations of X_{6k} hold. Set $r = x$ and $s = y$, so that $r^3 = x^3 = 1$, $s^2 = y^2 = 1$, and

$$rs = xy = yx^2 = sr^2 = sr^{-1}.$$

Now suppose the relations of D_6 hold. Then

$$x^n = r^{3k} = (r^3)^k = 1^k = 1, \quad y^2 = s^2 = 1$$

and

$$xy = rs = sr^{-1} = sr^2 = yx^2$$

Thus, the two presentations are equivalent, and $|X_{6k}| = |D_6| = 6$.

(b) Since $(3, n) = 1$, there exist integers a and b such that $3a + nb = 1$. From the relations of X_{2n}, we have $x^3 = 1$ and $x^n = 1$. Then

$$x = x^{3a+nb} = (x^3)^a(x^n)^b = 1^a1^b = 1$$

so that $X_{2n} = 1, y$ where $y^2 = 1$. It follows that $|X_{2n}| = 2$. ■

Exercise 1.2.18

Let Y be the group whose presentation is displayed in (1.3).

$$Y = \langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2\rangle.$$

(a) Show that $v^2 = v^{-1}$.
HINT. *Use the relation $v^3 = 1$.*
(b) Show that v commutes with u^3.
HINT. *Show that $v^2u^3v = u^3$ by writing the left-hand side as $(v^2u^2)(uv)$ and using the relations to reduce this to the right-hand side. Then use part (a).*
(c) Show that v commutes with u.
HINT. *Show that $u^9 = u$ and then use part (b).*
(d) Show that $uv = 1$.
HINT. *Use part (c) and the last relation.*
(e) Show that $u = 1$. Deduce that $v = 1$, and conclude that $Y = 1$.
HINT. *Use part (d) and the equation $u^4v^3 = 1$.*

Solution.

(a) $v^3 = 1 \implies v^2 = v^{-1}$.

(b) From $uv = v^2u^2$, we have $vuv = u^2$ so that $vu = u^2v^{-1} = u^2v^2$. It follows that

$$v^2u^3v = (v^2u^2)(uv) = (uv)(uv) = u(vu)v = u(u^2v^2)v = u^3.$$

(c) It follows that $u^9 = (u^4)^2u = u$. Then $uv = (u^3)^3v = v(u^3)^3 = vu$.

(d) $uv = v^2u^2 = u^2v^2$. Then $1 = uv$.

(e) $u^4v^3 = u(uv)^3 = u1^3 = 1$. Then $1v = 1$, and $u = v = 1$, so $Y = 1$. ■

1.3 Symmetric Groups

Exercise 1.3.1

Let σ be the permutation

$$1 \mapsto 3 \quad 2 \mapsto 4 \quad 3 \mapsto 5 \quad 4 \mapsto 2 \quad 5 \mapsto 1$$

and let τ be the permutation

$$1 \mapsto 5 \quad 2 \mapsto 3 \quad 3 \mapsto 2 \quad 4 \mapsto 4 \quad 5 \mapsto 1$$

Find the cycle decompositions of each of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution. We have $\sigma = (1\ 3\ 5)(2\ 4)$ and $\tau = (1\ 5)(2\ 3)$. The rest is straightforward. ■

Exercise 1.3.2

Let σ be the permutation

$$\begin{array}{ccccc} 1 \mapsto 13 & 2 \mapsto 2 & 3 \mapsto 15 & 4 \mapsto 14 & 5 \mapsto 10 \\ 6 \mapsto 6 & 7 \mapsto 12 & 8 \mapsto 3 & 9 \mapsto 4 & 10 \mapsto 1 \\ 11 \mapsto 7 & 12 \mapsto 9 & 13 \mapsto 5 & 14 \mapsto 11 & 15 \mapsto 8 \end{array}$$

and let τ be the permutation

$$\begin{array}{ccccc} 1 \mapsto 14 & 2 \mapsto 9 & 3 \mapsto 10 & 4 \mapsto 2 & 5 \mapsto 12 \\ 6 \mapsto 6 & 7 \mapsto 5 & 8 \mapsto 11 & 9 \mapsto 15 & 10 \mapsto 3 \\ 11 \mapsto 8 & 12 \mapsto 7 & 13 \mapsto 4 & 14 \mapsto 1 & 15 \mapsto 13 \end{array}$$

Find the cycle decompositions of the following permutations: $\sigma, \tau, \sigma^2, \sigma\tau, \tau\sigma$, and $\tau^2\sigma$.

Solution. $\sigma = (1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)$ and $\tau = (1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11)$. Again, the rest is straightforward. ■

Exercise 1.3.3

For each of the permutations whose cycle decompositions were computed in the preceding two exercises, compute its order.

Solution.

$$\begin{aligned} |\sigma| &= 6, 12 & |\tau| &= 2, 30 \\ |\sigma^2| &= 3, 6 & |\sigma\tau| &= 4, 6 \\ |\tau\sigma| &= 4, 6 & |\tau^2\sigma| &= 6, 13 \end{aligned}$$

■

Exercise 1.3.4

Compute the order of each of the elements in the following groups:
(a) S_3 (b) S_4

Solution.
(a)

Permutation	Order
1	1
(12)	2
(13)	2
(23)	2
(123)	3
(132)	3

(b)

Permutation	Order	Permutation	Order	Permutation	Order	Permutation	Order
1	1	(34)	2	(143)	3	(1342)	4
(12)	2	(123)	3	(234)	3	(1423)	4
(13)	2	(124)	3	(243)	3	(1432)	4
(14)	2	(132)	3	(1234)	4	(12)(34)	2
(23)	2	(134)	3	(1243)	4	(13)(24)	2
(24)	2	(142)	3	(1324)	4	(14)(23)	2

■

Exercise 1.3.5

Find the order of $(1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9)$.

Solution. There are cycles of lengths 5, 2, 3, and 2. Then the order is $\operatorname{lcm}(5, 2, 3, 2) = 30$. ■

Exercise 1.3.6

Write out the cycle decomposition of each element of order 4 in S_4.

Solution. See Exercise 1.3.4. ■

Exercise 1.3.7

Write out the cycle decomposition of each element of order 2 in S_4.

Solution. See Exercise 1.3.4. ■

Exercise 1.3.8

Prove that if Ω is a finite set, then S_Ω is a finite group. Prove that if $\Omega = \{1, 2, 3, \dots\}$, then S_Ω is an infinite group (do not say $\infty! = \infty$).

Solution. For each $n \in \mathbb{Z}^+$, consider the permutation $\sigma_n = (1\ n)$. Then $\sigma_n \in S_\Omega$ and $\sigma_n \neq \sigma_m$ for $n \neq m$, since they move different elements. Hence, S_Ω contains an infinite collection of distinct elements, so S_Ω is infinite. ■

Exercise 1.3.9

Let σ be an m-cycle. For which positive integers i is σ^i also an m-cycle?

(a) Let σ be the 12-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$. For which positive integers i is σ^i also a 12-cycle?

(b) Let τ be the 8-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8)$. For which positive integers i is τ^i also an 8-cycle?

(c) Let ω be the 14-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12\ 13\ 14)$. For which positive integers i is ω^i also a 14-cycle?

MANUAL NOTE. *The reader should notice a pattern: integers i such that $(m, i) = 1$ produce an m-cycle, while integers i such that $(m, i) = k$ produce k disjoint m/k-cycles.*

Solution.

(a) Computing the cycle for each integer 0 to 12:

$$\begin{aligned}
\sigma^1 &= \sigma & \sigma^7 &= (1\ 8\ 3\ 10\ 5\ 12\ 7\ 2\ 9\ 4\ 11\ 6)\\
\sigma^2 &= (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) & \sigma^8 &= (1\ 9\ 5)(2\ 10\ 6)(3\ 11\ 7)(4\ 12\ 8)\\
\sigma^3 &= (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) & \sigma^9 &= (1\ 10\ 7\ 4)(2\ 11\ 8\ 5)(3\ 12\ 9\ 6)\\
\sigma^4 &= (1\ 5\ 9)(2\ 6\ 10)(3\ 7\ 11)(4\ 8\ 12) & \sigma^{10} &= (1\ 11\ 9\ 7\ 5\ 3)(2\ 12\ 10\ 8\ 6\ 4)\\
\sigma^5 &= (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8) & \sigma^{11} &= (1\ 12\ 11\ 10\ 9\ 8\ 7\ 6\ 5\ 4\ 3\ 2)\\
\sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) & \sigma^{12} &= \mathrm{id}
\end{aligned}$$

Hence, the set of integers that produce a 12-cycle is the set $\{i + 12k \mid i \in \{1, 5, 7, 11\}, k \in \mathbb{Z}\}$.

(b) $\{i + 8k \mid i \in \{1, 3, 5, 7\}, k \in \mathbb{Z}\}$.

(c) $\{i + 14k \mid i \in \{1, 3, 5, 9, 11, 13\}, k \in \mathbb{Z}\}$. ■

Exercise 1.3.10

Prove that if σ is the m-cycle $(a_1\ a_2\ \dots\ a_m)$, then for all $i \in \{1, 2, \dots, m\}$, $\sigma^i(a_k) = a_{k+i}$, where $k + i$ is replaced by its least residue mod m when $k + i > m$. Deduce that $|\sigma| = m$.

Solution. Assume all subscripts of a are taken mod m. Inducting on $i = 1$, the base case is clear. If $\sigma^i(a_k) = a_{k+i}$ for $1 \le i \le m - 1$, then

$$\sigma^{i+1}(a_k) = \sigma(\sigma^i(a_k)) = \sigma(a_{k+i}) = a_{k+i+1}.$$

The result follows by induction.

To see that $|\sigma| = m$, note that $\sigma^m(a_k) = a_{k+m} = a_k$ for all $k \in \{1, 2, \dots, m\}$, so that $\sigma^m = \mathrm{id}$. If $|\sigma| = i < m$, then $\sigma^i(a_1) = a_{1+i} = a_1$, which implies that $i \equiv 0 \bmod m$. Hence, $|\sigma| = m$. ■

Exercise 1.3.11

Let σ be the m-cycle $(1\ 2\ \dots\ m)$. Show that σ^i is also an m-cycle if and only if i is relatively prime to m.

Solution.

($\Rightarrow$) Suppose $(m, i) = k > 1$. If $i > m$, replace i with i mod m since $\sigma^i = \sigma^{i \bmod m}$. Then $i = k\ell$ for $\ell \in \mathbb{Z}^+$. It is clear that $(\sigma^i)^{m/k} = \sigma^{\ell m} = \mathrm{id}$, so that $|\sigma^i| \le m/k < m$. Hence, σ^i is not an m-cycle.

($\Leftarrow$) Suppose that $(m, i) = 1$. Denote $x_m \equiv x \bmod m$. We claim that the integers

$$1_m, (1+i)_m, (1+2i)_m, \ldots, (1+(m-1)i)_m$$

are all distinct. Suppose $(1+xi)_m = (1+yi)_m$ for some $0 \le x, y \le m-1$ with $x \neq y$. Then $i(x-y) \equiv 0 \bmod m$. Since $(m, i) = 1$, then $m \mid (x-y)$. However, since $1-m \le x-y \le m-1$, then the only choice for $x-y$ is 0, or that $x = y$. Then we have exactly m distinct integers so that

$$\sigma^i = (1\ (1+i)_m\ (1+2i)_m\ \ldots\ (1+(m-1)i)_m).$$

Thus, σ^i is an m-cycle. ■

Exercise 1.3.12

(a) If $\tau = (1\,2)(3\,4)(5\,6)(7\,8)(9\,10)$, determine whether there is an n-cycle ($n \ge 10$) with $\tau = \sigma^k$ for some integer k.

(b) If $\tau = (1\ 2)(3\ 4\ 5)$, determine whether there is an n-cycle ($n \ge 5$) with $\tau = \sigma^k$ for some integer k.

Solution.

(a) We first observe that for any n-cycle σ, then σ^k sends any element a_i to a_{i+k}, where $i+k$ is replaced by its least residue mod n when $i+k > n$. In this specific problem, τ consists of 2-cycles, which means that σ^k must send each element to itself after two applications. Then $2k \equiv 0 \bmod n$, but $k \not\equiv 0 \bmod n$ since $\sigma^k \neq 1$. It follows that $n \mid 2k$ but $n \nmid k$, so it must be that $n = 2k$. We see that σ^k will be a product of k 2-cycles. However, τ is a product of 5 2-cycles, so $k = 5$ and $n = 10$. Thus, there exists an n-cycle σ such that $\sigma^5 = \tau$ when $n = 10$, and $\sigma = (1\ 6\ 2\ 7\ 3\ 8\ 4\ 9\ 5\ 10)$.

(b) We first prove that $(ac, bc) = c(a, b)$ for $a, b, c \in \mathbb{Z}$.

> ***Lemma.*** L
> *Proof.* et $d = (a, b)$ and $d' = (ac, bc)$. Then there exist $s, t \in \mathbb{Z}$ such that $d = as + bt$. Consider $cd = acs + bct$. Then $d' = (ac, bc)$ implies that $d' \mid cd$. Since $d \mid a$ and $d \mid b$, then $cd \mid ac$ and $cd \mid bc$, so that $cd \mid (ac, bc) = d'$. Then $d' = cd$ or $d' = c(a, b)$. ■

Assume for contradiction that there exists an n-cycle σ and k such that $\sigma^k = \tau$. Then $\tau^2(1) = 1$, hence $\sigma^{2k}(1) = 1$, or $2k \equiv 0 \bmod n$. Moreover, $\tau^3(3) = 3$, hence $\sigma^{3k}(3) = 3$, or $3k \equiv 0 \bmod n$. Then $n \mid 2k$ and $n \mid 3k$, so that $n \mid (2k, 3k)$. By the lemma, $(2k, 3k) = k(2, 3) = k$, so that $n \mid k$. However, $\sigma^k \neq 1$, so $n \nmid k$. This is a contradiction, so there does not exist an n-cycle σ such that $\sigma^k = \tau$. ■

Exercise 1.3.13

Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles.

Solution.

($\Rightarrow$) Let $\sigma \in S_n$ with $|\sigma| = 2$. Let $(a_1\ a_2\ \ldots\ a_m)$ be one of the cycles in its cycle decomposition. Since $\sigma \neq \mathrm{id}$ and $\sigma^2 = 1$, then $\sigma(a_1) = a_2$ and $\sigma(a_2) = a_1$. Then $m = 2$, and the cycle is a 2-cycle.

($\Leftarrow$) Suppose σ's cycle decomposition is a product of commuting 2-cycles $(a_1\ b_1)(a_2\ b_2)\ldots(a_k\ b_k)$. Since each 2-cycle has order 2, and the 2-cycles commute, then $\sigma^2 = \mathrm{id}$. Because $\sigma \neq \mathrm{id}$, then $|\sigma| = 2$. ■

Exercise 1.3.14

Let p be a prime. Show that an element has order p in S_n if and only if its cycle decomposition is a product of commuting p-cycles. Show by an explicit example that this need not be the case if p is not prime.

Solution.

($\Rightarrow$) Let $\sigma \in S_n$ with $|\sigma| = p$. Let $(a_1\ a_2\ \dots\ a_m)$ be one of the cycles in its cycle decomposition. By Exercise 1.3.10, we know $\sigma^i(a_i) = a_{1+i}$ where $1+i$ is taken mod m when $1+i > m$. Then $\sigma^p(a_1) = a_{1+p} = a_1$, so that $m \mid p$. Since p is prime, then $m = p$, hence any cycle in the cycle decomposition of σ is a p-cycle.

($\Leftarrow$) Let $\sigma = \tau_1\tau_2\dots\tau_k$ be the cycle decomposition of σ, where each τ_i is a p-cycle. Since the cycles are disjoint, they commute. Moreover, τ_i^t is a non-trivial p-cycle for $t < p$ by Exercise 1.3.11. Then $\sigma^t = \tau_1^t\tau_2^t\dots\tau_k^t$ is a product of non-trivial p-cycles for $t < p$, so that $\sigma^t \neq 1$. However, $\sigma^p = \tau_1^p\tau_2^p\dots\tau_k^p = 1$, so that $|\sigma| = p$.

For a counterexample when p is not prime, consider $\sigma = (1\ 2\ 3)(4\ 5)$ in S_5. Then $|\sigma| = 6$, but the cycle decomposition consists of a 3-cycle and a 2-cycle. ■

Exercise 1.3.15

Prove that the order of an element in S_n equals the least common multiple of the lengths of the cycles in its cycle decomposition.

Solution. Let $\sigma \in S_n$ have cycle decomposition $\tau_1\tau_2\dots\tau_k$, where each τ_i is a cycle of length m_i. Then $|\tau_i| = m_i$ by Exercise 1.3.10. Since the cycles are disjoint, they commute, so that $\sigma^t = \tau_1^t\tau_2^t\dots\tau_k^t$. Then $\sigma^t = 1$ if and only if $\tau_i^t = 1$ for all i, which occurs if and only if $m_i \mid t$ for all i. Hence, the order of σ is the least common multiple of the lengths of the cycles in its cycle decomposition. ■

Exercise 1.3.16

Show that if $n \geq m$ then the number of m-cycles in S_n is given by

$$\frac{n(n-1)(n-2)\dots(n-m+1)}{m}.$$

HINT. *Count the number of ways of forming an m-cycle and divide by the number of representations of a particular m-cycle.*

Solution. Note that in an m-cycle, there are m integers to write. The first integer has n choices, the second $n-1$ choices. Continuing on, then the m-th integer has $n-m+1$ choices. However, each choice of m integers has m equivalent representations (such as $(1\,2\,3\,4)$ being equivalent to $(2\,3\,4\,1)$ by shifting each integer to the left one place). Then take the product $n(n-1)(n-2)\dots(n-m+1)$ and divide by m. ■

Exercise 1.3.17

Show that if $n \geq 4$ then the number of permutations in S_n which are the product of two disjoint 2-cycles is

$$\frac{n(n-1)(n-2)(n-3)}{8}.$$

Solution. The first 2-cycle has $n(n-1)/2$ choices, and the second 2-cycle has $(n-2)(n-3)/2$ choices. Moreover, there are 2 representations for the same set of 2-cycles, so multiply the given quantities and divide by 2. ■

Exercise 1.3.18

Prove that if $n \geq 2m$ then the number of permutations in S_n which are the product of m disjoint 2-cycles is given by Find all numbers n such that S_5 contains an element of order n.

Solution. S_5 contains elements of order 1 through 5. Moreover, we can have a 2 and 3-cycle together whose LCM is 6. Then $n = 1, 2, 3, 4, 5, 6$. ■

Exercise 1.3.19

Find all numbers n such that S_7 contains a number an element of order n.

Solution. We have n from 1 to 7. We can have 2 and 5-cycles and 3 and 4-cycles. Then $n = 10$ and 12. ■

Exercise 1.3.20

Find a set of generators and relations for S_3.

Solution. Recall that $S_3 = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$. Moreover, no element in S_3 has order 6 so that no element generates S_3. It must be that there are at least 2 generators. Put $\alpha = (1\ 2)$ and $\beta = (1\ 3)$. Then $\alpha^2 = \beta^2 = 1$. Then $|\alpha| = |\beta| = 2$, and $\alpha\beta = (1\ 3\ 2), \beta\alpha = (1\ 2\ 3)$, and $\alpha\beta\alpha = (2\ 3)$. Since $\alpha^2 = \beta^2 = 1$ is not enough to determine the order of $\alpha\beta$ (because $\alpha\beta \neq \beta\alpha$), then we must add $(\alpha\beta)^2 = 1$. Then we have the presentation $S_3 = \langle \alpha, \beta \mid \alpha^2 = \beta^2 = (\alpha\beta)^3 = 1 \rangle$. ■

1.4 Matrix Groups

Let F be a field and let $n \in \mathbb{Z}^+$.

Exercise 1.4.1

Prove that $|\operatorname{GL}_2(\mathbb{F}_2)| = 6$.

Solution. Direct calculation gives:

$$\operatorname{GL}_2(\mathbb{F}_2) = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \right\}. \qquad \blacksquare$$

Exercise 1.4.2

Write out all the elements of $\operatorname{GL}_2(\mathbb{F}_2)$ and compute the order of each element.

Solution. By direct calculation, we obtain the following:

A	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$
$\lvert A \rvert$	1	2	2	2	3	3

$\blacksquare$

Exercise 1.4.3

Show that $\operatorname{GL}_2(\mathbb{F}_2)$ is non-Abelian.

Solution. It is trivial to see that the following two matrices do not commute, hence $\operatorname{GL}_2(\mathbb{F}_2)$ is non-Abelian.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \qquad \blacksquare$$

Exercise 1.4.4

Show that if n is not prime then $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Solution. Since n is not prime, it must be composite. Then there exist $a, b \in \mathbb{Z}^+$ such that $1 < a, b < n$ and $n = ab$. Moreover, $(a, n) \neq 1$ since $a \mid n$. Then a has no multiplicative inverse in $\mathbb{Z}/n\mathbb{Z}$, so $\mathbb{Z}/n\mathbb{Z}$ is not a field. $\blacksquare$

Exercise 1.4.5

Show that $\operatorname{GL}_n(F)$ is a finite group if and only if F has a finite number of elements.

Solution.

($\Rightarrow$) If $\operatorname{GL}_n(F)$ is finite, the subset $\{\alpha I \mid \alpha \in F\}$ must also be finite, which implies that F has a finite number of elements.

($\Leftarrow$) Suppose $|F| = q$. Then an $n \times n$ matrix over F has q^{n^2} possible entries. Since $\operatorname{GL}_n(F)$ is a subset of the set of all $n \times n$ matrices over F, it follows that $\operatorname{GL}_n(F)$ is finite. $\blacksquare$

Exercise 1.4.6

If $|F| = q$ is finite, prove that $|\operatorname{GL}_n(F)| < q^{n^2}$.

Solution. See ($\Leftarrow$) in Exercise 1.4.5. ■

Exercise 1.4.7

Let p be a prime. Prove that the order of $\operatorname{GL}_2(\mathbb{F}_p)$ is $p^4 - p^3 - p^2 + p$.
REMARK. *Do not just quote the order formula in this section.*
HINT. *Subtract the number of 2×2 matrices which are* not *invertible from the total number of 2×2 matrices over $\mathbb{F}_p$. You may use the fact that a 2×2 matrix is not invertible if and only if one row is a multiple of the other.*

Solution. Observe that there are p^4 total 2×2 matrices over $\mathbb{F}_p$. The number of non-invertible matrices is as follows:

- If the first row is $(0,0)$, then there are p^2 choices for the second row.
- If the first row is $(a,b) \neq (0,0)$, then there are p choices for the second row, since it must be a multiple of the first row. There are $p^2 - 1$ choices for the first row.

Then the number of non-invertible matrices is $p^2 + p(p^2-1) = p^3 + p^2 - p$. Subtracting from the total number of matrices, we have the desired number. ■

Exercise 1.4.8

Show that $\operatorname{GL}_n(F)$ is non-Abelian for any $n \geq 2$ and any F.

Solution. Consider the subset $S = \{0,1\}$ of F. Then $\operatorname{GL}_n(S)$ is a subset of $\operatorname{GL}_n(F)$. We proceed by induction on this subset. For $n = 2$, we have $\operatorname{GL}_2(S)$ is non-Abelian by Exercise 1.4.3. Suppose that $\operatorname{GL}_n(S)$ is non-Abelian for some $n \geq 2$. Then there exist $A, B \in \operatorname{GL}_n(S)$ such that $AB \neq BA$. Consider the matrices

$$A_0 = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, B_0 = \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix} \in \operatorname{GL}_{n+1}(S).$$

Observe that A_0B_0 contains AB as a block matrix, while B_0A_0 contains BA as a block matrix. Since $AB \neq BA$, then $A_0B_0 \neq B_0A_0$. Hence, $\operatorname{GL}_{n+1}(S)$ is non-Abelian. By induction, $\operatorname{GL}_n(S)$ is non-Abelian for all $n \geq 2$. Since $\operatorname{GL}_n(S) \subseteq \operatorname{GL}_n(F)$, then $\operatorname{GL}_n(F)$ is non-Abelian for all $n \geq 2$. ■

Exercise 1.4.9

Prove that the binary operation of matrix multiplication of 2×2 matrices with real number entries is associative.

Solution. The direct proof is laborious. Instead, associate with any 2×2 matrix some linear map. Since composition of linear maps is associative, the result follows. ■

Exercise 1.4.10

Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \,\middle|\, a, b, c \in \mathbb{R}, a \neq 0, c \neq 0 \right\}$$

(a) Compute the product of

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix} \text{ and } \begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix}$$

to show that G is closed under matrix multiplication.

(b) Find the matrix inverse of

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$$

and deduce that G is closed under inverses.

(c) Deduce that G is a subgroup of $\mathrm{GL}_2(\mathbb{R})$.

(d) Prove that the set of elements of G whose two diagonal entries are equal (i.e., $a = c$) is also a subgroup of $\mathrm{GL}_2(\mathbb{R})$.

Solution.

(a)

$$\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1c_2 \\ 0 & c_1c_2 \end{pmatrix}$$

Since $a_i \neq 0$ and $c_i \neq 0$, then $a_1a_2 \neq 0$ and $c_1c_2 \neq 0$ so that G is closed under matrix multiplication.

(b) The determinant is ac. We then have the inverse

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}^{-1} = \frac{1}{ac}\begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/ac \\ 0 & 1/c \end{pmatrix}$$

(c) Since G is closed under inverses and matrix multiplication, the result follows.

(d) Let $H = \{A \in G \mid a = c\}$. Then

$$\begin{pmatrix} a_1 & b_1 \\ 0 & a_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ 0 & a_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1a_2 \\ 0 & a_1a_2 \end{pmatrix} \in H$$

and

$$\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}^{-1} = \frac{1}{a^2}\begin{pmatrix} a & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} 1/a & -b/a^2 \\ 0 & 1/a \end{pmatrix} \in H$$

Then H is closed under inverses and matrix multiplication, hence it is a subgroup of $\mathrm{GL}_2(\mathbb{R})$. ■

Exercise 1.4.11

Let

$$H(F) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \middle| \, a, b, c \in F \right\},$$

called the **Heisenberg group** over F. Let

$$X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \text{ and } Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}$$

be elements of $H(F)$.

(a) Compute the matrix product XY and deduce that $H(F)$ is closed under matrix multiplication. Exhibit explicit matrices such that $XY \neq YX$ (so that $H(F)$ is always non-Abelian).
(b) Find an explicit formula for the matrix inverse X^{-1} and deduce that $H(F)$ is closed under inverses.
(c) Prove the associative law for $H(F)$ and deduce that $H(F)$ is a group of order $|F|^3$. (Do not assume that matrix multiplication is associative.)
(d) Find the order of each element of the finite group $H(\mathbb{Z}/2\mathbb{Z})$.
(e) Prove that every non-identity of the group $H(\mathbb{R})$ has infinite order.

Manual Note. *The choice of rewriting the matrix as a tuple is simply a time saver/ease of notation for the sake of computation. However, it does allow one to encode $H(F)$ in the multiplication law of the field.*

Solution.

(a) Calculating the product, we have

$$XY = \begin{pmatrix} 1 & d+a & e+af+b \\ 0 & 1 & f+c \\ 0 & 0 & 1 \end{pmatrix} \in H(F)$$

It is trivial to exhibit non-commutative matrices, hence $H(F)$ is non-Abelian.

(b) From part (a), we know G is closed under matrix multiplication. For ease of notation, represent a matrix as a tuple: $X = (a,b,c)$ and $Y = (d,e,f)$. Then we may represent matrix multiplication in $H(F)$ as

$$(a,b,c)(d,e,f) = (a+d, af+b+e, c+f)$$

Now we want to find (x,y,z) such that $(a,b,c)(x,y,z) = (0,0,0)$. Then $(a+x, az+b+y, c+z) = (0,0,0)$ implies $x = -a$, $y = -b+ac$, and $z = -c$. It follows that $X^{-1} = (-a, -b+ac, -c)$.

(c) For some $(a,b,c) \in H(F)$, we know that there are $|F|$ choices for each of a,b,c. Then $|H(F)| = |F|^3$. Now let $(a,b,c),(d,e,f),(g,h,i) \in H(F)$. Then we have

$$\begin{aligned}((a,b,c)(d,e,f))(g,h,i) &= (a+d, af+b+e, c+f)(g,h,i) \\ &= (a+d+g, (a+d)i+af+b+e+h, c+f+i) \\ &= (a+d+g, ai+di+af+b+e+h, c+f+i) \\ &= (a,b,c)(d+g, e+h, f+i) \\ &= (a,b,c)((d,e,f)(g,h,i))\end{aligned}$$

(d) The elements of $H(\mathbb{Z}/2\mathbb{Z})$ are

$$(0,0,0),(1,0,0),(0,1,0),(0,0,1),(1,1,0),(1,0,1),(0,1,1),(1,1,1).$$

Since $(a,b,c)^2 = (2a, 2b+ac, 2c) = (0, ac, 0)$ in $\mathbb{Z}/2\mathbb{Z}$, then we have two cases:

- If $ac = 0$, then at least one of a or c is 0, while b varies. These are the matrices that have order 2:
$$(1,0,0),(0,1,0),(0,0,1),(1,1,0),(0,1,1).$$
- If $ac = 1$, then both a and c are 1, while b varies. These are the matrices that have order 4:
$$(1,0,1),(1,1,1).$$

(e) Let $X \in H(\mathbb{R})$ be non-identity. It is easy to see by induction that

$$X^n = (na, n(n-1)ac/2 + nb, nc).$$

Since $X \neq I_3$, then at least one of a, b, c is nonzero. It is clear that $a \neq 0$ or $c \neq 0$ implies $X^n \neq I_3$. Now suppose $a = c = 0$ and $b \neq 0$. Then $X^n = (0, nb, 0)$. Since $b \neq 0$, then $nb \neq 0$ for all $n \in \mathbb{Z}^+$. Hence, $X^n \neq I_3$ for all $n \in \mathbb{Z}^+$. Therefore, every non-identity element of $H(\mathbb{R})$ has infinite order. ■

1.5 The Quaternion Group

Exercise 1.5.1

Compute the order of each of the elements in Q_8.

Solution.

a	1	-1	i	$-i$	j	$-j$	k	$-k$
$\lvert a\rvert$	1	2	4	4	4	4	4	4

■

Exercise 1.5.2

Write out the group tables for S_3, D_8, and Q_8.
MANUAL NOTE. *Tedious, but straightforward.*

Solution.
S_3:

$\circ$	1	$(1\,2)$	$(1\,3)$	$(2\,3)$	$(1\,2\,3)$	$(1\,3\,2)$
1	1	$(1\,2)$	$(1\,3)$	$(2\,3)$	$(1\,2\,3)$	$(1\,3\,2)$
$(1\,2)$	$(1\,2)$	1	$(1\,2\,3)$	$(1\,3\,2)$	$(1\,3)$	$(2\,3)$
$(1\,3)$	$(1\,3)$	$(1\,3\,2)$	1	$(1\,2\,3)$	$(2\,3)$	$(1\,2)$
$(2\,3)$	$(2\,3)$	$(1\,2\,3)$	$(1\,3\,2)$	1	$(1\,2)$	$(1\,3)$
$(1\,2\,3)$	$(1\,2\,3)$	$(2\,3)$	$(1\,2)$	$(1\,3)$	$(1\,3\,2)$	1
$(1\,3\,2)$	$(1\,3\,2)$	$(1\,3)$	$(2\,3)$	$(1\,2)$	1	$(1\,2\,3)$

D_8:

$\circ$	1	r	r^2	r^3	s	sr	sr^2	sr^3
1	1	r	r^2	r^3	s	sr	sr^2	sr^3
r	r	r^2	r^3	1	sr^3	s	sr	sr^2
r^2	r^2	r^3	1	r	sr^2	sr^3	s	sr
r^3	r^3	1	r	r^2	sr	sr^2	sr^3	s
s	s	sr	sr^2	sr^3	1	r	r^2	r^3
sr	sr	sr^2	sr^3	s	r^3	1	r	r^2
sr^2	sr^2	sr^3	s	sr	r^2	r^3	1	r
sr^3	sr^3	s	sr	sr^2	r	r^2	r^3	1

■

Q_8:

$\cdot$	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	j	$-j$
$-i$	$-i$	i	1	-1	$-k$	k	$-j$	j
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Exercise 1.5.3

Find a set of generators and relations for Q_8.
MANUAL NOTE. *One may notice I dropped k. Presentations should utilize minimal generators for efficiency, and since k can be expressed in terms of i and j, it is not necessary to include it.*

Solution. $\langle i, j \mid i^4 = 1, i^2 = j^2, ij = -ji \rangle$. ■

1.6 Homomorphisms and Isomorphisms

Exercise 1.6.1

Let $\varphi : G \to H$ be a homomorphism.

(a) Prove that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}^+$.

(b) Do part (a) for $n = -1$ and deduce that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}$.

Solution.

(a) Inducting on n, the base case of $n = 1$ is trivial. If the result holds for $n \in \mathbb{Z}^+$, then

$$\varphi(x^{n+1}) = \varphi(x^n x) = \varphi(x^n)\varphi(x) = \varphi(x)^n\varphi(x) = \varphi(x)^{n+1}.$$

(b) Let 1_G and 1_H be the identity elements of G and H, respectively. Then for $n = 0$, we have

$$\varphi(1_G) = \varphi(1_G \cdot 1_G) = \varphi(1_G)\varphi(1_G),$$

so that $\varphi(1_G) = 1_H$. For $n = -1$ and $x \in G$, we have

$$1_H = \varphi(1_G) = \varphi(xx^{-1}) = \varphi(x)\varphi(x^{-1})$$

so that $\varphi(x^{-1}) = \varphi(x)^{-1}$. Then for $n < -1$, we have

$$\varphi(x^n) = \varphi((x^{-1})^{-n}) = \varphi(x^{-1})^{-n} = \varphi(x)^{-n} = \varphi(x)^n.$$

■

Exercise 1.6.2

If $\varphi : G \to H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$. Is the result true if φ is only assumed to be a homomorphism?

Solution. Let $x \in G$. Let $|x| = m$ and $|\varphi(x)| = n$. Then

$$1_H = \varphi(1_G) = \varphi(x^m) = \varphi(x)^m$$

so that $n \leq m$. Similarly,

$$1_G = \varphi^{-1}(1_H) = \varphi^{-1}(\varphi(x)^n) = x^n$$

so that $m \leq n$. It follows that $m = n$, i.e., $|\varphi(x)| = |x|$. In particular, this shows that both x and $\varphi(x)$ both have either finite or infinite order. Moreover, since φ is bijective, it must be bijective between the elements of order n in G and the elements of order n in H. Hence, any two isomorphic groups have the same number of elements of order n for each $n \in \mathbb{Z}^+$.

The result is not true if φ is only assumed to be a homomorphism: the map $\varphi : G \to 1$ is a homomorphism for any group G, but G may have elements of various orders while 1 has only the identity element of order 1. ■

Exercise 1.6.3

If $\varphi : G \to H$ is an isomorphism, prove that G is Abelian if and only if H is Abelian. If $\varphi : G \to H$ is a homomorphism, what additional conditions on φ (if any) are sufficient to ensure that if G is Abelian, then so is H?

Solution. Note that φ is bijective, hence φ^{-1} exists and is also an isomorphism.

($\Rightarrow$) Suppose that G is Abelian. Then for $h_1 = \varphi(g_1), h_2 = \varphi(g_2) \in H$ for some $g_1, g_2 \in G$, we have

$$h_1h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) = \varphi(g_2g_1) = \varphi(g_2)\varphi(g_1) = h_2h_1.$$

Hence, H is Abelian.

($\Leftarrow$) Suppose that H is Abelian. Then for $g_1 = \varphi^{-1}(h_1), g_2 = \varphi^{-1}(h_2) \in G$ for some $h_1, h_2 \in H$, we have

$$g_1g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2) = \varphi^{-1}(h_1h_2) = \varphi^{-1}(h_2h_1) = \varphi^{-1}(h_2)\varphi^{-1}(h_1) = g_2g_1.$$

Hence, G is Abelian.

Let $\varphi : G \to H$ be a homomorphism. In the ($\Rightarrow$) direction, observe that the only property of φ used was its surjectivity to guarantee the existence of g_1 and g_2 such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Hence, if φ is surjective, then H is Abelian whenever G is Abelian. ∎

Exercise 1.6.4

Prove that the multiplicative groups $\mathbb{R} - \{0\}$ and $\mathbb{C} - \{0\}$ are not isomorphic.

Solution. Note that $i \in \mathbb{C} - \{0\}$ has order 4, but there is no element in $\mathbb{R} - \{0\}$ that has order 4. ∎

Exercise 1.6.5

Prove that the additive groups $\mathbb{R}$ and $\mathbb{Q}$ are not isomorphic.

Solution. $\mathbb{R}$ is uncountable, while $\mathbb{Q}$ is countable. But bijections preserve cardinality. ∎

Exercise 1.6.6

Prove that the additive groups $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic.

Solution. Every nonzero element of $\mathbb{Z}$ has infinite order. However, $1/k \in \mathbb{Q}$ has order k. ∎

Exercise 1.6.7

Prove that D_8 and Q_8 are not isomorphic.

Solution. D_8 has only 1 element of order 4, while Q_8 has 3 elements of order 4. ∎

Exercise 1.6.8

Prove that if $n \neq m$, then S_n and S_m are not isomorphic.

Solution. If $n \neq m$, then $n! \neq m!$ so that $|S_n| \neq |S_m|$. ∎

Exercise 1.6.9

Prove that D_{24} and S_4 are not isomorphic.

Solution. D_{24} has elements of order 12, but S_4 does not. ∎

Exercise 1.6.10

Fill in the details of the proof that the symmetric groups S_Δ and S_Ω are isomorphic if $|\Delta| = |\Omega|$ as follows: let $\theta : \Delta \to \Omega$ be a bijection. Define

$$\varphi : S_\Delta \to S_\Omega \quad \text{by} \quad \varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1} \quad \text{for all } \sigma \in S_\Delta$$

and prove the following:

(a) Prove that φ is well-defined, that is, if σ is a permutation of Δ then $\theta \circ \sigma \circ \theta^{-1}$ is a permutation of Ω.

(b) Prove that φ is a bijection from S_Δ onto S_Ω by finding a two-sided inverse for φ.

(c) Prove that φ is a homomorphism, that is, $\varphi(\sigma \circ \tau) = \varphi(\sigma) \circ \varphi(\tau)$.

REMARK. *Note the similarity to the change of basis or similarity transformations for matrices (we shall see the connections between these later in the text).*

Solution.

(a) Since $\theta : \Delta \to \Omega$ and $\theta^{-1} : \Omega \to \Delta$ are both bijections with σ a bijection, it follows their composition is a bijection. Moreover, $\theta \circ \sigma \circ \theta^{-1}$ is a permutation of Ω, so that φ is well-defined.

(b) Consider the mapping

$$\psi : S_\Omega \to S_\Delta \quad \text{by} \quad \psi(\tau) = \theta^{-1} \circ \tau \circ \theta \quad \text{for all } \tau \in S_\Omega$$

By the above discussion, ψ is well-defined. For any $\sigma \in S_\Delta$, we have

$$\psi(\varphi(\sigma)) = \theta^{-1} \circ (\theta \circ \sigma \circ \theta^{-1}) \circ \theta = \mathrm{id}_{S_\Delta} \circ \sigma \circ \mathrm{id}_{S_\Delta} = \sigma$$

and for any $\tau \in S_\Omega$, we have

$$\varphi(\psi(\tau)) = \theta \circ (\theta^{-1} \circ \tau \circ \theta) \circ \theta^{-1} = \mathrm{id}_{S_\Omega} \circ \tau \circ \mathrm{id}_{S_\Omega} = \tau$$

so that ψ is a two-sided inverse for φ. Hence, φ is a bijection from S_Δ onto S_Ω.

(c) Let $\sigma, \tau \in S_\Delta$. Then

$$\varphi(\sigma \circ \tau) = \theta \circ (\sigma \circ \tau) \circ \theta^{-1} = (\theta \circ \sigma) \circ (\tau \circ \theta^{-1}) = (\theta \circ \sigma \circ \theta^{-1}) \circ (\theta \circ \tau \circ \theta^{-1}) = \varphi(\sigma) \circ \varphi(\tau)$$

so that φ is a homomorphism. By the above, φ is an isomorphism, and hence $S_\Delta \cong S_\Omega$. ■

Exercise 1.6.11

Let A and B be groups. Prove that $A \times B \cong B \times A$.

Solution. The mapping

$$\varphi : A \times B \to B \times A \quad \text{given by} \quad (a, b) \mapsto (b, a)$$

for all $a \in A$ and $b \in B$ is clearly bijective. Moreover, for any $(a_1, b_1), (a_2, b_2) \in A \times B$, we have

$$\varphi((a_1, b_1)(a_2, b_2)) = \varphi((a_1a_2, b_1b_2)) = (b_1b_2, a_1a_2) = (b_1, a_1)(b_2, a_2) = \varphi(a_1, b_1)\varphi(a_2, b_2)$$

so that φ is a homomorphism. Therefore, φ is an isomorphism and $A \times B \cong B \times A$. ■

Exercise 1.6.12

Let A, B, and C be groups and let $G = A \times B$ and $H = B \times C$. Prove that $G \times C \cong A \times H$.

Solution. Consider the mapping

$$\varphi : G \times C \to A \times H \quad \text{by} \quad ((a,b),c) \mapsto (a,(b,c))$$

for all $a \in A$, $b \in B$, and $c \in C$. The proof is similar to the previous exercise, hence we omit the details to conclude that φ is an isomorphism and $G \times C \cong A \times H$. ■

Exercise 1.6.13

Let G and H be groups and let $\varphi : G \to H$ be a homomorphism. Prove that the image of φ, $\varphi(G)$, is a subgroup of H. Prove that if φ is injective then $G \cong \varphi(G)$.

Solution. Observe that $1_H \in \varphi(G)$ since $\varphi(1_G) = 1_H$, so that $\varphi(G)$ is nonempty. To show closure, let $h_1, h_2 \in \varphi(G)$. Then there exist $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Then

$$h_1h_2 = \varphi(g_1)\varphi(g_2) = \varphi(g_1g_2) \in \varphi(G)$$

so that $h_1h_2 \in \varphi(G)$. Finally,

$$h_1^{-1} = \varphi(g_1)^{-1} = \varphi(g_1^{-1}) \in \varphi(G)$$

so that $h_1^{-1} \in \varphi(G)$. Therefore, $\varphi(G)$ is a subgroup of H.

Suppose φ is injective. Then the mapping

$$\psi : G \to \varphi(G) \quad \text{by} \quad g \mapsto \varphi(g)$$

is well-defined. For any $g_1, g_2 \in G$,

$$\psi(g_1g_2) = \varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = \psi(g_1)\psi(g_2)$$

so that ψ is a homomorphism. Moreover, if $\psi(g_1) = \psi(g_2)$, then $\varphi(g_1) = \varphi(g_2)$, and since φ is injective, $g_1 = g_2$. Hence, ψ is injective. For any $h \in \varphi(G)$, there exists $g \in G$ such that $\varphi(g) = h$, and hence $\psi(g) = h$. Therefore, ψ is surjective, hence ψ is an isomorphism and $G \cong \varphi(G)$. ■

Exercise 1.6.14

Let G and H be groups and let $\varphi : G \to H$ be a homomorphism. Define the **kernel** of φ to be $\{g \in G \mid \varphi(g) = 1_H\}$ (so the kernel is the set of elements in G which map to the identity of H, i.e., is the fiber over the identity of H). Prove that the kernel of φ is a subgroup of G. Prove that φ is injective if and only if the kernel of φ is the identity subgroup of G.

Solution. Denote the kernel of φ by $\ker\varphi$. Since $\varphi(1_G) = 1_H$, then $1_G \in \ker\varphi$, so that $\ker\varphi$ is nonempty. To show closure, let $g_1, g_2 \in \ker\varphi$. Then

$$\varphi(g_1g_2) = \varphi(g_1)\varphi(g_2) = 1_H1_H = 1_H$$

so that $g_1g_2 \in \ker\varphi$. Finally,

$$\varphi(g_1^{-1}) = \varphi(g_1)^{-1} = 1_H^{-1} = 1_H$$

so that $g_1^{-1} \in \ker\varphi$. Therefore, $\ker\varphi$ is a subgroup of G.

($\Rightarrow$) If φ is injective, then for $g \in \ker\varphi$, we have $\varphi(g) = 1_H = \varphi(1_G)$. Then $g = 1_G$, so that $\ker\varphi = \{1_G\}$.

($\Leftarrow$) If $\ker\varphi = \{1_G\}$, then for $g_1, g_2 \in G$, if $\varphi(g_1) = \varphi(g_2)$, we have $\varphi(g_1)\varphi(g_2)^{-1} = 1_H$, so $g_1g_2^{-1} \in \ker\varphi$. Hence, $g_1g_2^{-1} = 1_G$, and thus $g_1 = g_2$. Therefore, φ is injective. ■

Exercise 1.6.15

Define a map $\pi : \mathbb{R}^2 \to \mathbb{R}$ by $\pi((x, y)) = x$. Prove that π is a homomorphism and find the kernel of π.

Solution. Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$. Then

$$\pi((x_1, y_1) + (x_2, y_2)) = \pi((x_1 + x_2, y_1 + y_2)) = x_1 + x_2 = \pi((x_1, y_1)) + \pi((x_2, y_2))$$

so that π is a homomorphism. Moreover, if $(x, y) \in \ker(\pi)$, then $\pi((x, y)) = x = 0$. Hence,

$$\ker(\pi) = \{(0, y) \mid y \in \mathbb{R}\}$$ ■

Exercise 1.6.16

Let A and B be groups and let G be their direct product, $A \times B$. Prove that the maps $\pi_1 : G \to A$ and $\pi_2 : G \to B$ defined by $\pi_1((a, b)) = a$ and $\pi_2((a, b)) = b$ are homomorphisms and find their kernels.

Solution. The proof is similar to the previous exercise, hence we omit the details to conclude that π_1 and π_2 are homomorphisms. Moreover, if $(a, b) \in \ker(\pi_1)$, then $\pi_1((a, b)) = a = 1_A$. If $(a, b) \in \ker(\pi_2)$, then $\pi_2((a, b)) = b = 1_B$. Hence,

$$\ker(\pi_1) = \{(1_A, b) \mid b \in B\}, \quad \text{and} \quad \ker(\pi_2) = \{(a, 1_B) \mid a \in A\}$$ ■

Exercise 1.6.17

Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^{-1}$ is a homomorphism if and only if G is Abelian.

Solution.

($\Rightarrow$) Suppose φ is a homomorphism. Then for $g, h \in G$, we have

$$(gh)^{-1} = \varphi(gh) = \varphi(g)\varphi(h) = g^{-1}h^{-1} = h^{-1}g^{-1}.$$

Since inverses are unique, then $hg = gh$ so that G is Abelian.

($\Leftarrow$) Suppose G is Abelian. Then for $g, h \in G$, we have

$$\varphi(gh) = (gh)^{-1} = h^{-1}g^{-1} = g^{-1}h^{-1} = \varphi(g)\varphi(h),$$

so that φ is a homomorphism. ■

Exercise 1.6.18

Let G be any group. Prove that the map from G to itself defined by $g \mapsto g^2$ is a homomorphism if and only if G is Abelian.

Solution. The proof is similar to Exercise 1.6.17, hence we omit the details. ■

Exercise 1.6.19

Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. Prove that for any fixed integer $k > 1$ the map from G to itself defined by $z \mapsto z^k$ is a surjective homomorphism but is not an isomorphism.

Solution. Since $G \subseteq \mathbb{C}$, then G is Abelian. Let $\varphi : z \to z^k$. Hence, for any $z, w \in G$, we have

$$\varphi(zw) = (zw)^k = z^k w^k = \varphi(z)\varphi(w)$$

so that φ is a homomorphism. To show surjectivity, let $w \in G$ such that $w^n = 1$ for some $n \in \mathbb{Z}^+$. Note that $w^{1/k} \in G$ because $(z^{1/k})^{nk} = z^n = 1$. Then

$$\varphi(z^{1/k}) = (z^{1/k})^k = z$$

so that φ is surjective. Finally, any $z = e^{2\pi i/k} \in G$ is such that $\varphi(z) = z^k = 1$ with $z \neq 1$ since $k > 1$. Hence, φ is not injective and thus not an isomorphism. ■

Exercise 1.6.20

Let G be a group and let $\mathrm{Aut}(G)$ be the set of all isomorphisms from G onto G. Prove that $\mathrm{Aut}(G)$ is a group under function composition (called the **automorphism group** of G and the elements of $\mathrm{Aut}(G)$ are called **automorphisms** of G).

Solution. Observe that $\mathrm{Aut}(G)$ is associative under function composition and has identity through the identity map $\mathrm{id} : G \to G$ given by $\mathrm{id}(g) = g$. Moreover, $\varphi \circ \psi$ for $\varphi, \psi \in \mathrm{Aut}(G)$ is a bijection. Then for any $g, h \in G$, we have

$$(\varphi \circ \psi)(gh) = \varphi(\psi(gh)) = \varphi(\psi(g)\psi(h)) = \varphi(\psi(g))\varphi(\psi(h)) = (\varphi \circ \psi)(g)(\varphi \circ \psi)(h),$$

so that $\varphi \circ \psi$ is a homomorphism. Hence, $\varphi \circ \psi \in \mathrm{Aut}(G)$, and $\mathrm{Aut}(G)$ is closed under function composition. Finally, for any $\varphi \in \mathrm{Aut}(G)$, then φ^{-1} exists and is a bijection. Then

$$\varphi^{-1}(gh) = \varphi^{-1}(\varphi(\varphi^{-1}(g))\varphi(\varphi^{-1}(h))) = \varphi^{-1}(\varphi(\varphi^{-1}(g)\varphi^{-1}(h))) = \varphi^{-1}(g)\varphi^{-1}(h),$$

so that φ^{-1} is a homomorphism. Hence, $\varphi^{-1} \in \mathrm{Aut}(G)$, and $(\mathrm{Aut}(G), \circ)$ is a group. ■

Exercise 1.6.21

Prove that for each fixed nonzero $k \in \mathbb{Q}$ the map from $\mathbb{Q}$ to itself defined by $q \mapsto kq$ is an automorphism of $\mathbb{Q}$.

Solution. Let $\varphi : \mathbb{Q} \to \mathbb{Q}$ be defined by $\varphi(q) = kq$ for all $q \in \mathbb{Q}$. For any $p, q \in \mathbb{Q}$, we have

$$\varphi(p + q) = k(p + q) = kp + kq = \varphi(p) + \varphi(q),$$

so that φ is a homomorphism. Bijectivity is clear, hence $\varphi \in \mathrm{Aut}(\mathbb{Q})$. ■

Exercise 1.6.22

Let A be an Abelian group and fix some $k \in \mathbb{Z}$. Prove that the map $a \mapsto a^k$ is a homomorphism from A to itself. If $k = -1$ prove that this homomorphism is an isomorphism (i.e., is an automorphism of A).

Solution. Let $\varphi : A \to A$ be defined by $\varphi(a) = a^k$ for all $a \in A$. For any $a_1, a_2 \in A$, we have

$$\varphi(a_1 a_2) = (a_1 a_2)^k = a_1^k a_2^k = \varphi(a_1)\varphi(a_2)$$

so that φ is a homomorphism. For $k = -1$, see Exercise 1.6.17 to conclude that $\varphi \in \operatorname{Aut}(A)$. ■

Exercise 1.6.23

Let G be a finite group which possesses an automorphism σ such that $\sigma(g) = g$ if and only if $g = 1$. If σ^2 is the identity map from G to G, prove that G is Abelian (such an automorphism σ is called **fixed point free** of order 2).
HINT. *Show that every element of G can be written in the form $x^{-1}\sigma(x)$ and apply σ to such an expression.*

Solution. Define the map

$$f : G \to G \quad \text{by} \quad x \mapsto x^{-1}\sigma(x)$$

for all $x \in G$. If $f(x) = f(y)$ for $x, y \in G$, then $x^{-1}\sigma(x) = y^{-1}\sigma(y)$ so that $yx^{-1} = \sigma(yx^{-1})$. Since σ is fixed point free, then $yx^{-1} = 1$ and $x = y$. Hence, f is injective. Since G is finite, then f is also surjective, and every element of G may be written in the form $x^{-1}\sigma(x)$ for some $x \in G$.

Let $g \in G$ be arbitrary. Then there exists $x \in G$ such that $g = x^{-1}\sigma(x)$. Applying σ to both sides, we have

$$\sigma(g) = \sigma(x^{-1}\sigma(x)) = \sigma(x^{-1})\sigma^2(x) = \sigma(x)^{-1}x = (x^{-1}\sigma(x))^{-1} = g^{-1}.$$

Then for any $g, h \in G$, we have

$$\sigma(gh) = (gh)^{-1} = h^{-1}g^{-1} = \sigma(h)\sigma(g) = \sigma(hg).$$

Since σ is injective, then $gh = hg$ and G is Abelian. ■

Exercise 1.6.24

Let G be a finite group and let x and y be distinct elements of order 2 in G that generate G. Prove that $G \cong D_{2n}$, where $n = |xy|$.

Solution. To show that G is isomorphic to D_{2n}, we first need to produce an element $t \in G$ of order n along with an element $x \in G$ of order 2 such that $xt = t^{-1}x$. The clear choice is to let $t = xy$ and keep x as is. Note that $|t| = |xy| = n$ by assumption, and $|x| = 2$ by assumption. Moreover,

$$xt = x(xy) = (xx)y = y = (xy)^{-1}x = t^{-1}x$$

so that the elements $x, t \in G$ satisfy the relations of D_{2n}. Additionally, observe that $y = xt$ so that t and x generate G. Following this, it is easy to see that every element of G can be written in the form $x^i t^j$ for $i \in \{0, 1\}$ and $0 \leq j < n$. For $x = 0$, we have the elements in the cyclic subgroup generated by t, and for $x = 1$, we have the elements $\{x, xt, xt^2, \ldots, xt^{n-1}\}$. Since $|t| = n$, then these are all distinct elements of G. Hence, $|G| = 2n$.

Since the relations of D_{2n} are satisfied by $x, t \in G$, G is generated by x and t, and $|G| = |D_{2n}| = 2n$, it follows that the map

$$\varphi : D_{2n} \to G \quad \text{by} \quad r \mapsto t, s \mapsto x$$

extends to an isomorphism from D_{2n} onto G. Therefore, $G \cong D_{2n}$. ■

Exercise 1.6.25

Let $n \in \mathbb{Z}^+$, let r and s be the usual generators of D_{2n} and let $\theta = 2\pi/n$.

(a) Prove that the matrix
$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$
is the matrix of the linear transformation which rotates the x, y-plane about the origin in a counterclockwise direction by θ radians.

(b) Prove that the map $\varphi : D_{2n} \to \mathrm{GL}_2(\mathbb{R})$ defined on generators by
$$\varphi(r) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \text{and} \quad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
extends to a homomorphism of D_{2n} into $\mathrm{GL}_2(\mathbb{R})$.

(c) Prove that the homomorphism φ defined above is injective.

Solution.

(a) Let $\mathbb{R}^2$ be generated by the standard basis vectors $\mathbf{e}_1 = (1 \quad 0)^T$ and $\mathbf{e}_2 = (0 \quad 1)^T$. Let T denote the linear transformation which rotates the x, y-plane about the origin in a counterclockwise direction by θ radians. Then
$$T(\mathbf{e}_1) = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \quad \text{and} \quad T(\mathbf{e}_2) = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$
Then the matrix of T with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ is exactly the desired matrix.

(b) It is easy to show by induction that for any positive k, then $\varphi(r)^k$ is given by the following matrix:
$$\varphi(r)^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix}$$
so that $\varphi(r)^n = I$. It is fairly straightforward to see that the set of vectors $\{T(\mathbf{e}_1), T(\mathbf{e}_2)\}$ is orthonormal, so $\varphi(r)$ is an orthogonal matrix, and hence $\varphi(r)^{-1} = \varphi(r)^T$. Moreover, we can calculate that $\varphi(s)^2 = I$. Lastly, we calculate that
$$\varphi(s)\varphi(r) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \varphi(r)^{-1}\varphi(s)$$
so that $\varphi(s)\varphi(r) = \varphi(r)^{-1}\varphi(s)$. Since these relations match those of D_{2n}, then φ extends to a homomorphism from D_{2n} to $\mathrm{GL}_2(\mathbb{R})$.

(c) To show that φ is injective, we show that $\ker\varphi$ is trivial. Let $g \in \ker\varphi$.
- If $g = r^k$ for some $0 \le k < n$, then $\varphi(r^k) \neq I$, since $\varphi(r)^k$ is a rotation matrix and only $\varphi(r)^n = I$.
- If $g = sr^k$ for some $0 \le k < n$, then $\varphi(sr^k) = \varphi(s)\varphi(r)^k$. Note that $\varphi(s)$ has determinant -1 while $\varphi(r)^k$ has determinant 1 since it is a rotation matrix. Hence, $\varphi(sr^k)$ has determinant -1 and cannot be the identity matrix.

Therefore, the only element in $\ker\varphi$ is the identity element of D_{2n}, so that φ is injective. ■

Exercise 1.6.26

Let i and j be the generators of Q_8 described in Section 5. Prove that the map φ from Q_8 to $GL_2(\mathbb{C})$ defined on generators by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \quad \text{and} \quad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism. Prove that φ is injective.

Solution. Matrix multiplication shows that the relations described in Exercise 1.5.3 are satisfied, so that φ extends to a homomorphism from Q_8 to $GL_2(\mathbb{C})$. Moreover, φ is injective since the only element that maps to the identity matrix is $1 \in Q_8$. Therefore, φ is an injective homomorphism. ■

1.7 Group Actions

Exercise 1.7.1

Let F be a field. Show that the multiplicative group of nonzero elements of F (denoted by $F^\times$) acts on the set F by $g \cdot a = ga$, where $g \in F^\times$, $a \in F$ and ga is the usual product in F of the two field elements (state clearly which axioms in the definition of a field are used).

Solution. Let $g, h \in F^\times$ and $f \in F$. Then

$$g \cdot (h \cdot f) = g \cdot (hf) = g(hf) = (gh)f = (gh) \cdot f$$

where the associativity of multiplication in F was used to conclude that $g(hf) = (gh)f$. Moreover, $1 \cdot f = 1f = f$ where 1 is the multiplicative identity in F. Therefore, the group action axioms are satisfied and $F^\times$ acts on F. ■

Exercise 1.7.2

Show that the additive group $\mathbb{Z}$ acts on itself by $z \cdot a = z + a$ for all $z, a \in \mathbb{Z}$.

Solution. Let $z, w \in \mathbb{Z}$ and $a \in \mathbb{Z}$. Then

$$z \cdot (w \cdot a) = z \cdot (w + a) = z + (w + a) = (z + w) + a = (z + w) \cdot a$$

Moreover, $0 \cdot a = 0 + a = a$. Then $\mathbb{Z}$ acts on itself. ■

Exercise 1.7.3

Show that the additive group $\mathbb{R}$ acts on the x, y plane $\mathbb{R} \times \mathbb{R}$ by $r \cdot (x, y) = (x + r, y)$.

Solution. Let $r, s \in \mathbb{R}$ and $(x, y) \in \mathbb{R}^2$. Then

$$r \cdot (s \cdot (x, y)) = r \cdot (x + s, y) = (x + s + r, y) = (x + (r + s), y) = (r + s) \cdot (x, y)$$

Moreover, $0 \cdot (x, y) = (x + 0, y) = (x, y)$. Then $\mathbb{R}$ acts on $\mathbb{R}^2$. ■

Exercise 1.7.4

Let G be a group acting on a set A and fix some $a \in A$. Show that the following sets are subgroups of G:

(a) the kernel of the action,

(b) $\{g \in G \mid ga = a\}$–this subgroup is called the **stabilizer** of a in G.

Solution.

(a) Let K denote the kernel of the action. If $g, h \in K$, then for any $a \in A$, we have

$$gh \cdot a = g \cdot (h \cdot a) = g \cdot a = a$$

so that $gh \in K$. Also, for any $g \in K$ and any $a \in A$, we have

$$g^{-1} \cdot a = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1 \cdot a = a$$

so that $g^{-1} \in K$. Therefore, K is a subgroup of G.

(b) Let G_a denote the stabilizer of a. If $g, h \in G_a$, then for any $a \in A$, we have

$$gh \cdot a = g \cdot (h \cdot a) = g \cdot a = a$$

so that $gh \in G_a$. Also, for any $g \in G_a$, we have

$$g^{-1} \cdot a = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1 \cdot a = a$$

so that $g^{-1} \in G_a$. Therefore, G_a is a subgroup of G. ■

Exercise 1.7.5

Prove that the kernel of an action of the group G on the set A is the same as the kernel of the corresponding permutation representation $G \to S_A$ (cf. Exercise 1.6.14).

Solution. Let K denote the kernel of the action, and let K' denote $\ker \varphi$, where $\varphi : G \to S_A$ is the permutation representation corresponding to the action. Suppose $g \in K$. Then $g \cdot a = a$ for every $a \in A$, hence $\varphi(g)$ is the identity permutation in S_A and $g \in K'$. Conversely, suppose $g \in K'$. Then $\varphi(g)$ is the identity permutation in S_A, so that for every $a \in A$, we have $\varphi(g)(a) = a$. But $\varphi(g)(a) = g \cdot a$ by definition of the permutation representation, so that $g \cdot a = a$ for every $a \in A$ and $g \in K$. Therefore, $K = K'$. ■

Exercise 1.7.6

Prove that a group G acts faithfully on a set A if and only if the kernel of the action is the set consisting only of the identity.

Solution.

($\Rightarrow$) Suppose G acts faithfully on A with kernel K. Certainly, $1 \cdot a = a$ for all $a \in A$, hence $1 \in K$. If $g \in K$, then $g \cdot a = a$. Faithfulness of the action implies that $g = 1$, so that $K = \{1\}$.

($\Leftarrow$) Suppose the kernel of the action is the set consisting only of the identity. Let $g, h \in G$ be such that $g \cdot a = h \cdot a$ for every $a \in A$. Then

$$h^{-1}g \cdot a = h^{-1} \cdot (g \cdot a) = h^{-1} \cdot (h \cdot a) = (h^{-1}h) \cdot a = 1 \cdot a = a$$

so that $h^{-1}g$ is in the kernel of the action. By assumption, then $h^{-1}g$ is the identity element of G, and $g = h$. Therefore, the action is faithful. ■

Exercise 1.7.7

Prove that in Example 2 in this section the action is faithful.

Solution. Let V be a vector space over a field F. Let $F^\times$ act on V by $r \cdot v = rv$ for all $r \in F^\times$ and $v \in V$. Suppose $r \in F^\times$ is in the kernel of the action. Then for every $v \in V$, we have $r \cdot v = v$, so that $rv = v$. Since V is a vector space, then it contains the nonzero vector $v = 1$. Hence, $r(1) = 1$ so that $r = 1$. Therefore, the kernel of the action is the set consisting only of the identity, and by the previous exercise, the action is faithful. ■

Exercise 1.7.8

Let A be a nonempty set and let k be a positive integer with $k \le |A|$. The symmetric group S_A acts on the set B consisting of all subsets of A of cardinality k by

$$\sigma(\{a_1, \dots, a_k\}) = \{\sigma(a_1), \dots, \sigma(a_k)\}.$$

(a) Prove that this is a group action.
(b) Describe explicitly how the elements $(1\ 2)$ and $(1\ 2\ 3)$ act on the six 2-element subsets of $\{1, 2, 3, 4\}$.

Solution.

(a) Let $\sigma, \tau \in S_A$. Then for any cardinality k subset $\{x_1, \dots, x_k\}$ of A, we have

$$\begin{aligned} \sigma \cdot (\tau \cdot \{x_1, \dots, x_k\}) &= \sigma \cdot \{\tau(x_1), \dots, \tau(x_k)\} \\ &= \{\sigma(\tau(x_1)), \dots, \sigma(\tau(x_k))\} \\ &= (\sigma \circ \tau) \cdot \{x_1, \dots, x_k\} \end{aligned}$$

Moreover, $1 \cdot \{x_1, \dots, x_k\} = \{1(x_1), \dots, 1(x_k)\} = \{x_1, \dots, x_k\}$. Then it is a group action.

(b)

A	$(1\ 2) \cdot A$	$(1\ 2\ 3) \cdot A$
$\{1, 2\}$	$\{2, 1\}$	$\{2, 3\}$
$\{1, 3\}$	$\{2, 3\}$	$\{2, 1\}$
$\{1, 4\}$	$\{2, 4\}$	$\{2, 4\}$
$\{2, 3\}$	$\{1, 3\}$	$\{3, 1\}$
$\{2, 4\}$	$\{1, 4\}$	$\{3, 4\}$
$\{3, 4\}$	$\{3, 4\}$	$\{1, 4\}$

■

Exercise 1.7.9

Do both parts of the preceding exercise with "ordered k-tuples" in place of "k-element subsets," where the action on k-tuples is defined as above but with set braces replaced by parentheses.
REMARK. *Note that, for example, the 2-tuples $(1, 2)$ and $(2, 1)$ are different even though the sets $\{1, 2\}$ and $\{2, 1\}$ are the same, so the sets being acted upon are different.*

Solution.

(a) The proof is similar to that of Exercise 1.7.8.
(b) Since order matters, there are twice as many cases to check:

A	$(1\ 2) \cdot A$	$(1\ 2\ 3) \cdot A$
$(1, 2)$	$(2, 1)$	$(2, 3)$
$(1, 3)$	$(2, 3)$	$(2, 1)$
$(1, 4)$	$(2, 4)$	$(2, 4)$
$(2, 3)$	$(1, 3)$	$(3, 1)$
$(2, 4)$	$(1, 4)$	$(3, 4)$
$(3, 4)$	$(3, 4)$	$(1, 4)$

A	$(1\ 2) \cdot A$	$(1\ 2\ 3) \cdot A$
$(2, 1)$	$(1, 2)$	$(3, 2)$
$(3, 1)$	$(3, 2)$	$(1, 3)$
$(4, 1)$	$(4, 2)$	$(4, 3)$
$(3, 2)$	$(3, 1)$	$(1, 3)$
$(4, 2)$	$(4, 1)$	$(4, 3)$
$(4, 3)$	$(4, 3)$	$(1, 3)$

■

Exercise 1.7.10

With reference to the preceding two exercises determine:
(a) for which values of k the action of S_n on k-element subsets is faithful, and
(b) for which values of k the action of S_n on ordered k-tuples is faithful.

Solution.
(a) Let $\sigma \in S_n$ be non-identity. Then there exists $a \in A$ such that $\sigma(a) \neq a$. If $k < |A|$, then we can choose a k-element subset S of A such that $a \in S$ and $\sigma(a) \notin S$. Then $\sigma(S) \neq S$, hence no non-identity permutation fixes every k-element subset of A. Therefore, the action is faithful for all $1 \leq k < |A|$. If $k = |A|$, then any k-element subset of A is A itself, which is fixed by all permutations, so the action is not faithful for $k = |A|$.
(b) Let $\sigma \in S_n$ be a non-identity permutation. Then there exists some $a \in A$ such that $\sigma(a) \neq a$. Consider the ordered k-tuple $T = (a, x_2, x_3, \ldots, x_k)$ where $x_2, x_3, \ldots, x_k$ are distinct elements of A different from a and $\sigma(a)$. Then $\sigma(T) = (\sigma(a), \sigma(x_2), \sigma(x_3), \ldots, \sigma(x_k)) \neq T$ because the first element is different. Hence, no non-identity permutation fixes every ordered k-tuple of elements of A. Therefore, the action is faithful for all $1 \leq k \leq |A|$. ■

Exercise 1.7.11

Write out the cycle decomposition of the eight permutations in S_4 corresponding to the elements of D_8 given by the action of D_8 on the vertices of a square (where the vertices of the square are labeled as in Section 2).

Solution. Let $\varphi : D_8 \to S_4$ be the permutation representation of the action of D_8 on the vertices of a square $\{1, 2, 3, 4\}$, where the vertices are labeled in order around the square starting in the top left corner, then going clockwise. Then the cycle decompositions are as follows:

$$\begin{aligned} \varphi(1) &= 1 & \varphi(r) &= (1\,2\,3\,4) \\ \varphi(r^2) &= (1\,3)(2\,4) & \varphi(r^3) &= (1\,4\,3\,2) \\ \varphi(s) &= (2\,4) & \varphi(sr) &= (1\,4)(2\,3) \\ \varphi(sr^2) &= (1\,3) & \varphi(sr^3) &= (1\,2)(3\,4). \end{aligned}$$ ■

Exercise 1.7.12

Assume n is an even positive integer and show that D_{2n} acts on the set consisting of pairs of opposite vertices of a regular n-gon. Find the kernel of this action (label vertices as usual).

Solution. Let the n-gon have the usual labeling of vertices. Put P_k as the pair of opposite vertices

$$P_k = \{k, k + n/2\} \quad \text{defined for all } 1 \leq k \leq n/2,$$

with P denoting the set of all opposite pairs of vertices. Let D_{2n} act on P by

$$g \cdot P_i = g(P_i) = P_j$$

where P_j is the pair of opposite vertices containing the images of the vertices in P_i under the action of g on the vertices of the n-gon. Since elements of D_{2n} are symmetries of the n-gon, then they will map pairs of opposite vertices to other pairs of opposite vertices. Hence, D_{2n} acts on P.

We now consider the elements of the kernel. If $r^k \cdot P_i = P_i$ for some given $P_i = \{i, i + n/2\}$, then r^k must map i to either i or $i+n/2$. Similarly, it must map $i+n/2$ to either i or $i+n/2$. This is only possible if $k = 0$ or $k = n/2$, corresponding to the identity rotation and the rotation by 180°, respectively.

If $s \cdot P_i = P_i$, then s must map the vertices in P_i to themselves or to each other. However, a reflection in D_{2n} will only fix a pair of opposite vertices if the axis of reflection passes through that pair. Since there are multiple pairs of opposite vertices, no single reflection can fix all pairs simultaneously. Therefore, no reflection can be in the kernel of this action. ■

Exercise 1.7.13

Find the kernel of the left regular action.

Solution. If g is in the kernel of the left regular action of G on itself, then $g \cdot 1 = 1$ in particular. Then $g1 = 1$, implying that $g = 1$. Then the kernel of the left regular action is trivial. ■

Exercise 1.7.14

Let G be a group and let $A = G$. Show that if G is non-Abelian then the maps defined by

$$g \cdot a = ag \quad \text{for all } g, a \in G$$

do not satisfy the axioms of a (left) group action of G on itself.

Solution. Since G is non-Abelian, there exists $g, h \in G$ such that $gh \neq hg$. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (ah) = (ah)g = a(hg)$$

On the other hand, $(gh) \cdot a = a(gh)$. Since $gh \neq hg$, then $a(hg) \neq a(gh)$ for some $a \in A$. Then the maps do not satisfy the axioms of a left group action. ■

Exercise 1.7.15

Let G be any group and let $A = G$. Show that the maps defined by

$$g \cdot a = ag^{-1} \quad \text{for all } g, a \in G$$

do satisfy the axioms of a (left) group action of G on itself.

Solution. Let $g, h \in G$. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (ah^{-1}) = a(h^{-1}g^{-1}) = a(gh)^{-1} = (gh) \cdot a$$

Moreover, $1 \cdot a = a1^{-1} = a1 = a$. Then the maps define a left group action. ■

Exercise 1.7.16

Let G be any group and let $A = G$. Show that the maps defined by

$$g \cdot a = gag^{-1} \quad \text{for all } g, a \in G$$

do satisfy the axioms of a (left) group action. This action is called **conjugation**.

Solution. Let $g, h \in G$. Then for any $a \in A$, we have

$$g \cdot (h \cdot a) = g \cdot (hah^{-1}) = g(hah^{-1})g^{-1} = (gh)a(gh)^{-1} = (gh) \cdot a$$

Moreover, $1 \cdot a = 1a1^{-1} = a$. Then the maps define a left group action. ■

Exercise 1.7.17

Let G be a group and let G act on itself by left conjugation, so each $g \in G$ maps G to G by

$$x \mapsto gxg^{-1}.$$

For fixed $g \in G$, prove that conjugation by g is an isomorphism from G onto itself. Deduce that x and gxg^{-1} have the same order for all $x \in G$ and that for any subset A of G, $|A| = |gAg^{-1}|$ (here $gAg^{-1} = \{gag^{-1} \mid a \in A\}$).

Solution. Let $\varphi : x \mapsto gxg^{-1}$. Let $x, y \in G$. Then

$$\varphi(xy) = g(xy)g^{-1} = (gxg^{-1})(gyg^{-1}) = \varphi(x)\varphi(y)$$

Moreover, $\varphi(1) = g1g^{-1} = 1$, so φ preserves the identity. To show injectivity, suppose $\varphi(x) = \varphi(y)$ for some $x, y \in G$. Then $gxg^{-1} = gyg^{-1}$, implying $x = y$. Surjectivity is clear from conjugation by g^{-1}, so φ is a bijection. Therefore, φ is an isomorphism from G onto itself.

From Exercise 1.6.2 and the fact that $\varphi \in \mathrm{Aut}(G)$ shows that $|x| = |gxg^{-1}|$. Moreover, the bijection φ restricts to a bijection from A to gAg^{-1}, so $|A| = |gAg^{-1}|$. ■

Exercise 1.7.18

Let H be a group acting on a set A. Prove that the relation $\sim$ on A defined by

$$a \sim b \quad \text{if and only if} \quad a = hb \text{ for some } h \in H$$

is an equivalence relation. (For each $x \in A$ the equivalence class of x under $\sim$ is called the **orbit** of x under the action of H. The orbits under the action of H partition the set A.)

Solution.

- Let $a \in A$. Then $a = 1a$, hence $a \sim a$.
- Let $a, b \in A$ such that $a \sim b$. Then $a = bh$ for some $h \in H$. Then $b = h^{-1}a$, hence $b \sim a$.
- Let $a, b, c \in A$ such that $a \sim b$ and $b \sim c$. Then $a = bh_1$ and $b = ch_2$ for some $h_1, h_2 \in H$. Then $a = (ch_2)h_1 = c(h_2h_1)$, hence $a \sim c$. ■

Exercise 1.7.19

Let H be a subgroup of the finite group G and let H act on G (here $A = G$) by left multiplication. Let $x \in G$ and let $\mathcal{O}$ be the orbit of x under this action of H. Prove that the map

$$H \to \mathcal{O}, \qquad h \mapsto hx$$

is a bijection (hence all orbits have cardinality $|H|$). From this and the preceding exercise deduce **Lagrange's Theorem**:

If G is a finite group and H is a subgroup of G then $|H|$ divides $|G|$.

Solution. Let $\varphi : H \to O$ be the map defined by $\varphi(h) = hx$ for all $h \in H$. We show that φ is a bijection. To show injectivity, suppose $\varphi(h_1) = \varphi(h_2)$ for some $h_1, h_2 \in H$. Then

$$h_1x = h_2x$$

implying that $h_1 = h_2$. Any $y \in O$ has some $h \in H$ such that $y = hx$, hence $\varphi(h) = y$ and φ is surjective. Therefore, φ is a bijection and $|O| = |H|$.

To deduce Lagrange's Theorem, note by the previous exercise that the orbits of the action of H on G partition G. Since each orbit has cardinality $|H|$, then $|G|$ is a sum of multiples of $|H|$. Therefore, $|H|$ divides $|G|$. ■

Exercise 1.7.20

Show that the group of rigid motions of a tetrahedron is isomorphic to a subgroup of S_4.

Solution. Let G be the group of rigid motions of a tetrahedron with vertices labeled $1, 2, 3, 4$. Then each $\alpha \in G$ sends some vertex to another vertex so that G acts on the set of vertices $A = \{1, 2, 3, 4\}$.

Since G acts on A, this gives rise to a homomorphism

$$\varphi : G \longrightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(i) = \alpha(i) \quad \text{for all} \quad i \in A.$$

To see that φ is injective, suppose $\varphi(\alpha)$ is the identity permutation in S_4 for some $\alpha \in G$. Then for every vertex $i \in A$, we have $\varphi(\alpha)(i) = i$, so that $\alpha(i) = i$. Since a rigid motion that fixes all vertices must be the identity rigid motion, then α is the identity element of G. Therefore, φ is injective and G is isomorphic to a subgroup of S_4. ■

Exercise 1.7.21

Show that the group of rigid motions of a cube is isomorphic to S_4. (This group acts on the set of four pairs of opposite vertices.)

Solution. Recall that $|G| = 24$ from Exercise 1.2.10. Let G be the group of rigid motions of a cube, and let $A = \{a_1, a_2, a_3, a_4\}$ be the set of pairs of opposite vertices of the cube. Then each $\alpha \in G$ sends some pair of opposite vertices to another pair of opposite vertices so that G acts on A.

Using similar reasoning as in the exercise, we have an injective homomorphism

$$\varphi : G \longrightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(a_i) = \alpha(a_i) \quad \text{for all} \quad a_i \in A.$$

Since $|G| = 24 = |S_4|$, it follows that φ is surjective as well, hence $G \cong S_4$. ■

Exercise 1.7.22

Show that the group of rigid motions of an octahedron is isomorphic to a subgroup of S_4. (This group acts on the set of four pairs of opposite faces.) Deduce that the groups of rigid motions of a cube and of an octahedron are isomorphic. (These two groups are isomorphic because these solids are "dual.")

Solution. Let G be the group of rigid motions of an octahedron. Let $A = \{f_1, f_2, f_3, f_4\}$ be the set of pairs of opposite faces of the octahedron. Then each $\alpha \in G$ sends some pair of opposite faces to another pair of opposite faces so that G acts on A.

Using similar reasoning as in the previous exercises, we have an injective homomorphism

$$\varphi : G \longrightarrow S_4 \quad \text{where} \quad \varphi(\alpha)(f_i) = \alpha(f_i) \quad \text{for all} \quad f_i \in A.$$

Therefore, G is isomorphic to a subgroup of S_4. From Exercise 1.2.11, we know that $|G| = 24$, hence $G \cong S_4$. From the previous exercise, we know that the group of rigid motions of a cube is also isomorphic to S_4, so the groups of rigid motions of a cube and of an octahedron are isomorphic. ■

Exercise 1.7.23

Explain why the action of the group of rigid motions of a cube on the set of three pairs of opposite faces is not faithful. Find the kernel of this action.

Solution. The group of rigid motions of a cube has 24 elements, while permutations of the set of three pairs of opposing faces has 6 elements, so the action cannot be faithful as no homomorphism can be injective between finite groups of different sizes (if there was some injective homomorphism, it would imply bijectivity between the two groups of different cardinality, which is impossible).

Consider a cube such that its center is at the origin of $\mathbb{R}^3$ space. Then any rotation around the axes fixes a pair of faces, while sends the other two pairs back to themselves (to visualize this explicitly, consider the cube with vertices $(\pm1, \pm1, \pm1)$ and the z-axis. A rotation around the z-axis would fix the faces $(\pm1, \pm1, 1)$ and $(\pm1, \pm1, -1)$, while it rotates the pair of faces $(1, \pm1, \pm1)$ and $(-1, \pm1, \pm1)$ back to each other—these are the faces that vary along the x-axis. One can construct the same for the pair of faces that lie along the y-axis and deduce the same thing). There are exactly 3 of these 180° rotations, so the kernel of this action consists of these rotations and the identity. ■

2 Subgroups

2.1 Definitions and Examples

Exercise 2.1.1

In each of (a) – (e) prove that the specified subset is a subgroup of the given group.

(a) The set of complex numbers of the form $a + ai$, $a \in \mathbb{R}$ (under addition).

(b) The set of complex numbers of absolute value 1, i.e., the unit circle in the complex plane (under multiplication).

(c) For fixed $n \in \mathbb{Z}^+$, the set of rational numbers whose denominators divide n (under addition).

(d) For fixed $n \in \mathbb{Z}^+$, the set of rational numbers whose denominators are relatively prime to n (under addition).

(e) The set of nonzero real numbers whose square is a rational number (under multiplication).

Solution. Let G be the group and H be the subset in question.

(a) Since $0 + 0i \in H$, then H is nonempty. For $a + ai, b + bi \in H$,

$$(a + ai) - (b + bi) = (a - b) + (a - b)i \in H$$

so that $H \leq G$.

(b) Since $|1| = 1$, then $1 \in H$ so that it is nonempty. For $z, w \in H$, we first note that $|w^{-1}| = |\overline{w}|/|w|^2 = 1/1 = 1$, where $\overline{w}$ is the complex conjugate of w. Then

$$|zw^{-1}| = |z||w^{-1}| = 1 \cdot 1 = 1$$

so that $zw^{-1} \in H$, hence $H \leq G$.

(c) Fix $n \in \mathbb{Z}^+$. Since $0 = 0/k$ for some k such that $k \mid n$, then $0 \in H$ so that H is nonempty. Let $a/b, c/d \in H$ such that $(a, c) = (b, d) = 1$, and $b \mid n$ and $d \mid n$. Then there exists $x, y \in \mathbb{Z}^+$ such that $bx = n$ and $dy = n$. Then

$$\frac{a}{b} - \frac{c}{d} = \frac{ax - cy}{n}.$$

Reducing this fraction still yields a denominator that divides n. Then $H \leq G$.

(d) Fix $n \in \mathbb{Z}^+$. Since $0 = 0/k$ for some k such that $(k, n) = 1$, then $0 \in H$ so that H is nonempty. Let $a/b, c/d \in H$ such that $(a, c) = 1$, $(b, n) = 1$, and $(d, n) = 1$. Then

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

If $(bd, n) \neq 1$, this would imply some $p \mid bd$ and $p \mid n$. Then $p \mid b$ or $p \mid d$, contradicting our assumption. Hence, $(bd, n) = 1$, so that $H \leq G$.

(e) Since $1 = 1^1 = 1/1$, then $1 \in H$ so that H is nonempty. For $x, y \in H$, then $x^2, y^2 \in \mathbb{Q}$. Then

$$\left(\frac{x}{y}\right)^2 = \frac{x^2}{y^2} \in \mathbb{Q}$$

so that $x/y \in H$, hence $H \leq G$. ∎

Exercise 2.1.2

In each of (a) – (e) prove that the specified subset is *not* a subgroup of the given group.

(a) The set of 2-cycles in S_n for $n \geq 3$.
(b) The set of reflections in D_{2n} for $n \geq 3$.
(c) For n a composite integer > 1 and G a group containing an element of order n, the set $\{x \in G \mid |x| = n\} \cup \{1\}$.
(d) The set of (positive and negative) odd integers in $\mathbb{Z}$ together with 0.
(e) The set of real numbers whose square is a rational number (under addition).

Solution. Let G be the group and H be the subset in question.

(a) For $n = 3$, then $(1\ 2), (1\ 3) \in H$, but $(1\ 2)(1\ 3) = (1\ 3\ 2) \notin H$.
(b) For $n = 3$, then $s, sr \in H$, but $s(sr) = r \notin H$.
(c) Let $n = ab$ for some $a, b \in \mathbb{Z}^+$ with $1 < a, b < n$. Let $x \in G$ with $|x| = n$. Then $x \in H$ so that $x^a \in H$, but $(x^a)^b = x^n = 1$ implies that $|x^a| \leq b$ so that $x^a \notin H$.
(d) Observe that $1 \in H$, but $1 + 1 = 2 \notin H$.
(e) Observe that $\sqrt{2}, \sqrt{3} \in H$, but $(\sqrt{2} + \sqrt{3})^2 \notin \mathbb{Q}$. ■

Exercise 2.1.3

Show that the following subsets of the dihedral group D_8 are actually subgroups:

(a) $\{1, r^2, s, sr^2\}$
(b) $\{1, r^2, sr, sr^3\}$

Solution.

(a)

$\circ$	1	r^2	s	sr^2
1	1	r^2	s	sr^2
r^2	r^2	1	sr^2	s
s	s	sr^2	1	r^2
sr^2	sr^2	s	r^2	1

(b)

$\circ$	1	r^2	sr	sr^3
1	1	r^2	sr	sr^3
r^2	r^2	1	sr^3	sr
sr	sr	sr^3	1	r^2
sr^3	sr^3	sr	r^2	1

Both tables show closure, since no product is an element outside the subset. ■

Exercise 2.1.4

Give an explicit example of a group G and an infinite subset H of G that is closed under the group operation but is not a subgroup of G.

Solution. Let $G = \mathbb{Z}$ and $H = \mathbb{Z}^+$. Then for any $m, n \in \mathbb{Z}^+$, we have $m + n \in \mathbb{Z}^+$ but $m - n \notin \mathbb{Z}^+$ when $m < n$. Hence, H is not a subgroup of G. ■

Exercise 2.1.5

Prove that G cannot have a subgroup H with $|H| = n - 1$, where $n = |G| > 2$.

Solution. If $H \leq G$ with $|H| = n - 1$, then there exists $k \in \mathbb{Z}^+$ such that $k(n-1) = n$ by Lagrange's Theorem. This implies $kn - k = n$, or equivalently $n = k/(k-1)$. Since $k = 2$ implies $n = 2$ and $k \geq 3$ implies $n \notin \mathbb{Z}^+$, then there is no such subgroup H. ■

Exercise 2.1.6

Let G be an Abelian group. Prove that $\{g \in G \mid |g| < \infty\}$ is a subgroup of G (called the **torsion subgroup** of G). Give an explicit example where this set is not a subgroup when G is non-Abelian.

Solution. Let the torsion subgroup be denoted as $\mathrm{Tor}(G)$. Since $|1| = 1$, then $1 \in \mathrm{Tor}(G)$. If $g, h \in \mathrm{Tor}(G)$ such that $|g| = m$ and $|h| = n$, we have

$$(gh^{-1})^{mn} = g^{mn}(h^{-1})^{mn} = (g^m)^n((h^n)^{-1})^m = 1^n(1^{-1})^m = 1,$$

where the first equality follows from G being Abelian. Then $|gh^{-1}| \leq mn$, hence $gh^{-1} \in \mathrm{Tor}(G)$, and $\mathrm{Tor}(G) \leq G$.

Consider $G = \mathrm{Aut}(\mathbb{R})$ where $f(x) = -x$ and $g(x) = 1 - x$. Then $|f| = |g| = 2$, but

$$(fg)(x) = f(g(x)) = f(1-x) = x - 1$$

By induction, $(fg)^n(x) = x - n$ for all $n \in \mathbb{Z}^+$, so $|fg| = \infty$. Hence, $fg \notin \mathrm{Tor}(G)$. ■

Exercise 2.1.7

Fix some $n \in \mathbb{Z}$ with $n > 1$. Find the torsion subgroup of $\mathbb{Z} \times (\mathbb{Z}/n\mathbb{Z})$. Show that the set of elements of infinite order together with the identity is *not* a subgroup of this direct product.

Solution. The only finite order element in $\mathbb{Z}$ is 0, while $\mathbb{Z}/n\mathbb{Z}$ is finite. Then

$$\mathrm{Tor}(\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) = \{(0, \bar{x}) \mid \bar{x} \in \mathbb{Z}/n\mathbb{Z}\}.$$

Consider $H = \{(x, \bar{y}) \in \mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \mid x \neq 0\} \cup \{(0, \bar{0})\}$. Then H contains the identity and all elements of infinite order. However, $(1, \bar{1}), (-1, \bar{0}) \in H$, but $(0, \bar{1}) = (1, \bar{1}) + (-1, \bar{0}) \notin H$, since it has finite order. Hence, H is not a subgroup of $\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$. ■

Exercise 2.1.8

Let H and K be subgroups of G. Prove that $H \cup K$ is a subgroup if and only if either $H \subseteq K$ or $K \subseteq H$.

Solution.

($\Rightarrow$) Suppose $H \cup K$ is a subgroup. If $H \subseteq K$, then we are done. Suppose that $H \not\subseteq K$. Then there exists some $h \in H$ such that $h \notin K$. Since $H \cup K$ is a subgroup, then for any $k \in K$, we have $hk \in H \cup K$.

Observe that $k^{-1} \in K$ since $K \leq G$. If $hk \in K$, then $h = (hk)k^{-1} \in K$, contradicting our assumption. It must be that $hk \in H$. Since $h \in H$, then $h^{-1} \in H$ as $H \leq G$. Then $h^{-1}(hk) = k \in H$, hence $K \subseteq H$.

($\Leftarrow$) If $H \subseteq K$, then $H \cup K = K \leq G$. Similarly, if $K \subseteq H$, then $H \cup K = H \leq G$. ■

Exercise 2.1.9

Let $G = \mathrm{GL}_n(F)$, where F is any field. Define

$$\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) \mid \det(A) = 1\}$$

(called the **special linear group**). Prove that $\mathrm{SL}_n(F) \leq \mathrm{GL}_n(F)$.

Solution. Since $\det(I_n) = 1$, then $I_n \in \mathrm{SL}_n(F)$. Suppose $A, B \in \mathrm{SL}_n(F)$. Then

$$\det(AB^{-1}) = \det(A)\det(B^{-1}) = \frac{\det(A)}{\det(B)} = \frac{1}{1} = 1$$

so that $AB^{-1} \in \mathrm{SL}_n(F)$. It follows that $\mathrm{SL}_n(F) \leq \mathrm{GL}_n(F)$. ■

Exercise 2.1.10

(a) Prove that if H and K are subgroups of G then so is their intersection $H \cap K$.
(b) Prove that the intersection of an arbitrary nonempty collection of subgroups of G is again a subgroup of G (do not assume the collection is countable).

Solution. We prove (b) and note that (a) is a special case with $I = \{1, 2\}$ and $H_1 = H$, $H_2 = K$. Let I be an indexing set, and consider the collection of subgroups $\{H_i\}_{i \in I}$ of G. Let

$$H = \bigcap_{i \in I} H_i$$

Since $1 \in H_i$ for all $i \in I$, then $1 \in H$. Suppose $a, b \in H$. Then $a, b \in H_i$ for all $i \in I$. Since each H_i is a subgroup of G, then $ab^{-1} \in H_i$ for all $i \in I$. Hence, $ab^{-1} \in H$, and $H \leq G$. ■

Exercise 2.1.11

Let A and B be groups. Prove that the following sets are subgroups of the direct product $A \times B$:
(a) $\{(a, 1) \mid a \in A\}$
(b) $\{(1, b) \mid b \in B\}$
(c) $\{(a, a) \mid a \in A\}$, where we assume $B = A$ (called the diagonal subgroup)

Solution. Let C be the set in each part.

(a) Since $1 \in A$, then $(1, 1) \in C$ so that C is nonempty. Suppose $(a_1, 1), (a_2, 1) \in C$. Then

$$(a_1, 1)(a_2, 1)^{-1} = (a_1, 1)(a_2^{-1}, 1) = (a_1 a_2^{-1}, 1) \in C$$

since $a_1 a_2^{-1} \in A$. Hence, $C \leq A \times B$.

(b) Since $1 \in B$, then $(1, 1) \in C$ so that C is nonempty. Suppose $(1, b_1), (1, b_2) \in C$. Then

$$(1, b_1)(1, b_2)^{-1} = (1, b_1)(1, b_2^{-1}) = (1, b_1 b_2^{-1}) \in C$$

since $b_1 b_2^{-1} \in B$. Hence, $C \leq A \times B$.

(c) Since $1 \in A$, then $(1, 1) \in C$ so that C is nonempty. Suppose $(a_1, a_1), (a_2, a_2) \in C$. Then

$$(a_1, a_1)(a_2, a_2)^{-1} = (a_1, a_1)(a_2^{-1}, a_2^{-1}) = (a_1 a_2^{-1}, a_1 a_2^{-1}) \in C$$

since $a_1 a_2^{-1} \in A$. Hence, $C \leq A \times B$. ■

Exercise 2.1.12

Let A be an Abelian group and fix some $n \in \mathbb{Z}$. Prove that the following sets are subgroups of A.
(a) $\{a^n \mid a \in A\}$.
(b) $\{a \in A \mid a^n = 1\}$.

Solution. Let B be the sets in question.
(a) Since $1 = 1^n$, then $1 \in B$ so that B is nonempty. Suppose $a^n, b^n \in B$. Then
(b)

$$(a^n)(b^n)^{-1} = a^n b^{-n} = (ab^{-1})^n \in B$$

where the last equality follows from the fact that A is Abelian. Hence, $B \leq A$.
(c) Since $1^n = 1$, then $1 \in B$ so that B is nonempty. Suppose $a, b \in B$. Then

$$(ab^{-1})^n = a^n(b^{-1})^n = a^n(b^n)^{-1} = 1 \cdot 1^{-1} = 1$$

where the second equality follows from the fact that A is Abelian. Hence, $B \leq A$. ■

Exercise 2.1.13

Let H be a subgroup of the additive group of rational numbers with the property that $1/x \in H$ for every nonzero element x of H. Prove that $H = 0$ or $\mathbb{Q}$.

Solution. Let $H \leq \mathbb{Q}$. Then $0 \in H$. If no other elements are in H, then $H = 0$, and we are done. Suppose there exists some nonzero $a/b \in H$ where $a, b \in \mathbb{Z}$ and $b \neq 0$. Moreover, we may assume without loss of generality that $a/b > 0$, for otherwise we may take $-a/b$ since H contains additive inverses. We now use a/b to generate $\mathbb{Q}$.

Since $a/b \in H$, then $b(a/b) = a \in H$ so that $1/a \in H$. Then $a(1/a) = 1 \in H$. From this, we can see that $\mathbb{Z} \subseteq H$. Now let $c/d \in \mathbb{Q}$ for some $c, d \in \mathbb{Z}$ with $d \neq 0$. Since $1 \in H$, then $d(1) = d \in H$ so that $1/d \in H$. Then $c(1/d) = c/d \in H$. Since c/d was arbitrary, then $\mathbb{Q} \subseteq H$. Hence, $H = \mathbb{Q}$. ■

Exercise 2.1.14

Show that $\{x \in D_{2n} \mid x^2 = 1\}$ is *not* a subgroup of D_{2n} (here $n \geq 3$).

Solution. Let H be the set in question. All reflections are in H. In particular, s and sr are in H, but $r = s(sr) \notin H$ since $|r| = n > 2$ ■

Exercise 2.1.15

Let $H_1 \subseteq H_2 \subseteq \cdots$ be an ascending chain of subgroups of G. Prove that

$$\bigcup_{i=1}^{\infty} H_i$$

is a subgroup of G.

Solution. Let H be the union in question. From $H_i \leq G$ for each $i \in \mathbb{Z}^+$, then $1 \in H_i$ so that $1 \in H$. Let $a, b \in H$ so that there exists $m, n \in \mathbb{Z}^+$ where $a \in H_m$ and $b \in H_n$. Put $k = \max(m, n)$. It follows that $H_m \subseteq H_k$ and $H_n \subseteq H_k$, so that $a, b \in H_k$. Since $H_k \leq G$, then $ab^{-1} \in H_k$, hence $ab^{-1} \in H$. Therefore, $H \leq G$. ■

Exercise 2.1.16

Let $n \in \mathbb{Z}^+$ and let F be a field. Prove that the set $\{(a_{ij}) \in \mathrm{GL}_n(F) \mid a_{ij} = 0 \text{ for all } i > j\}$ is a subgroup of $\mathrm{GL}_n(F)$ (called the group of upper triangular matrices).

Solution. Let $\mathrm{UT}_n(F)$ denote the set of $n \times n$ upper triangular matrices with entries from F. Observe that I_n contains 0's everywhere except the diagonal, hence $I_n \in \mathrm{UT}_n(F)$ so that $\mathrm{UT}_n(F)$ is nonempty. Suppose $A, B \in \mathrm{UT}_n(F)$, and consider $C = AB$. Then the (i, j)-th entry of C is given by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Consider the case when $i > j$, which is below the main diagonal.

- If $i > k$, then $a_{ik} = 0$ since A is upper triangular.
- If $k > j$, then $b_{kj} = 0$ since B is upper triangular.

Hence, for all k, either $a_{ik} = 0$ or $b_{kj} = 0$, so that $c_{ij} = 0$. It follows that $C \in \mathrm{UT}_n(F)$.

Let $A \in \mathrm{UT}_n(F)$. Since $A \in \mathrm{GL}_n(F)$, we know that the entries on the main diagonal are all non-zero, for otherwise A is not invertible. By back-substitution, we may solve the matrix equation $AX = I_n$ for X, and the solution will also be upper triangular. Hence, $A^{-1} \in \mathrm{UT}_n(F)$. ∎

Exercise 2.1.17

Let $n \in \mathbb{Z}^+$ and let F be a field. Prove that the set $\{(a_{ij}) \in \mathrm{GL}_n(F) \mid a_{ij} = 0 \text{ for all } i > j, \text{ and } a_{ii} = 1 \text{ for all } i\}$ is a subgroup of $\mathrm{GL}_n(F)$.

Solution. Let H be the set in question. Since the diagonal of I_n is only 1's, then $I_n \in H$. Suppose $A, B \in H$. By the previous exercise, $A, B \in \mathrm{UT}_n(F)$, so it remains to show that $a_{ii} = b_{ii} = 1$ for all $1 \le i \le n$. Then

$$(AB)_{ii} = \sum_{k=1}^{n} a_{ik} b_{ki}$$

Since $a_{ik} = 0$ for $k < i$ and $b_{ki} = 0$ for $k > i$, then the sum degrades to $a_{ii} b_{ii} = 1$. Then H is closed under multiplication. Moreover, suppose $D \in \mathrm{UT}_n(F)$ such that $DA = I_n$. Then

$$1 = (DA)_{ii} = d_{ii} a_{ii}$$

where we use the above to collapse the ii-th term of DA. Then $d_{ii} = 1$, hence $D \in H$, and H is closed under inverses. Hence, $H \le \mathrm{GL}_n(F)$. ∎

2.2 Centralizers and Normalizers, Stabilizers and Kernels

Exercise 2.2.1

Prove that $C_G(A) = \{g \in G \mid g^{-1}ag = a \text{ for all } a \in A\}$.

Solution. $g \in C_G(A) \iff ga = ag$ for all $a \in A \iff g^{-1}ga = g^{-1}ag \iff a = g^{-1}ag$ for all $a \in A$. ■

Exercise 2.2.2

Prove that $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$.

Solution. By definition, $C_G(Z(G)) \subseteq G$. Now let $g \in G$ and $z \in Z(G)$. Then $gz = zg$, or $gzg^{-1} = z$. Then $g \in C_G(Z(G))$, hence $G \subseteq C_G(Z(G))$. It follows that $C_G(Z(G)) = G$. Since $C_G(Z(G)) \leq N_G(Z(G)) \leq G$, then $N_G(Z(G)) = G$. ■

Exercise 2.2.3

Prove that if A and B are subsets of G with $A \subseteq B$ then $C_G(B)$ is a subgroup of $C_G(A)$.

Solution. Pick $g \in C_G(B)$. Then $gbg^{-1} = b$ for all $b \in B$. In particular, $gag^{-1} = a$ for all $a \in A$ because $A \subseteq B$. Then $C_G(B) \subseteq C_G(A)$. Since $C_G(B)$ and $C_G(A)$ are subgroups of G, then $C_G(B) \leq C_G(A)$. ■

Exercise 2.2.4

For each of S_3, D_8, and Q_8 compute the centralizers of each element and find the center of each group. Does Lagrange's theorem simplify your work?

Solution. Note that $Z(G)$ is the intersection of all centralizers of G, so that we can compute $Z(G)$ after computing all centralizers. Note that $C_G(1) = G$ for any group G, so we only need to compute the centralizers of the non-identity elements. Beginning with S_3, we know by Exercise 1.3.20 that $(1\ 2)$ and $(1\ 3)$ generate S_3, hence $C_{S_3}((1\ 2))$ cannot be the whole group. Since $\{1, (1\ 2)\} \leq C_{S_3}((1\ 2))$, then $|C_{S_3}((1\ 2))| = 2$. By symmetry, the same argument holds for $(1\ 3)$ and $(2\ 3)$, so that

$$C_{S_3}((1\ 3)) = \{1, (1\ 3)\}, \quad C_{S_3}((2\ 3)) = \{1, (2\ 3)\}.$$

A similar argument for $(1\ 2\ 3)$ along with the observation that $(1\ 2)(1\ 2\ 3) \neq (1\ 2\ 3)(1\ 2)$ shows that $C_{S_3}((1\ 2\ 3)) = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$. By symmetry, the same argument holds for $(1\ 3\ 2)$. Then $C_{S_3}((1\ 3\ 2)) = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$. Finally, $Z(S_3) = \{1\}$.

The following are the centralizers for each element in D_8:

$$\begin{aligned} C_{D_8}(1) &= D_8 & C_{D_8}(r) &= \{1, r, r^2, r^3\} \\ C_{D_8}(r^3) &= \{1, r, r^2, r^3\} & C_{D_8}(r^2) &= D_8 \\ C_{D_8}(sr) &= \{1, r^2, sr, sr^3\} & C_{D_8}(s) &= \{1, r^2, s, sr^2\} \\ C_{D_8}(sr^3) &= \{1, r^2, sr, sr^3\} & C_{D_8}(sr^2) &= \{1, r^2, s, sr^2\} \end{aligned}$$

and $Z(D_8) = \{1, r^2\}$. Lastly, the centralizers of Q_8 are

$$\begin{aligned} C_{Q_8}(1) &= Q_8 & C_{Q_8}(-1) &= Q_8 \\ C_{Q_8}(i) &= \{1,-1,i,-i\} & C_{Q_8}(-i) &= \{1,-1,i,-i\} \\ C_{Q_8}(j) &= \{1,-1,j,-j\} & C_{Q_8}(-j) &= \{1,-1,j,-j\} \\ C_{Q_8}(k) &= \{1,-1,k,-k\} & C_{Q_8}(-k) &= \{1,-1,k,-k\} \end{aligned}$$

and $Z(Q_8) = \{1, -1\}$. ■

Exercise 2.2.5

In each of parts (a) to (c) show that for the specified group G and subgroup A of G, $C_G(A) = A$ and $N_G(A) = G$.

(a) $G = S_3$ and $A = \{1, (1\ 2\ 3), (1\ 3\ 2)\}$.
(b) $G = D_8$ and $A = \{1, s, r^2, sr^2\}$.
(c) $G = D_{10}$ and $A = \{1, r, r^2, r^3, r^4\}$.

Solution.

(a) Observe that A is generated by $(1\ 2\ 3)$. From Exercise 2.2.4, we know $C_{S_3}(A) = A$. Then 3 divides $|N_G(A)|$ and $|N_G(A)|$ divides 6 by Lagrange's Theorem. Consider $(1\ 2) \in S_3$. Then

$$(1\ 2)A(1\ 2) = \{(1\ 2)(1)(1\ 2), (1\ 2)(1\ 2\ 3)(1\ 2), (1\ 2)(1\ 3\ 2)(1\ 2)\} = \{1, (1\ 3\ 2), (1\ 2\ 3)\} = A$$

so that $(1\ 2) \in N_G(A)$. Hence, $|N_G(A)| = 6$ and $N_G(A) = G$.

(b) Since $|A| = 4$, then 4 divides $|C_G(A)|$ and $|C_G(A)|$ divides 8 by Lagrange's Theorem. Observe that $r \notin C_G(A)$ since $rs = sr^3 \neq sr$. Then $|C_G(A)| \neq 8$, hence $|C_G(A)| = 4$ and $C_G(A) = A$. Then 4 divides $|N_G(A)|$ and $|N_G(A)|$ divides 8 by Lagrange's Theorem. Consider $r \in D_8$. Then

$$rAr^{-1} = \{r(1)r^{-1}, r(s)r^{-1}, r(r^2)r^{-1}, r(sr^2)r^{-1}\} = \{1, sr^3, r^2, sr\} = A$$

so that $r \in N_G(A)$. Then $|N_G(A)| = 8$ and $N_G(A) = G$.

(c) Note 5 divides $|C_G(A)|$ and $|C_G(A)|$ divides 10 by Lagrange's Theorem. Since $s \notin C_G(A)$ because $sr \neq rs$, then $C_G(A) = A$. Then 5 divides $|N_G(A)|$ which divides 10 by Lagrange's. A similar computation shows $sAs^{-1} = \{1, r^4, r^3, r^2, r\} = A$, hence $N_G(A) = G$. ■

Exercise 2.2.6

Let H be a subgroup of the group G.

(a) Show that $H \leq N_G(H)$. Give an example to show that this is not necessarily true if H is not a subgroup.
(b) Show that $H \leq C_G(H)$ if and only if H is Abelian.

Solution.

(a) Let $h \in H$ and hkh^{-1} for $k \in H$. Then $hkh^{-1} \in hHh^{-1}$ by closure. Conversely, for $h^{-1}yh \in hHh^{-1}$ for $y \in H$ implies that $y = h(h^{-1}yh)h^{-1} \in H$, which gives equality.

As a counter example, consider the set $G = D_6$ and $H = \{1, r, s\}$. Then

$$rHr^{-1} = \{1, r, sr^2\} \neq H$$

so that r does not normalize H, and H is not a subgroup of $N_G(H)$.

(b) (⟹) Suppose $H \leq C_G(H)$. Then for any $h_1, h_2 \in H$, we have $h_1 h_2 = h_2 h_1$ since $h_1 \in C_G(H)$. Hence, H is Abelian.

(⟸) Suppose H is Abelian, and let $h_1 \in H$. Then for any $h_2 \in H$, we have $h_1 h_2 = h_2 h_1$, so that $h_1 \in C_G(H)$. Since h_1 was arbitrary, then $H \leq C_G(H)$. ■

Exercise 2.2.7

Let $n \in \mathbb{Z}$ with $n \geq 3$. Prove the following:
(a) $Z(D_{2n}) = \{1\}$ if n is odd.
(b) $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$.

Solution. We claim that $Z(D_{2n})$ contains no reflections and contains r^i if and only if $n \mid 2i$. To see the first part, suppose $sr^j \in Z(D_{2n})$ for some $0 \leq j < n$. Since it must commute with r, then

$$sr^j r = rsr^j \implies sr^{j+1} = r^{n-1}sr^j \implies sr^{j+1} = sr^{j-1} \implies r^{j+1} = r^{j-1} \implies r^2 = 1$$

which is a contradiction since $n \geq 3$. Hence, no reflections are in $Z(D_{2n})$.

Now consider $r^i \in Z(D_{2n})$ for some $0 \leq i < n$. Since it must commute with s, then

$$r^i s = sr^i \implies r^i s = sr^{n-i} \implies r^i = r^{n-i} \implies r^{2i} = 1$$

so that $n \mid 2i$.
(a) If n is odd, then $n \mid 2i$ implies that $n \mid i$. Since $0 \leq i < n$, then $i = 0$ and $Z(D_{2n}) = \{1\}$.
(b) If $n = 2k$, then $n \mid 2i$ implies that $2k \mid 2i$. Simplifying, we have $k \mid i$. Since $0 \leq i < 2k$, then $i = 0$ or $i = k$, so that $Z(D_{2n}) = \{1, r^k\}$. ■

Exercise 2.2.8

Let $G = S_n$, fix an $i \in \{1, 2, \ldots, n\}$ and let $G_i = \{\sigma \in G \mid \sigma(i) = i\}$ (the stabilizer of i in G). Use group actions to prove that G_i is a subgroup of G. Find $|G_i|$.

Solution. Since $\mathrm{id}(1) = 1$, then $\mathrm{id} \in G_i$ so that G_i is nonempty. Suppose $\sigma \in G_i$. Then $\sigma(i) = i$. Then

$$\sigma^{-1}(i) = \sigma^{-1}(\sigma(i)) = i$$

so that $\sigma^{-1} \in G_i$. Now suppose $\tau \in G_i$. Then $\tau(i) = i$. It follows that

$$(\sigma\tau)(i) = \sigma(\tau(i)) = \sigma(i) = i$$

so that $\sigma\tau \in G_i$. Hence, $G_i \leq G$.

To find $|G_i|$, observe that any $\sigma \in G_i$ is completely determined by its action on the set $\{1, 2, \ldots, n\} \setminus \{i\}$, which has $n-1$ elements. Since σ is a permutation, then there are $(n-1)!$ ways to permute these $n-1$ elements. It follows that $|G_i| = (n-1)!$. ■

Exercise 2.2.9

For any subgroup H of G and any nonempty subset A of G define $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Show that $N_H(A) = N_G(A) \cap H$ and deduce that $N_H(A)$ is a subgroup of H (note that A need not be a subset of H).

Solution. Suppose $h \in N_H(A)$. By definition, $hAh^{-1} = A$ and $h \in H$. Then $h \in N_G(A)$ and $h \in H$, so that $h \in N_G(A) \cap H$. It follows that $N_H(A) \subseteq N_G(A) \cap H$.

Now suppose $h \in N_G(A) \cap H$. Then $hAh^{-1} = A$ and $h \in H$. By definition, then $h \in N_H(A)$. It follows that $N_G(A) \cap H \subseteq N_H(A)$. Hence, $N_H(A) = N_G(A) \cap H$. Since $N_G(A) \leq G$ and $H \leq G$, and $N_H(A)$ is the intersection of these two subgroups, then $N_H(A) \leq H$ by Exercise 2.1.10. ■

Exercise 2.2.10

Let H be a subgroup of order 2 in G. Show that $N_G(H) = C_G(H)$. Deduce that if $N_G(H) = G$ then $H \leq Z(G)$.

Solution. By assumption, $H = \{1, h\}$ for some $h \in G$ with $h \neq 1$. Since $C_G(H) \leq N_G(H)$, it suffices to show that $N_G(H) \subseteq C_G(H)$. To that end, pick $g \in N_G(H)$. Then $gHg^{-1} = H$, so that

$$gHg^{-1} = \{g(1)g^{-1}, ghg^{-1}\} = \{1, ghg^{-1}\} = H = \{1, h\}$$

It follows that $ghg^{-1} = h$, so that $g \in C_G(H)$. Since g was arbitrary, then $N_G(H) \subseteq C_G(H)$, hence $N_G(H) = C_G(H)$.

Now suppose $N_G(H) = G$. Then $C_G(H) = G$, so that for any $g \in G$, we have $ghg^{-1} = h$. It follows that $hg = gh$ for all $g \in G$, hence $H \leq Z(G)$. ■

Exercise 2.2.11

Prove that $Z(G) \leq N_G(A)$ for any subset A of G.

Solution. Let $z \in Z(G)$. For any $zaz^{-1} \in zAz^{-1}$, then $z \in Z(G)$ implies $zaz^{-1} = a \in A$. Hence $zAz^{-1} \subseteq A$. Conversely, for any $a \in A$, we have $a = zaz^{-1}$ for some $a \in A$ since $z \in Z(G)$. Hence $A \subseteq zAz^{-1}$. It follows that $zAz^{-1} = A$, so that $z \in N_G(A)$. Since z was arbitrary, then $Z(G) \leq N_G(A)$. ■

Exercise 2.2.12

Let R be the set of all polynomials with integer coefficients in the independent variables x_1, x_2, x_3, x_4 i.e., the members of R are finite sums of elements of the form $ax_1^{r_1}x_2^{r_2}x_3^{r_3}x_4^{r_4}$ where a is any integer and $r_1, \ldots, r_4$ are nonnegative integers. For example,

$$12x_1^5x_2^7x_4 - 18x_2^3x_3 + 11x_1^6x_2x_3^3x_4^{23} \qquad (*)$$

is a typical element of R. Each $\sigma \in S_4$ gives a permutation of $\{x_1, \ldots, x_4\}$ by defining $\sigma \cdot x_i = x_{\sigma(i)}$. This may be extended to a map from R to R by defining

$$\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all $p(x_1, x_2, x_3, x_4) \in R$ (i.e., σ simply permutes the indices of the variables).

(a) Let $p = p(x_1, \ldots, x_4)$ be the polynomial in $(*)$ above, let $\sigma = (1\ 2\ 3\ 4)$ and let $\tau = (1\ 2\ 3)$. Compute $\sigma \cdot p, \tau \cdot (\sigma \cdot p), (\tau \circ \sigma) \cdot p$, and $(\sigma \circ \tau) \cdot p$.

(b) Prove that these definitions give a (left) group action of S_4 on R.

(c) Exhibit all permutations in S_4 that stabilize x_4 and prove that they form a subgroup isomorphic to S_3.

(d) Exhibit all permutations in S_4 that stabilize the element $x_1 + x_2$ and prove that they form an Abelian subgroup of order 4.

(e) Exhibit all permutations in S_4 that stabilize the element $x_1x_2 + x_3x_4$ and prove that they form a subgroup isomorphic to the dihedral group of order 8.

(f) Show that the permutations in S_4 that stabilize the element $(x_1 + x_2)(x_3 + x_4)$ are exactly the same as those found in part (e).

Remark. *The two polynomials appearing in parts (e) and (f) and the subgroup that stabilizes them will play an important role in the study of roots of quartic equations in Section 14.6.*

Solution.

(a) Note that $\tau \circ \sigma = (1\ 3\ 4\ 2)$ and $\sigma \circ \tau = (1\ 3\ 2\ 4)$. Then

$$\begin{aligned}
\sigma \cdot p &= 12x_1x_2^5x_3^7 - 18x_3^3x_4 + 11x_1^{23}x_2^6x_3x_4^3 \\
\tau \cdot (\sigma \cdot p) &= 12x_1^7x_2x_3^5 - 18x_1^3x_4 + 11x_1x_2^{23}x_3^6x_4^3 \\
(\tau \circ \sigma) \cdot p &= 12x_1^7x_2x_3^5 - 18x_1^3x_4 + 11x_1x_2^{23}x_3^6x_4^3 \\
(\sigma \circ \tau) \cdot p &= 12x_1x_3^5x_4^7 - 18x_2x_4^3 + 11x_1^{23}x_2^3x_3^6x_4
\end{aligned}$$

(b) Note that $1 \cdot p = p$ for any $p \in R$. Let $\sigma, \tau \in S_4$, then

$$\begin{aligned}
\sigma \cdot (\tau \cdot p(x_1, x_2, x_3, x_4)) &= \sigma \cdot p(x_{\tau(1)}, x_{\tau(2)}, x_{\tau(3)}, x_{\tau(4)}) \\
&= p(x_{\sigma(\tau(1))}, x_{\sigma(\tau(2))}, x_{\sigma(\tau(3))}, x_{\sigma(\tau(4))}) \\
&= (\sigma \circ \tau) \cdot p(x_1, x_2, x_3, x_4)
\end{aligned}$$

(c) The permutations that stabilize x_4 fix 4 and permute $\{1, 2, 3\}$:

$$G = \{1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\} \cong S_3.$$

(d) The permutations that stabilize $x_1 + x_2$ swap 1 and 2 or fix them:

$$H = \{1, (1\ 2), (3\ 4), (1\ 2)(3\ 4)\}$$

which is Abelian since every non-identity element has order 2. Moreover, $|H| = 4$.

(e) The permutations that stabilize $x_1x_2 + x_3x_4$ swap 1 and 2 or fix them, swap 3 and 4 or fix them, swap the pairs $(1\ 2)$ and $(3\ 4)$, or swap all four elements:

$$K = \{1, (1\ 2), (3\ 4), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3), (1\ 3\ 2\ 4), (1\ 4\ 2\ 3)\}.$$

It is straightforward to see that K is generated by $(1\ 2)$ and $(1\ 3\ 2\ 4)$. Let $\varphi : D_8 \to K$ be defined by

$$\varphi(r) = (1\ 3\ 2\ 4), \quad \varphi(s) = (1\ 2)$$

Since $\varphi(r)^4 = \varphi(s)^2 = 1$ and $\varphi(s)\varphi(r) = (1\ 2)(1\ 3\ 2\ 4) = (1\ 4)(2\ 3) = \varphi(r)^{-1}\varphi(s)$, then φ is a homomorphism. Since φ is clearly surjective and $|D_8| = |K| = 8$, then φ is an isomorphism, so that $K \cong D_8$.

(f) The permutations that stabilize $(x_1+x_2)(x_3+x_4)$ must either swap 1 and 2 or fix them, and either swap 3 and 4 or fix them. Additionally, it may swap the pairs $(1\ 2)$ and $(3\ 4)$. These are exactly the same permutations found in part (e). ■

Exercise 2.2.13

Let n be a positive integer and let R be the set of all polynomials with integer coefficients in the independent variables $x_1, x_2, \ldots, x_n$, i.e., the members of R are finite sums of elements of the form $ax_1^{r_1}x_2^{r_2}\cdots x_n^{r_n}$ where a is any integer and $r_1, \ldots, r_n$ are nonnegative integers. For each $\sigma \in S_n$ define a map

$$\sigma : R \to R \quad \text{by} \quad \sigma \cdot p(x_1, x_2, \ldots, x_n) = p(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)})$$

Prove that this defines a (left) group action of S_n on R.

Solution. Since $\mathrm{id}(i) = i$ for all $1 \le i \le n$, then $\mathrm{id} \cdot p = p$ for any $p \in R$. Let $\sigma, \tau \in S_n$, then

$$\begin{aligned}\sigma \cdot (\tau \cdot p(x_1, x_2, \ldots, x_n)) &= \sigma \cdot p(x_{\tau(1)}, x_{\tau(2)}, \ldots, x_{\tau(n)}) \\ &= p(x_{\sigma(\tau(1))}, x_{\sigma(\tau(2))}, \ldots, x_{\sigma(\tau(n))}) \\ &= (\sigma \circ \tau) \cdot p(x_1, x_2, \ldots, x_n)\end{aligned}$$

Hence, this defines a (left) group action of S_n on R. ■

Exercise 2.2.14

Let $H(F)$ be the Heisenberg group over the field F introduced in Exercise 1.4.11. Determine which matrices lie in the center of $H(F)$ and prove that $Z(H(F))$ is isomorphic to the additive group F.

Solution. Let $X, Y \in H(F)$ be written as

$$X = (a, b, c) \quad \text{and} \quad Y = (d, e, f)$$

where $X \in Z(H(F))$. Then $XY = YX$, implying that

$$XY = (a + d, b + e + af, c + f) = (d + a, e + b + dc, f + c) = YX$$

so that $af + b + e = dc + e + b$. Simplifying, we have $af = dc$. Since this must hold for all $d, f \in F$, then $a = 0$ and $c = 0$. It follows that

$$Z(H(F)) = \{(0, x, 0) \mid x \in F\}$$

Moreover, the mapping $\varphi : F \to Z(H(F))$ defined by

$$\varphi(x) = (0, x, 0)$$

is clearly surjective. If $\varphi(x) = (0, 0, 0)$, then $x = 0$, so that φ is injective. Finally, for any $x, y \in F$, we have

$$\varphi(x + y) = (0, x + y, 0) = (0, x, 0)(0, y, 0) = \varphi(x)\varphi(y).$$

Hence, φ is an isomorphism, so that $Z(H(F)) \cong F$. ■

2.3 Cyclic Groups and Cyclic Subgroups

Exercise 2.3.1

Find all subgroups of $Z_{45} = \langle x \rangle$, giving a generator for each. Describe the containments between these subgroups.

Solution. Recall that the containment relation is the following: $\langle x^a \rangle \leq \langle x^b \rangle \iff (b, 45) \mid (a, 45)$.

$$\begin{aligned} Z_{45} = \langle x \rangle &> \langle x^3 \rangle, \langle x^5 \rangle, \langle x^9 \rangle, \langle x^{15} \rangle, 1 \\ \langle x^3 \rangle &> \langle x^9 \rangle, \langle x^{15} \rangle \\ \langle x^5 \rangle &> \langle x^{15} \rangle \\ \langle x^9 \rangle &> 1 \\ \langle x^{15} \rangle &> 1 \\ 1 &= \langle x^0 \rangle \end{aligned}$$

■

Exercise 2.3.2

If x is an element of the finite group G and $|x| = |G|$, prove that $G = \langle x \rangle$. Give an explicit example to show that this result need not be true if G is an infinite group.

Solution. For some $x \in G$ where $|x| = |G| = n$, then Proposition 2.2 says that $1, x, x^2, \dots, x^{n-1}$ are all distinct elements of G. Since G has n elements only, it follows that these are the elements of G, hence $G = \langle x \rangle$. Moreover, this is not true if G is infinite, since $|\mathbb{Z}| = |2| = \infty$, but $\langle 2 \rangle$ generates only the even integers. ■

Exercise 2.3.3

Find all generators for $\mathbb{Z}/48\mathbb{Z}$.

Solution. Using Proposition 2.5 and $|\mathbb{Z}/48\mathbb{Z}| = 48$, then $\langle \bar{a} \rangle$ generates $\mathbb{Z}/48\mathbb{Z}$ when $(a, 48) = 1$. We have the integers

$$a \in \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47\}$$

■

Exercise 2.3.4

Find all generators for $\mathbb{Z}/202\mathbb{Z}$.

Solution. Note that $202 = 2 \cdot 101$, which are both prime numbers. Then its generators is every number between 1 and 202, except 101 and even numbers. ■

Exercise 2.3.5

Find the number of generators for $\mathbb{Z}/49000\mathbb{Z}$.

Solution. Let φ denote the Euler-φ function. Then

$$\begin{aligned} \varphi(49000) &= \varphi(2^3)\varphi(5^3)\varphi(7^2) \\ &= 2^2(2-1)5^2(5-1)7(7-1) \\ &= 16800 \end{aligned}$$

■

Exercise 2.3.6

In $\mathbb{Z}/48\mathbb{Z}$ write out all elements of $\langle \bar{a} \rangle$ for every $\bar{a}$. Find all inclusions between subgroups in $\mathbb{Z}/48\mathbb{Z}$.

Solution. We know $\langle \bar{1} \rangle = \mathbb{Z}/48\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \dots, \overline{46}, \overline{47}\}$. The proper subgroups are:

$$\begin{aligned}
&\bar{2} = \{\bar{0}, \bar{2}, \dots, \overline{44}, \overline{46}\} &&\bar{6} = \{\bar{0}, \bar{6}, \dots, \overline{36}, \overline{42}\} &&\overline{16} = \{\bar{0}, \overline{16}, \overline{32}\}\\
&\bar{3} = \{\bar{0}, \bar{3}, \dots, \overline{42}, \overline{45}\} &&\bar{8} = \{\bar{0}, \bar{8}, \overline{16}, \overline{24}, \overline{32}, \overline{40}\} &&\overline{24} = \{\bar{0}, \overline{24}\}\\
&\bar{4} = \{\bar{0}, \bar{4}, \dots, \overline{40}, \overline{44}\} &&\overline{12} = \{\bar{0}, \overline{12}, \overline{24}, \overline{36}\} &&\bar{0} = \{\bar{0}\}
\end{aligned}$$

Moreover, the subgroup inclusions are clear from applying the fact that $\langle \bar{a} \rangle \leq \langle \bar{b} \rangle$ if and only if $b \mid a$. ■

Exercise 2.3.7

Let $Z_{48} = \langle x \rangle$ and use the isomorphism $\mathbb{Z}/48\mathbb{Z} \cong Z_{48}$ given by $\bar{1} \mapsto x$ to list all subgroups of Z_{48} as computed in the preceding exercise.

Solution. The subgroups of Z_{48} are $\langle x \rangle$, $\langle x^2 \rangle$, $\langle x^3 \rangle$, $\langle x^4 \rangle$, $\langle x^6 \rangle$, $\langle x^8 \rangle$, $\langle x^{12} \rangle$, $\langle x^{16} \rangle$, $\langle x^{24} \rangle$, and 1. ■

Exercise 2.3.8

Let $Z_{48} = \langle x \rangle$. For which integers a does the map φ_a defined by $\varphi_a : \bar{1} \mapsto x^a$ extend to an isomorphism from $\mathbb{Z}/48\mathbb{Z}$ to Z_{48}?

Solution. We know that $\bar{1}$ generates $\mathbb{Z}/48\mathbb{Z}$. By Theorem 2.4, φ_a extends to an isomorphism if and only if x^a generates Z_{48}. By Proposition 2.5, this occurs if and only if $(a, 48) = 1$. Hence, the integers a such that φ_a extends to an isomorphism are the a from Exercise 2.3.3. ■

Exercise 2.3.9

Let $Z_{36} = \langle x \rangle$. For which integers a does the map ψ_a defined by $\psi_a : \bar{1} \mapsto x^a$ extend to a *well-defined homomorphism* from $\mathbb{Z}/48\mathbb{Z}$ into Z_{36}. Can ψ_a ever be a surjective homomorphism?

Solution. Suppose $\bar{r}, \bar{s} \in \mathbb{Z}/48\mathbb{Z}$ such that $\bar{r} = \bar{s}$. Then $r \equiv s \bmod 48$. If ψ_a is well-defined, then

$$\psi_a(\bar{r}) = \psi_a(r \cdot \bar{1}) = x^{ar} = x^{as} = \psi_a(s \cdot \bar{1}) = \psi_a(\bar{s})$$

so that $x^{a(r-s)} = 1$. By $|x| = 36$, we know $36 \mid a(r-s)$. Since $48 \mid (r-s)$, then there exists k such that $36 \mid 48ak$, or $4 \mid 3ak$. Since k is arbitrary, this must hold for all integers k. Hence, $3 \mid a$.

ψ_a can never be surjective, since that would imply $(a, 36) = 1$. ■

Exercise 2.3.10

What is the order of $\overline{30}$ in $\mathbb{Z}/54\mathbb{Z}$? Write out all the elements and their orders in $\langle \overline{30} \rangle$.

Solution. By Proposition 2.5, we have $|\bar{1}| = 54$. Since $(54, 30) = 6$, then $|\overline{30}| = 9$. Then $\overline{30} = \bar{6}$, hence

$$\langle \overline{30} \rangle = \{\bar{0}, \bar{6}, \overline{12}, \overline{18}, \overline{24}, \overline{30}, \overline{36}, \overline{42}, \overline{48}\}.$$

Moreover, the orders of each $\bar{x} \in \overline{30}$ is just $54/(54, x)$. ■

Exercise 2.3.11

Find all cyclic subgroups of D_8. Find a proper subgroup of D_8 which is not cyclic.

Solution. The cyclic subgroups of D_8 are

$$\begin{aligned}\langle 1\rangle &= \{1\}\\ \langle r\rangle = \langle r^3\rangle &= \{1, r, r^2, r^3\}\\ \langle r^2\rangle &= \{1, r^2\}\\ \langle sr^k\rangle &= \{1, sr^k\}\end{aligned}$$

Moreover, a proper subgroup that is not cyclic is $\langle r^2, s\rangle = \{1, r^2, s, sr^2\}$. ■

Exercise 2.3.12

Prove that the following groups are *not* cyclic:
(a) $Z_2 \times Z_2$ (b) $Z_2 \times \mathbb{Z}$ (c) $\mathbb{Z} \times \mathbb{Z}$

Solution.
(a) Put $Z_2 = \langle x\rangle$. Then $Z_2 \times Z_2 = \{(1,1), (1,x), (x,1), (x,x)\}$. No element generates all four elements, hence $Z_2 \times Z_2$ is not cyclic.
(b) If (a, b) generates $Z_2 \times \mathbb{Z}$, then it must be one of the forms $(1, \pm 1)$ or $(x, \pm 1)$, since $\langle \pm 1\rangle = \mathbb{Z}$. But $(1, \pm 1)$ generates elements whose first component is only 1. If we consider $(x, \pm 1)$, then this also doesn't generate $Z_2 \times \mathbb{Z}$ since $(1, 1) \notin \langle (x, \pm 1)\rangle$.
(c) The only candidates for generators of $\mathbb{Z} \times \mathbb{Z}$ is $(\pm 1, \pm 1)$. But any subgroup generated by $(\pm 1, \pm 1)$ contain elements that differ only in sign as $(x, y) \notin \langle (\pm 1, \pm 1)\rangle$ when $|x| \neq |y|$. ■

Exercise 2.3.13

Prove that the following pairs of groups are *not* isomorphic:
(a) $\mathbb{Z} \times Z_2$ and $\mathbb{Z}$ (b) $\mathbb{Q} \times Z_2$ and $\mathbb{Q}$.

Solution.
(a) $(0, x) \in \mathbb{Z} \times Z_2$ has order 2, but no element in $\mathbb{Z}$ has order 2.
(b) Same reason as above. ■

Exercise 2.3.14

Let $\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$. For each of the following integers a compute σ^a: $a = 13, 65, 626, 1195, -6, -81, -570$, and -1211.

Solution. Refer to Exercise 1.3.9 to see the distinct powers of σ. We may use the division algorithm to see that each $a = 12q + r$ for integers q and r such that $0 \le r < 12$. Then $\sigma^a = \sigma^r$.

$$\begin{aligned}\sigma^{13} &= \sigma^{12+1} & \sigma^{-6} &= \sigma^{-1(12)+6}\\ \sigma^{65} &= \sigma^{5(12)+5} & \sigma^{-81} &= \sigma^{-7(12)+3}\\ \sigma^{626} &= \sigma^{52(12)+2} & \sigma^{-570} &= \sigma^{-48(12)+6}\\ \sigma^{1195} &= \sigma^{99(12)+7} & \sigma^{-1211} &= \sigma^{-101(12)+1}\end{aligned}$$ ■

Exercise 2.3.15

Prove that $\mathbb{Q} \times \mathbb{Q}$ is not cyclic.

Solution. If it were cyclic, then all subgroups are also cyclic, by Theorem 2.7. But $\mathbb{Z}\times\mathbb{Z}$ is not cyclic. ■

Exercise 2.3.16

Assume $|x| = n$ and $|y| = m$. Suppose that x and y *commute*: $xy = yx$. Prove that $|xy|$ divides the least common multiple of m and n. Need this be true of x and y do *not* commute? Give an example of commuting elements x and y such that the order of xy is not equal to the least common multiple of $|x|$ and $|y|$.

Solution. Let ℓ be the least common multiple of m and n. Then there exist $a, b \in \mathbb{Z}$ such that $\ell = am = bn$. Then

$$(xy)^\ell = x^\ell y^\ell = x^{bn} y^{am} = 1 \cdot 1 = 1$$

so that by Proposition 2.3, then $|xy|$ divides ℓ. Moreover, this is not true if x and y do not commute. If we consider $r, s \in D_8$, then $|r| = 4$ and $|s| = 2$, but $|rs| = 2$ which 4 does not divide. Moreover, consider $r^2, s \in D_8$. Then $|r^2| = 2$ and $|s| = 2$, but $|r^2 s| = 4 \neq \operatorname{lcm}(2, 2)$. ■

Exercise 2.3.17

Find a presentation for Z_n with one generator.

Solution. $Z_n = \langle x \mid x^n = 1\rangle$. ■

Exercise 2.3.18

Show that if H is any group and h is an element of H with $h^n = 1$, then there is a unique homomorphism from $Z_n = \langle x \rangle \to H$ such that $x \mapsto h$.

Solution. Define the map $\varphi : x^k \mapsto h^k$ so that $\varphi(x) = h$. Suppose $x^a = x^b$ for some $a, b \in \mathbb{Z}$. Then $n \mid (a - b)$, or $a - b = nk$ for some $k \in \mathbb{Z}$. It follows that

$$\varphi(x^a) = h^a = h^{b+nk} = h^b (h^n)^k = h^b \cdot 1 = h^b = \varphi(x^b)$$

so that φ is well-defined. Moreover,

$$\varphi(x^{a+b}) = h^{a+b} = h^a h^b = \varphi(x^a)\varphi(x^b)$$

so that φ is a homomorphism. Lastly, if $\psi : Z_n \to H$ is another homomorphism such that $\psi(x) = h$, it must satisfy $\psi(x^k) = \psi(x)^k = h^k$. This shows that $\psi = \varphi$. ■

Exercise 2.3.19

Show that if H is any group and h is an element of H, then there is a unique homomorphism from $\mathbb{Z}$ to H such that $1 \mapsto h$.

Solution. Define the map $\varphi : n \mapsto h^n$ so that $\varphi(1) = h$. The homomorphism and uniqueness follow similarly to Exercise 2.3.18. ■

Exercise 2.3.20

Let p be a prime and let n be a positive integer. Show that if x is an element of the group G such that $x^{p^n} = 1$, then $|x| = p^m$ for some $m \leq n$.

Solution. If $x^{p^n} = 1$, then Proposition 2.3 says that $|x|$ divides p^n. Since p is prime, then $|x|$ must also be a power of p that is at most n. ■

Exercise 2.3.21

Let p be an odd prime and let n be a positive integer. Use the Binomial Theorem to show that $(1+p)^{p^{n-1}} \equiv 1 \bmod p^n$ but $(1+p)^{p^{n-2}} \not\equiv 1 \bmod p^n$. Deduce that $1+p$ is an element of order p^{n-1} in the multiplicative group $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

Solution. By the Binomial Theorem, then

$$(1+p)^{p^{n-1}} = \sum_{k=0}^{p^{n-1}} \binom{p^{n-1}}{k} p^k = 1 + p^n + \sum_{k=2}^{p^{n-1}-2} \binom{p^{n-1}}{k} p^k.$$

Observe that for $k \geq 2$, the binomial coefficient is

$$\binom{p^{n-1}}{k} = \frac{p^{n-1}(p^{n-1}-1)\cdots(p^{n-1}-k+1)}{k!}$$

Lemma. $k!$ contains at most p^{k-1} as a factor.

Proof. Using the formula from Exercise 0.2.8, we obtain

$$\sum_{i=1}^{\infty} \left\lfloor \frac{k}{p^i} \right\rfloor \leq \sum_{i=1}^{\infty} \frac{k}{p^i} = k \cdot \frac{1/p}{1-1/p} = \frac{k}{p-1} \leq \frac{k}{2} \leq k-1.$$ ■

Then the numerator contains at least p^{n-1} as a factor, so that the entire term contains at least $p^{n-1-(k-1)} = p^{n-k}$ as a factor. Since this is multiplied by p^k in the summation, then each term for $k \geq 2$ contains at least p^n as a factor, so that $(1+p)^{p^{n-1}} \equiv 1 \bmod p^n$.

Now consider $(1+p)^{p^{n-2}}$. Using the binomial theorem, this expands to $1 + p^{n-1} +$ terms that are divisible by p^n. This is clearly not $1 \bmod p^n$, so the order of $1+p$ in $(\mathbb{Z}/p^n\mathbb{Z})^\times$ is p^{n-1}. ■

Exercise 2.3.22

Let n be an integer ≥ 3. Use the Binomial Theorem to show that $(1+2^2)^{2^{n-2}} \equiv 1 \bmod 2^n$ but $(1+2^2)^{2^{n-3}} \not\equiv 1 \bmod 2^n$. Deduce that 5 is an element of order 2^{n-2} in the multiplicative group $(\mathbb{Z}/2^n\mathbb{Z})^\times$.

Solution. The process is similar to the previous exercise, where we use the Binomial Theorem to expand $(1+2^2)^{2^{n-2}}$ and see that $k=0$ and $k=1$ terms are 1 and 2^n respectively. For $k \geq 2$, we have the summand as

$$\binom{2^{n-2}}{k} 2^{2k} = \frac{2^{n-2}(2^{n-2}-1)\cdots(2^{n-2}-k+1)}{k!} 2^{2k}$$

Using a similar formulation as above, we know that

$$\sum_{i=1}^{n}\left\lfloor \frac{k}{2^i}\right\rfloor < \sum_{i=1}^{\infty}\left\lfloor \frac{k}{2^i}\right\rfloor \leq \sum_{i=1}^{\infty}\frac{k}{2^i} = k$$

Since infinity is never achieved, then we may take this as strict, hence the greatest power of 2 is at most $k-1$. With 2^{2k}, the entire term contains at least $2^{n-1-(k-1)+2k} = 2^{n+k}$ as a factor. Since $k \geq 2$, then $n+k \geq n+2 > n$, so that each term for $k \geq 2$ contains at least 2^n as a factor. Hence, $(1+2^2)^{2^{n-2}} \equiv 1 \bmod 2^n$ as desired. Moreover, considering $(1+2^2)^{2^{n-3}}$, the $k = 1$ term is 2^{n-1} which is not divisible by 2^n. Hence, $(1+2^2)^{2^{n-3}} \not\equiv 1 \bmod 2^n$. Since the only integers that divide 2^n are $1, 2, 2^2, \ldots, 2^n$, and the first power that results in $1 \bmod 2^n$ is 2^{n-2}, it follows that the order of 5 in $(\mathbb{Z}/2^n\mathbb{Z})^\times$ is 2^{n-2}. ■

Exercise 2.3.23

Show that $(\mathbb{Z}/2^n\mathbb{Z})^\times$ is not cyclic for any $n \geq 3$.
HINT. *Find two distinct subgroups of order 2.*

Solution. By Theorem 2.7, we must have one subgroup of order 2 in $(\mathbb{Z}/2^n\mathbb{Z})^\times$ if it were cyclic. However,

$$(2^n-1)^2 = (-1)^2 \equiv 1 \bmod 2^n, \quad \text{and} \quad (2^{n-1}+1)^2 = 2^{2n-2} + 2^n + 1 \equiv 1 \bmod 2^n.$$

Since $n \geq 3$, then $2^n - 1 \neq 2^{n-1} + 1$ but both have order 2 in $(\mathbb{Z}/2^n\mathbb{Z})^\times$. Hence, it cannot be cyclic. ■

Exercise 2.3.24

Let G be a finite group and let $x \in G$.

(a) Prove that if $g \in N_G(\langle x\rangle)$ then $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

(b) Prove conversely that if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$ then $g \in N_G(\langle x\rangle)$.
HINT. *Show first that $gx^kg^{-1} = (gxg^{-1})^k = x^{ak}$ for any integer k so that $g\langle x\rangle g^{-1} \leq \langle x\rangle$. If x has order n, show that the elements gx^ig^{-1}, $i = 0, 1, \ldots, n-1$ are distinct, so that $|g\langle x\rangle g^{-1}| = |\langle x\rangle| = n$ and conclude that $g\langle x\rangle g^{-1} = \langle x\rangle$.*

REMARK. *Note that this cuts down some work in commuting normalizers of cyclic subgroups since one does not have to check $ghg^{-1} \in \langle x\rangle$ for every $h \in \langle x\rangle$.*

Solution.

(a) If $g \in N_G(\langle x\rangle)$, then $g\langle x\rangle g^{-1} = \langle x\rangle$. In particular, $gxg^{-1} \in \langle x\rangle$, so there exists some $a \in \mathbb{Z}$ such that $gxg^{-1} = x^a$.

(b) Suppose $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$. We first show that $gx^kg^{-1} = (gxg^{-1})^k$ by induction. For $k = 1$, the result is trivial. Suppose it holds for some $k \geq 1$. Then

$$gx^{k+1}g^{-1} = gx^kg^{-1}gxg^{-1} = (gxg^{-1})^k(gxg^{-1}) = (gxg^{-1})^{k+1}$$

so that the result holds for all positive integers k. For negative integers, observe that $(gxg^{-1})^{-1} = gx^{-1}g^{-1}$. Using the above result for all positive integers, we may extend it to negative integers as well.

Now, for any integer k, we have

$$gx^kg^{-1} = (gxg^{-1})^k = (x^a)^k = x^{ak}$$

so that $g\langle x\rangle g^{-1} \leq \langle x\rangle$. Since conjugation is an isomorphism, it is trivial to see that $|x| = |gxg^{-1}|$. If $|x| = n$, then the elements gx^ig^{-1} for $i = 0, 1, \ldots, n-1$ are distinct since if $gx^ig^{-1} = gx^jg^{-1}$ for some $0 \leq i < j \leq n-1$, then $x^i = x^j$ which contradicts the order of x. Hence, $|g\langle x\rangle g^{-1}| = |\langle x\rangle| = n$, so that $g\langle x\rangle g^{-1} = \langle x\rangle$. Therefore, $g \in N_G(\langle x\rangle)$. ■

Exercise 2.3.25

Let G be a cyclic group of order n and let k be an integer relatively prime to n. Prove that the map $x \mapsto x^k$ is surjective. Use Lagrange's Theorem to prove that the same is true for any finite group of order n. (For such k each element has a kth root in G. It follows from Cauchy's Theorem in Section 3.2 that if k is not relatively prime to the order of G then the map $x \mapsto x^k$ is not surjective.)

Solution. Fix $k \in \mathbb{Z}^+$ such that $(k, n) = 1$. Let $\varphi : x \mapsto x^k$. Since G is cyclic, then $G = \langle x \rangle$ for some $x \in G$. For any $y \in G$, there exists some integer m such that $y = x^m$. Since $(k, n) = 1$, there exist integers s, t such that $ks + nt = 1$. Then

$$\varphi(x^{ms}) = x^{kms} = x^{m(1-nt)} = x^m(x^n)^{-mt} = x^m \cdot 1 = y$$

so that φ is surjective.

Now let G be any finite group of order n. For any $y \in G$, consider the subgroup $\langle y \rangle$. If $|\langle y \rangle| = m$, then by Lagrange's Theorem, $m \mid n$. Since $(k, n) = 1$, then $(k, m) = 1$ as well. By the previous result, there exists some $z \in \langle y \rangle$ such that $z^k = y$. Hence, the map $x \mapsto x^k$ is surjective. ∎

Exercise 2.3.26

Let Z_n be a cyclic group of order n and for each integer a let

$$\sigma_a : Z_n \to Z_n \quad \text{by} \quad \sigma_a(x) = x^a \text{ for all } x \in Z_n$$

(a) Prove that σ_a is an automorphism of Z_n if and only if a and n are relatively prime.
(b) Prove that $\sigma_a = \sigma_b$ if and only if $a \equiv b \bmod n$.
(c) Prove that *every* automorphism of Z_n is equal to σ_a for some integer a.
(d) Prove that $\sigma_a \circ \sigma_b = \sigma_{ab}$. Deduce that the map $\bar{a} \mapsto \sigma_a$ is an isomorphism of $(\mathbb{Z}/n\mathbb{Z})^\times$ onto the automorphism group of Z_n (so $\operatorname{Aut}(Z_n)$ is an Abelian group of order $\varphi(n)$).

Solution.

(a) ($\Rightarrow$) Suppose $\sigma_a \in \operatorname{Aut}(Z_n)$, and let $d = (n, a)$. Assume, by way of contradiction, that $d \neq 1$. Then there are $s, t \in \mathbb{Z}$ such that $n = ds$ and $a = dt$. Let $Z_n = \langle x \rangle$. Then

$$\sigma_a(x^s) = x^{as} = x^{dts} = x^{ns} = 1$$

If $x^s \neq 1$, then $\ker \sigma_a$ would be non-trivial, contradicting that σ_a is an automorphism. Hence, $x^s = 1$, and since $|x| = n$ and $n \mid s$, then $s = n$ so that $d = 1$.

($\Leftarrow$) Suppose that $(a, n) = 1$. By the previous exercise, then σ_a is a surjective map. Since Z_n is finite, then σ_a is also bijective. Moreover,

$$\sigma_a(xy) = (xy)^a = x^a y^a = \sigma_a(x)\sigma_a(y)$$

for some $x, y \in Z_n$. Then σ_a is also a homomorphism, hence $\sigma_a \in \operatorname{Aut}(Z_n)$.

(b) ($\Rightarrow$) Suppose $\sigma_a = \sigma_b$. Let $Z_n = \langle x \rangle$. Then

$$x^a = \sigma_a(x) = \sigma_b(x) = x^b$$

so that $n \mid (a - b)$, or $a \equiv b \bmod n$.

($\Leftarrow$) Suppose $a \equiv b \bmod n$. Then there exists some integer k such that $a - b = nk$. Let $Z_n = \langle x \rangle$. For any $x^m \in Z_n$, we have

$$\sigma_a(x^m) = x^{am} = x^{(b+nk)m} = x^{bm}(x^n)^{km} = x^{bm} \cdot 1 = \sigma_b(x^m)$$

so that $\sigma_a = \sigma_b$.

(c) Let $Z_n = \langle x \rangle$ and let $\varphi \in \operatorname{Aut}(Z_n)$. Since Z_n is cyclic, there exists some integer a such that $\varphi(x) = x^a$. For any $x^m \in Z_n$, we have

$$\varphi(x^m) = \varphi(x)^m = (x^a)^m = x^{am} = \sigma_a(x^m)$$

so that $\varphi = \sigma_a$.

(d) Let $Z_n = \langle x \rangle$. For any $x^m \in Z_n$, we have

$$\sigma_a \circ \sigma_b(x^m) = \sigma_a(x^{bm}) = x^{abm} = \sigma_{ab}(x^m)$$

so that $\sigma_a \circ \sigma_b = \sigma_{ab}$. Consider the map

$$\varphi : (\mathbb{Z}/n\mathbb{Z})^\times \to \operatorname{Aut}(Z_n) \quad \text{by} \quad \varphi(\bar{a}) = \sigma_a$$

By part (b), φ is well-defined. Part (a) shows that φ is injective, while part (c) shows that φ is surjective. The first part of (d) shows that φ is a homomorphism. Hence, φ is an isomorphism of $(\mathbb{Z}/n\mathbb{Z})^\times$ onto $\operatorname{Aut}(Z_n)$. ∎

2.4 Subgroups Generated by Subsets of a Group

Exercise 2.4.1

Prove that if H is a subgroup of G then $\langle H\rangle = H$.

Solution. It is clear that $H \subseteq \langle H\rangle$. Recall that $\langle H\rangle$ is the intersection of all subgroups of G that contain H. Since H is itself a subgroup of G that contains H, then $\langle H\rangle \subseteq H$. Hence, $\langle H\rangle = H$. ■

Exercise 2.4.2

Prove that if A is a subset of B then $\langle A\rangle \leq \langle B\rangle$. Give an example where $A \subseteq B$ with $A \neq B$ but $\langle A\rangle = \langle B\rangle$.

Solution. Since $\langle B\rangle$ is the smallest subgroup of G that contains B, and $A \subseteq B$, then $A \subseteq \langle B\rangle$. Since $\langle A\rangle$ is the smallest subgroup of G that contains A, then $\langle A\rangle \subseteq \langle B\rangle$, so that $\langle A\rangle \leq \langle B\rangle$.

Take $G = \mathbb{Z}$ with $A = \{1\}$ and $B = \{1,2\}$. Then $A \subseteq B$ with $A \neq B$, but $\langle A\rangle = \langle B\rangle = \mathbb{Z}$. ■

Exercise 2.4.3

Prove that if H is an Abelian subgroup of a group G then $\langle H, Z(G)\rangle$ is Abelian. Give an explicit example of an Abelian subgroup H of a group G such that $\langle H, C_G(H)\rangle$ is not Abelian.

Solution. Let $g, h \in \langle H, Z(G)\rangle$. Using Proposition 2.9, we may put them as follows:

$$g = g_1^{\epsilon_1} g_2^{\epsilon_2} \dots g_m^{\epsilon_m}, \quad h = h_1^{\delta_1} h_2^{\delta_2} \dots h_n^{\delta_n}$$

where $\epsilon_i = \delta_i = \pm 1$, and $g_i, h_i \in H \cup Z(G)$ for all i. Since elements of H commute amongst each other, and elements of $Z(G)$ commute with every element, we obtain

$$gh = g_1^{\epsilon_1} \dots g_m^{\epsilon_m} h_1^{\delta_1} \dots h_n^{\delta_n} = h_1^{\delta_1} \dots h_n^{\delta_n} g_1^{\epsilon_1} \dots g_m^{\epsilon_m} = hg$$

so that $\langle H, Z(G)\rangle$ is Abelian.

Consider non-Abelian groups with non-trivial centers. Let $G = D_8$ and $H = Z(G) = \{1, r^2\}$. Then $C_G(H) = G$, but $\langle H, C_G(H)\rangle = \langle Z(G), G\rangle = G$ which is non-Abelian. ■

Exercise 2.4.4

Prove that if H is a subgroup of G then H is generated by the set $H - \{1\}$.

Solution. If H is trivial, then $H - \{1\} = \varnothing$ so that $\langle \varnothing\rangle = 1 = H$.

If H is non-trivial, let $h \in H$. If $h = 1$, then $h \in \langle H - \{1\}\rangle$. If $h \neq 1$, then $h \in H - \{1\}$ so that $h \in \langle H - \{1\}\rangle$, so $H \leq \langle H - \{1\}\rangle$. Since $H - \{1\} \subseteq H$, we have $\langle H - \{1\}\rangle \leq H$. Then $\langle H - \{1\}\rangle = H$. ■

Exercise 2.4.5

Prove that the subgroup generated by any two distinct elements of order 2 in S_3 is all of S_3.

Solution. See Exercise 1.3.20 for a discussion on why a pair of 2-cycles generate S_3. Calculations can be repeated for any other pair of distinct 2-cycles in S_3. ■

Exercise 2.4.6

Prove that the subgroup of S_4 generated by $(1\ 2)$ and $(1\ 2)(3\ 4)$ is a noncyclic group of order 4.

Solution. Let $a = (1\ 2)$ and $b = (3\ 4)$. Then $\langle a, ab\rangle = \{1, a, b, ab\}$ where $ab = ba$ since they are disjoint cycles. Moreover, $|a| = |b| = |ab| = 2$ so that it is noncyclic. ■

Exercise 2.4.7

Prove that the subgroup of S_4 generated by $(1\ 2)$ and $(1\ 3)(2\ 4)$ is isomorphic to the dihedral group of order 8.

Solution. Let $\alpha = (1\ 2)$ and $\beta = (1\ 3)(2\ 4)$. Let $\gamma = \alpha\beta$. It is routine to verify $\gamma\alpha = \alpha\gamma^{-1}$ and that $|\gamma| = 4$. By Exercise 1.2.6, these satisfy the relations of D_8. Moreover, $\langle\alpha, \beta\rangle$ contains $\langle\gamma\rangle$ and the distinct elements $\alpha\gamma, \alpha\gamma^2, \alpha\gamma^3$, along with the identity, giving 8 distinct elements. Hence, $\langle\alpha, \beta\rangle$ has order 8 and is isomorphic to D_8. ■

Exercise 2.4.8

Prove that $S_4 = \langle(1\ 2\ 3\ 4), (1\ 2\ 4\ 3)\rangle$.

Solution. Let $\alpha = (1\ 2\ 3\ 4)$ and $\beta = (1\ 2\ 4\ 3)$. Since $|\alpha| = 4$ and $|\alpha\beta| = |(1\ 3\ 2)| = 3$, then $\langle\alpha, \beta\rangle$ must have order 12 or 24 by Lagrange's. Observe that $\alpha^2 = (1\ 3)(2\ 4)$ and $\alpha^3\beta\alpha^3 = (1\ 2)$. Then $\langle\alpha, \beta\rangle$ has a subgroup of order 8 by Exercise 2.4.7, so $\langle\alpha, \beta\rangle$ must have order 24 and hence equals S_4. ■

Exercise 2.4.9

Prove that $\mathrm{SL}_2(\mathbb{F}_3)$ is the subgroup of $\mathrm{GL}_2(\mathbb{F}_3)$ generated by

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

HINT. *Recall from Exercise 2.1.9 that* $\mathrm{SL}_2(\mathbb{F}_3)$ *is the subgroup of matrices of determinant* 1. *You may assume this subgroup has order 24—this will be an exercise in Section 3.2.*

Solution. Consider the 12 distinct matrices in $\mathrm{SL}_2(\mathbb{F}_3)$ along with I_2:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \qquad A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$B^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \qquad AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \qquad BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$(AB)^2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \qquad ABA^2 = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} \qquad A^2B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$

$$B^2A = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \qquad ABA = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \qquad BAB = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Since $|\langle A, B\rangle| > 12$ and $\langle A, B\rangle \leq \mathrm{SL}_2(\mathbb{F}_3)$ which has order 24, we must have $\langle A, B\rangle = \mathrm{SL}_2(\mathbb{F}_3)$. ■

Exercise 2.4.10

Prove that the subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$ generated by

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is isomorphic to the quaternion group of order 8.
HINT. *Use a presentation for Q_8.*

Solution. Let

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

By direct computation, we have that $A^4 = B^4 = I_2$, and $A^2 = B^2 = -I_2$. Moreover, $AB = -BA$. Hence, the subgroup generated by A and B satisfies the relations of the quaternion group Q_8. Then the map

$$\varphi : Q_8 \to \langle A, B \rangle \quad \text{by} \quad i \mapsto A, \quad j \mapsto B$$

extends to a homomorphism since the relations are satisfied. Moreover, φ maps generators of Q_8 to generators of $\langle A, B \rangle$ so that φ is surjective. In particular, this implies $|\langle A, B \rangle| \leq |Q_8| = 8$. Since $B \notin \langle A \rangle$, we must have $|\langle A, B \rangle| = 8$, and hence φ is an isomorphism. ■

Exercise 2.4.11

Show that $\mathrm{SL}_2(\mathbb{F}_3)$ and S_4 are two non-isomorphic groups of order 24.

Solution. Recall that Q_8 and S_4 both have 6 elements of order 4. However, no two elements in Q_8 can generate the entirety of $\mathrm{SL}_2(\mathbb{F}_3)$ since $|Q_8| = 8$, while $S_4 = \langle (1\ 2\ 3\ 4), (1\ 2\ 4\ 3) \rangle$, hence they cannot be isomorphic. ■

Exercise 2.4.12

Prove that the subgroup of upper triangular matrices in $\mathrm{GL}_3(\mathbb{F}_2)$ is isomorphic to the dihedral group of order 8.
HINT. *First find the order of this subgroup.*

Solution. Let $\mathrm{UT}_3(\mathbb{F}_2)$ denote the upper triangular matrices in $\mathrm{GL}_3(\mathbb{F}_2)$. Note each matrix has 1's on the diagonal, and the 3 entries above the diagonal can each be either 0 or 1. Therefore, there are $2^3 = 8$ such matrices, so $|\mathrm{UT}_3(\mathbb{F}_2)| = 8$. Using the notation from Exercise 1.4.11, we must find $X, Y \in \mathrm{UT}_3(\mathbb{F}_2)$ such that $X^4 = Y^2 = I_3$ and $XY = YX^3$.

From Exercise 1.4.11 (d), pick out elements of order 4 and 2, namely $X = (1, 0, 1)$ and $Y = (1, 0, 0)$. The first set of relations are easily verified, while the relation $XY = YX^3$ can be checked by direct computation:

$$XY = (1, 0, 1)(1, 0, 0) = (0, 1, 1) = (1, 0, 0)(1, 1, 1) = YX^3.$$

Moreover, $Y \notin \langle X \rangle$, so $\langle X, Y \rangle$ has more than 4 elements. Since $\mathrm{UT}_3(\mathbb{F}_2)$ has 8 elements, it follows that $\langle X, Y \rangle = \mathrm{UT}_3(\mathbb{F}_2)$, and thus $\mathrm{UT}_3(\mathbb{F}_2) \cong D_8$. ■

Exercise 2.4.13

Prove that the multiplicative group of positive rational numbers is generated by the set $\{1/p \mid p \text{ is prime}\}$.

Solution. Let P be the set in question and $\mathbb{Q}_+^\times$ be the multiplicative group of positive rationals. Since $\langle P \rangle \leq \mathbb{Q}_+^\times$, then $(1/p)^{-1} = p \in \langle P \rangle$, so $p^k \in \langle P \rangle$ for $k \in \mathbb{Z}$. Then for any $q \in \mathbb{Q}_+^\times$ with $q = m/n$ for $m, n \in \mathbb{Z}^+$, we may write m and n as products of primes. Then $m/n = q \in \langle P \rangle$, hence $\mathbb{Q}_+^\times \leq \langle P \rangle$. ■

Exercise 2.4.14

A group H is called **finitely generated** if there is a finite set A such that $H = \langle A \rangle$.

(a) Prove that every finite group is finitely generated.
(b) Prove that $\mathbb{Z}$ is finitely generated.
(c) Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic. [If H is a finitely generated subgroup of $\mathbb{Q}$, show that $H \leq \langle 1/k \rangle$, where k is the product of all denominators appearing in a set of generators for H.]
(d) Prove that $\mathbb{Q}$ is not finitely generated.

Solution.

(a) Since G is finite and generates itself, then $G = \langle G \rangle$.

(b) $\mathbb{Z} = \langle 1 \rangle$.

(c) Let $H = \langle A \rangle$, where

$$A = \left\{ \frac{p_1}{q_1}, \frac{p_2}{q_2}, \dots, \frac{p_n}{q_n} \right\}$$

where $(p_i, q_i) = 1$ for all $1 \leq i \leq n$. Then every element $h \in H$ is of the form

$$h = \sum_{i=1}^{n} a_i \frac{p_i}{q_i}, \quad a_i \in \mathbb{Z}$$

since $\mathbb{Q}$ is Abelian. Define the quantities

$$k = \prod_{i=1}^{n} q_i, \quad k_j = k/q_j$$

so that k is the product of all denominators in A, and k_j is the product of all denominators except the denominator in the j-th fraction of A. We may then rewrite h as

$$h = \frac{1}{k} \sum_{i=1}^{n} a_i p_i k_i$$

so that $h \in \langle 1/k \rangle$, which is a cyclic subgroup of $\mathbb{Q}$. Then $H \leq \langle 1/k \rangle$, hence it is cyclic.

(d) Suppose $\mathbb{Q}$ was finitely generated. Then $\mathbb{Q} = \langle p/q \rangle$ where $(p, q) = 1$, by the previous part. Let $r \in \mathbb{Z}$ such that r does not divide q. Then there is some $n \in \mathbb{Z}$ such that

$$n \frac{p}{q} = \frac{1}{r}$$

Then $q = npr$, contradicting that r doesn't divide q. ■

Exercise 2.4.15

Exhibit a proper subgroup of $\mathbb{Q}$ which is not cyclic.

Solution. By the previous exercise, such a subgroup cannot be finitely generated. Consider the set

$$A = \left\{ \frac{1}{2^k} \middle| k \in \mathbb{Z}^+ \cup \{0\} \right\}$$

where $\langle A \rangle \leq \mathbb{Q}$. Note that $\langle A \rangle < \mathbb{Q}$ since $1/3 \notin \langle A \rangle$. Moreover, if $\langle A \rangle = \langle p/q \rangle$, then $q = 2^m$ since every element of A has a power of 2. Then $1/2^{m+1} \notin \langle p/q \rangle$, but $1/2^{m+1} \in \langle A \rangle$, hence $\langle A \rangle$ cannot be cyclic. ∎

Exercise 2.4.16

A subgroup M of a group G is called a **maximal subgroup** if $M \neq G$ and the only subgroups of G which contain M are M and G.

(a) Prove that if H is a proper subgroup of the finite group G then there is a maximal subgroup of G containing H.
(b) Show that the subgroup of all rotations in a dihedral group is a maximal subgroup.
(c) Show that if $G = \langle x \rangle$ is a cyclic group of order $n \geq 1$ then a subgroup H is maximal if and only if $H = \langle x^p \rangle$ for some prime p dividing n.

Solution.

(a) If H is maximal, we are done. If not, there exists $H_1 < G$ such that $H < H_1$. If H_1 is maximal, then we are done. Otherwise, construct subgroups H_i such that $H_i < H_{i+1} < G$. Since G is finite, this process must terminate at some H_k which is maximal. Then H_k is a maximal subgroup with $H < H_k$.

(b) Since $s \notin \langle r \rangle$, then $\langle r \rangle < D_{2n}$. If $\langle r \rangle$ is not maximal, there exists $H < G$ such that $sr^k \in H$ for $1 \leq k < n$. Since $r^{n-k} \in H$, then $sr^k r^{n-k} = s \in H$, which implies $H = D_{2n}$, contradicting that H is a proper subgroup. It follows that $\langle r \rangle$ is maximal.

(c) Suppose H is maximal, and put $H = \langle x^k \rangle$ and $d = (n, k)$. Note that $d > 1$ for the subgroup to be proper. Let p be a prime that divides n. If $k = p$, then we are done. If $k \neq p$, consider $\langle x^p \rangle$. Then $H < \langle x^p \rangle$ since $p \mid d$. Since H is maximal, then $\langle x^p \rangle = G$. Then $(p, n) = 1$, but this contradicts that $p \mid n$. Then $k = p$.

Suppose that $H = \langle x^p \rangle$ for prime $p \mid n$. If H is not maximal, there exists $\langle x^d \rangle$ such that $d \mid p$. Since p is prime, then either $d = 1$ or p. If $d = 1$, then $\langle x^d \rangle = G$ which shows that $\langle x^d \rangle$ is not a proper subgroup of G. If $d = p$, then $\langle x^d \rangle = \langle x^p \rangle$ so that H is not a proper subgroup of $\langle x^d \rangle$. It follows that H is maximal. ∎

Exercise 2.4.17

This is an exercise involving Zorn's Lemma. Prove that every nontrivial finitely generated group possesses maximal subgroups. Let G be a finitely generated group, say $G = \langle g_1, g_2, \ldots, g_n \rangle$, and let $\mathcal{S}$ be the set of all proper subgroups of G. Then $\mathcal{S}$ is partially ordered by inclusion. Let C be a chain in $\mathcal{S}$.

(a) Prove that the union H of all the subgroups in C is a subgroup of G.
(b) Prove that H is a proper subgroup.
(c) Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup).

Solution.

(a) Let $C \subseteq \mathcal{S}$ be a chain. Put

$$H = \bigcup_{K \in C} K$$

Since at least one subgroup is in C and subgroups are nonempty, then C is nonempty. Suppose $g, h \in H$. Then $g \in K_1$ and $h \in K_2$ for some $K_1, K_2 \in C$ such that $K_1 \le K_2$ or $K_2 \le K_1$, or both. Without loss of generality, suppose $K_1 \le K_2$. Then $g \in K_2$ as well so that $gh^{-1} \in K_2 \subseteq H$. Then $H \le G$.

(b) If H was not a proper subgroup, then H must contain all generators g_i. Associate each generator with a (not necessarily distinct) subgroup K_i so that $g_i \in K_i$ for every $1 \le i \le n$. Since C is a chain, then we may order the subgroups such that $K_j \le K_{j+1}$ for all $1 \le j \le n-1$. It follows that K_n contains every generator g_i so that $K_n = G$, contradicting that C is a chain of proper subgroups of G.

(c) Because G is nontrivial, then $\{1\} \in \mathcal{S}$ so that $\mathcal{S}$ is nonempty. Moreover, $H \in \mathcal{S}$ by the previous part, and for any $K \in C$, we have $K \le H$ so that H is an upper bound for C. Then every chain in $\mathcal{S}$ has an upper bound, so by Zorn's Lemma, $\mathcal{S}$ must have a maximal element. ∎

Exercise 2.4.18

Let p be a prime and let $Z = \{z \in \mathbb{C} \mid z^{p^n} = 1 \text{ for some } n \in \mathbb{Z}^+\}$ (so Z is the multiplicative group of all p-power roots of unity in $\mathbb{C}$). For each $k \in \mathbb{Z}^+$ let $H_k = \{z \in Z \mid z^{p^k} = 1\}$ (the group of p^kth roots of unity). Prove:

(a) $H_k \le H_m$ if and only if $k \le m$.
(b) H_k is cyclic for all k.
(c) Every proper subgroup of Z equals H_k for some k. In particular, every proper subgroup of Z is finite and cyclic.
(d) Z is not finitely generated.

Solution.

(a) Note that $|H_k| = p^k$ for any k as there are exactly p^k roots of unity. Now, if $H_k \le H_m$, then $p^k \mid p^m$ by Lagrange's Theorem. Then $p^k \le p^m$, or $k \le m$ since $p > 0$.

If $k \le m$, then $p^k \le p^m$. In particular, $p^k \mid p^m$. Then for any $z \in H_k$, we have

$$z^{p^m} = (z^{p^k})^{p^{m-k}} = 1$$

so that $z \in H_m$. Then $H_k \le H_m$.

(b) Let $\theta = e^{2\pi i/p^k}$. Note that $\langle\theta\rangle$ has order p^k, and $\langle\theta\rangle \subseteq H_k$. Now for some $z \in H_k$, we have $z = e^{2\pi i x/p^k}$ for some $0 \le x < p^k$. Then $z = (e^{2\pi i/p^k})^x \in \langle\theta\rangle$ so that $\langle\theta\rangle = H_k$.

(c) Let H be a proper subgroup of Z. Note that every element in Z has order p^i for some $i \in \mathbb{Z}^+$, so elements of H will have similar orders. Define the set

$$S = \{n \in \mathbb{Z}^+ \mid |h| = p^n \text{ for some } h \in H\}$$

so that S is the set of integers n such that H contains a p^n-th root of unity. Moreover, H is trivially nonempty if we define $H_0 = \{z \in Z \mid z^{p^0} = z = 1\}$ so that the trivial proper subgroup $\{1\}$ of Z allows $1 \in H$.

Suppose now that S is infinite. For any $n \in \mathbb{Z}^+$, there exists $h \in H$ and $m \in \mathbb{Z}^+$ such that $|h| = p^m > p^n$, for otherwise it would be that every $h \in H$ has order $|h| \le p^n$, meaning that n is an upper bound for S. Then $|h| = p^m$ so that $H_m = \langle h \rangle$, and $H_m \leq H$. Since $H_n \leq H_m$ for every $n \in \mathbb{Z}^+$, then

$$Z = \bigcup_{n \in \mathbb{Z}^+} H_n \subseteq H$$

which shows $Z \leq H$. But $H \leq Z$, hence $Z = H$, which contradicts that $H < Z$. It must be that S is finite, so it has some maximal element s. Because $s \in S$, then there is $h_0 \in H$ where $|h_0| = p^s$ so that $H_s = \langle h_0 \rangle \leq H$. Suppose $h \in H$ with $|h| = p^k$ for some $k \in \mathbb{Z}^+$. Then $k \in S$ where $k \leq s$ so that $H_k \leq H_s$. Since $h \in H_k$, then $h \in H_s$ so $H \leq H_s$. Hence, $H = H_s$.

(d) Put $Z = \langle z_1, z_2, \dots, z_n \rangle$ for some $n \in \mathbb{Z}^+$ such that $|z_i| = p^{x_i}$. Let $x = \max(x_1, x_2, \dots, x_n)$. Then $z_i \in H_x$ for every $1 \leq i \leq n$ so that $Z \leq H_x$. But recall that Z comprises every p-power roots of unity so that Z is not a subgroup of H_x, contradicting part (a). It must be that Z is infinitely generated. ■

Exercise 2.4.19

A nontrivial Abelian group A (written multiplicatively) is called **divisible** if for each element $a \in A$ and each nonzero integer k there is an element $x \in A$ such that $x^k = a$.

(a) Prove that the additive group of rational numbers $\mathbb{Q}$ is divisible.

(b) Prove that no finite Abelian group is divisible.

Solution.

(a) Suppose $p/q \in \mathbb{Q}$. Then $(p/qn)n = p/q$, where $p/qn \in \mathbb{Q}$.

(b) Let G be finite Abelian with $|G| = n$. For some nontrivial $g \in G$, we have $g^n = 1$. If G was divisible, then there would be some $x \in G$ such that $x^n = g$. But then $g = x^n = 1$, contradicting that g is nontrivial. ■

Exercise 2.4.20

Prove that if A and B are nontrivial Abelian groups, then $A \times B$ is divisible if and only if both A and B are divisible.

Solution. Suppose $A \times B$ is divisible, and let $a \in A$, $b \in B$, and $k \in \mathbb{Z}^+$. Then there exists $(c, d) \in A \times B$ such that $(c, d)^k = (c^k, d^k) = (a, b)$. But then $c^k = a$ and $d^k = b$ so that A and B are divisible.

If A and B are both divisible, pick $(a, b) \in A \times B$ and let $k \in \mathbb{Z}^+$. Then there exists $c \in A$ and $d \in B$ such that $c^k = a$ and $d^k = b$. Then $(c, d)^k = (c^k, d^k) = (a, b)$ so that $A \times B$ is divisible. ■

2.5 The Lattice of Subgroups of a Group

Exercise 2.5.1

Let H and K be subgroups of G. Exhibit all possible sublattices which show only $G, 1, H, K$ and their joins and intersections. What distinguishes the different drawings?

Solution. If H and K are distinct and neither is contained in the other, then the sublattice is a diamond shape. Otherwise, one of H or K is contained in the other, or they are equal, leading to a chain structure.

$$
\begin{tikzcd}[arrows=-]
 & G \arrow[d] & \\
 & \langle H, K\rangle \arrow[dl] \arrow[dr] & \\
H \arrow[dr] & & K \arrow[dl] \\
 & H \cap K \arrow[d] & \\
 & 1 &
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}[arrows=-]
G \arrow[d] \\
H \arrow[d] \\
K \arrow[d] \\
1
\end{tikzcd}
$$

■

Exercise 2.5.2

In each of (a) to (d) list all subgroups of D_{16} that satisfy the given condition.

(a) Subgroups that are contained in $\langle sr^2, r^4\rangle$
(b) Subgroups that are contained in $\langle sr^7, r^4\rangle$
(c) Subgroups that contain $\langle r^4\rangle$
(d) Subgroups that contain $\langle s\rangle$.

Solution. Using the subgroup lattice of D_{16}, we find:

(a) $\langle sr^2, r^4\rangle, \langle sr^6\rangle, \langle sr^2\rangle, \langle r^4\rangle, 1$.
(b) $\langle sr^3, r^4\rangle, \langle r^4\rangle, \langle sr^3\rangle, \langle sr^7\rangle, 1$.
(c) $\langle r^4\rangle, \langle sr^2, r^4\rangle, \langle s, r^4\rangle, \langle r^2\rangle, \langle sr^3, r^4\rangle, \langle sr^5, r^4\rangle, \langle s, r^2\rangle, \langle r\rangle, \langle sr, r^2\rangle, D_{16}$.
(d) $\langle s\rangle, \langle s, r^4\rangle, \langle s, r^2\rangle, D_{16}$. ■

Exercise 2.5.3

Show that the subgroup $\langle s, r^2\rangle$ of D_8 is isomorphic to V_4.

Solution. Note that $\langle s, r^2\rangle = \{1, s, r^2, sr^2\}$. Consider the mapping $\varphi : V_4 \to \langle s, r^2\rangle$ defined by

$$\varphi(1) = 1, \quad \varphi(a) = s, \quad \varphi(b) = r^2, \quad \varphi(c) = sr^2.$$

To see that φ is a homomorphism, we can check a pair of non-identity elements, for example:

$$\varphi(a)\varphi(b) = s \cdot r^2 = sr^2 = \varphi(c) = \varphi(ab).$$

Similar checks for the other pairs show that φ preserves the group operation. Since $\langle s, r^2\rangle$ has order 4 and φ is surjective, it follows that φ is also injective, and hence an isomorphism. Therefore, $\langle s, r^2\rangle \cong V_4$. ■

Exercise 2.5.4

Use the given lattice to find all pairs of elements that generate D_8 (there are 12 pairs).

Solution. Note that $D_8 = \langle s, r\rangle$. Moreover, $s \neq rs$, and $r \in \langle s, rs\rangle$ so that $\langle s, rs\rangle = D_8$. Since $r = r^3$, we also have $\langle s, r^3\rangle = D_8$. We may also replace the reflection s with sr^2, since combinations of a reflection with an odd rotation and a reflection with an even rotation contain r and thus generates D_8. It follows that the pairs of elements that generate D_8 are:

$$\langle s,r\rangle, \langle s,r^3\rangle, \langle s,rs\rangle, \langle s,r^3s\rangle, \langle r^2s,r\rangle, \langle r^2s,r^3\rangle, \langle r^2s,rs\rangle, \langle r^2s,r^3s\rangle, \langle r,rs\rangle, \langle r^3,rs\rangle, \langle r,r^3s\rangle, \langle r^3,r^3s\rangle \qquad \blacksquare$$

Exercise 2.5.5

Use the given lattice to find all elements $x \in D_{16}$ such that $D_{16} = \langle x, s\rangle$ (there are 8 such elements x).

Solution. Note that $\langle r\rangle = \langle r^3\rangle = \langle r^5\rangle = \langle r^7\rangle$. We then pair each generator with an s so that we obtain just the rotation to then generate D_{16}: $x = r, r^3, r^5, r^7, sr, sr^3, sr^5, sr^7$. ■

Exercise 2.5.6

Use the given lattices to help find the centralizers of every element in the following groups:
(a) D_8 (b) Q_8 (c) S_3 (d) D_{16}.

Solution.

(a) To find $C_G(a)$ for some $a \in G$, we look for elements in $g \in G$ such that $g \notin \langle a\rangle$, but $ga = ag$, then look for the smallest subgroup containing both $\langle a\rangle$ and g.

$$\begin{aligned} C_{D_8}(1) &= D_8 & C_{D_8}(s) &= \langle s, r^2\rangle \\ C_{D_8}(r) &= \langle r\rangle & C_{D_8}(rs) &= \langle rs, r^2\rangle \\ C_{D_8}(r^2) &= D_8 & C_{D_8}(r^2s) &= \langle s, r^2\rangle \\ C_{D_8}(r^3) &= \langle r\rangle & C_{D_8}(r^3s) &= \langle sr, r^2\rangle \end{aligned}$$

(b) Note that $1, -1 \in Z(Q_8)$, while none of i, j, and k commute with anything but themselves. Then:

$$\begin{aligned} C_{Q_8}(1) = C_{Q_8}(-1) &= Q_8 \\ C_{Q_8}(i) = C_{Q_8}(-i) &= \langle i\rangle \\ C_{Q_8}(j) = C_{Q_8}(-j) &= \langle j\rangle \\ C_{Q_8}(k) = C_{Q_8}(-k) &= \langle k\rangle \end{aligned}$$

(c) Note that no element of S_3 commutes with other elements, then $C_{S_3}(a) = \langle a\rangle$ for $a \in S_3 - \{1\}$.

(d) Use similar reasoning as in part (a) to obtain the centralizers:

$$\begin{aligned} C_{D_{16}}(1) = C_{D_{16}}(r^4) &= D_{16} \\ C_{D_{16}}(r^k) &= \langle r\rangle \text{ for all } k = 1, 2, 3, 5, , 6, \\ C_{D_{16}}(s) = C_{D_{16}}(sr^4) &= \langle s, r^4\rangle \\ C_{D_{16}}(sr) = C_{D_{16}}(sr^5) &= \langle sr^5, r^4\rangle \\ C_{D_{16}}(sr^2) = C_{D_{16}}(sr^6) &= \langle sr^2, r^4\rangle \\ C_{D_{16}}(sr^3) = C_{D_{16}}(sr^7) &= \langle sr^3, r^4\rangle \end{aligned} \qquad \blacksquare$$

Exercise 2.5.7

Find the center of D_{16}.

Solution. $Z(D_{16}) = \{1, r^4\}$ by Exercise 2.2.7. ∎

Exercise 2.5.8

In each of the following groups find the normalizer of each subgroup:
(a) S_3 (b) Q_8.

Solution.

(a) From the subgroup lattice of S_3, each nontrivial subgroup is cyclic and maximal. Then $N_{S_3}(\langle\alpha\rangle) = \langle\alpha\rangle$ or S_3. For any 2-cycle α, note that $\alpha\beta \neq \beta\alpha$ for any other 2-cycle β, hence $N_{S_3}(\langle\alpha\rangle) = \langle\alpha\rangle$. For the 3-cycle $\alpha = (1\ 2\ 3)$, any 2-cycle β satisfies $\beta\alpha\beta^{-1} \in \langle\alpha\rangle$, so $N_{S_3}(\langle\alpha\rangle) = S_3$.

(b) Since $Z(Q_8) = \langle -1\rangle$, then $N_{Q_8}(\langle -1\rangle) = Q_8$. Similarly, $N_{Q_8}(\{1\}) = Q_8$. For the subgroups of order 4, note that $ji(-j) = -i$ and $j(-i)(-j) = i$, so that $j \in N_{Q_8}(\langle i\rangle)$. Then $N_{Q_8}(\langle i\rangle) = Q_8$. By symmetry, the same holds for $\langle j\rangle$ and $\langle k\rangle$. ∎

Exercise 2.5.9

Draw the lattices of subgroups of the following groups:
(a) $\mathbb{Z}/16\mathbb{Z}$ (b) $\mathbb{Z}/24\mathbb{Z}$ (c) $\mathbb{Z}/48\mathbb{Z}$.

Solution.

$$\begin{tikzcd}[arrows=dash]
\mathbb{Z}/16\mathbb{Z} \arrow[d] \\
\langle 2\rangle \arrow[d] \\
\langle 4\rangle \arrow[d] \\
\langle 8\rangle \arrow[d] \\
\langle 16\rangle
\end{tikzcd}
\qquad
\begin{tikzcd}[arrows=dash]
& \mathbb{Z}/24\mathbb{Z} \arrow[dl] \arrow[dr] & & & \\
\langle 3\rangle \arrow[dr] & & \langle 2\rangle \arrow[dl] \arrow[dr] & & \\
& \langle 6\rangle \arrow[dr] & & \langle 4\rangle \arrow[dl] \arrow[dr] & \\
& & \langle 12\rangle \arrow[dr] & & \langle 8\rangle \arrow[dl] \\
& & & \langle 24\rangle &
\end{tikzcd}
\qquad
\begin{tikzcd}[arrows=dash]
& \mathbb{Z}/48\mathbb{Z} \arrow[dl] \arrow[dr] & & & & \\
\langle 3\rangle \arrow[dr] & & \langle 2\rangle \arrow[dl] \arrow[dr] & & & \\
& \langle 6\rangle \arrow[dr] & & \langle 4\rangle \arrow[dl] \arrow[dr] & & \\
& & \langle 12\rangle \arrow[dr] & & \langle 8\rangle \arrow[dl] \arrow[dr] & \\
& & & \langle 24\rangle \arrow[dr] & & \langle 16\rangle \arrow[dl] \\
& & & & \langle 48\rangle &
\end{tikzcd}$$

∎

Exercise 2.5.10

Classify groups of order 4 by proving that if $|G| = 4$, then $G \cong Z_4$ or $G \cong V_4$.
HINT. *cf.* *Exercise 1.1.36.*

Solution. Let $G = \{1, a, b, c\}$. If G has an element of order 4, then $G \cong Z_4$ by Theorem 2.4.

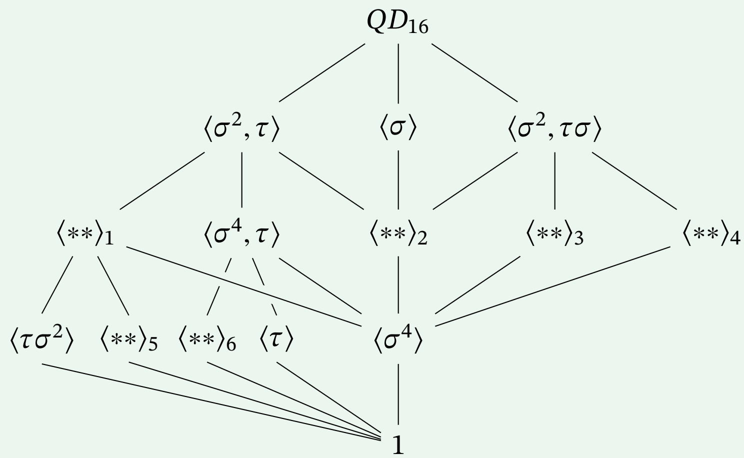
Suppose G is not cyclic. Then every nonidentity element has order 2 by Lagrange's Theorem. In particular, $a^2 = b^2 = c^2 = 1$. Note that $ab \neq 1$, since otherwise $b = a^{-1} = a$. Similarly, $ab \neq a$ and $ab \neq b$, so that $ab = c$. By similar reasoning, we have $bc = a$ and $ca = b$. This is precisely the definition of V_4, so that $G \cong V_4$. ∎

Exercise 2.5.11

Consider the group of order 16 with the following presentation:

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1, \sigma\tau = \tau\sigma^3 \rangle$$

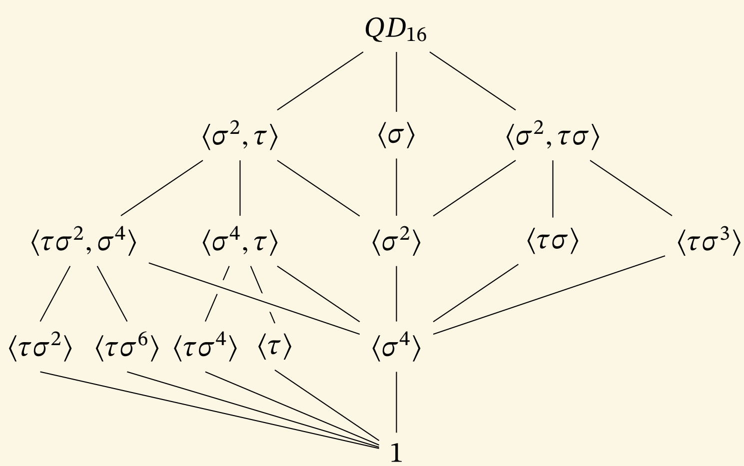
(called the **quasidihedral** or **semidihedral** group of order 16). This group has three subgroups of order 8: $\langle \tau, \sigma^2 \rangle \cong D_8$, $\langle \sigma \rangle \cong Z_8$, and $\langle \sigma^2, \sigma\tau \rangle \cong Q_8$, and every proper subgroup is contained in one of these three subgroups. Fill in the missing subgroups in the lattice of all subgroups of the quasidihedral group, exhibiting each subgroup with at most two generators.



Solution. The above has had subscripts added to distinguish the unknown subgroups. It is clear that $\langle ** \rangle_2$ must be $\langle \sigma^2 \rangle$ since it is the only subgroup contained in both $\langle \sigma \rangle$ and $\langle \sigma^2, \tau \rangle$. To find the other subgroups, we calculate the cyclic subgroups of the form $\langle \tau\sigma^k \rangle$:

$$\langle \tau\sigma \rangle = \{1, \tau\sigma, \sigma^4, \tau\sigma^5\} \qquad \langle \tau\sigma^3 \rangle = \{1, \tau\sigma^3, \sigma^4, \tau\sigma^7\}$$
$$\langle \tau\sigma^4 \rangle = \{1, \tau\sigma^4\} \qquad \langle \tau\sigma^6 \rangle = \{1, \tau\sigma^6\}$$

It becomes clear that 6 must be $\langle \tau\sigma^4 \rangle$, since the subgroup above contains both τ and σ^4. 3 and 4 must be $\langle \tau\sigma \rangle$ and $\langle \tau\sigma^3 \rangle$ respectively, since $\langle \sigma^2, \tau\sigma \rangle$ both contain $\tau\sigma$ and σ^2 which multiply to $\tau\sigma^3$. Certainly, 1 cannot be $\langle \tau\sigma^6 \rangle$ as it does not contain $\langle \tau\sigma^2 \rangle$, so it must be 5. Then 1 must be $\langle \tau\sigma^2, \sigma^4 \rangle$ as it contains both subgroups. Hence, the completed diagram is



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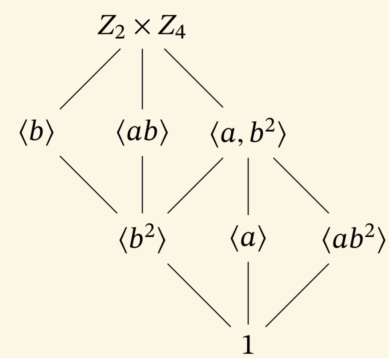
Exercise 2.5.12

The group $A = Z_2 \times Z_4 = \langle a, b \mid a^2 = b^4 = 1, ab = ba\rangle$ has order 8 and has three subgroups of order 4: $\langle a, b^2\rangle \cong V_4$, $\langle b\rangle \cong Z_4$, and $\langle ab\rangle \cong Z_4$ and every proper subgroup is contained in one of these three. Draw the lattice of all subgroups of A, giving each subgroup in terms of at most two generators.

Solution. Since A is the direct product of two Abelian groups, it itself is Abelian, and we may write:

$$A = \{1, b, b^2, b^3, a, ab, ab^2, ab^3\}$$

Using the isomorphisms given, we construct the lattice.

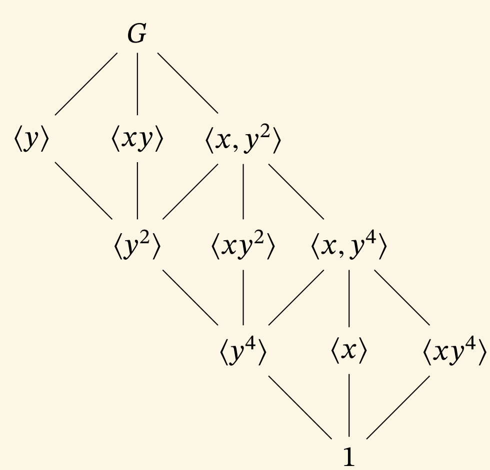


■

Exercise 2.5.13

The group $G = Z_2 \times Z_8 = \langle x, y \mid x^2 = y^8 = 1, xy = yx\rangle$ has order 16 and has three subgroups of order 8: $\langle x, y^2\rangle \cong Z_2 \times Z_4$, $\langle y\rangle \cong Z_8$, and $\langle xy\rangle \cong Z_8$, and every proper subgroup is contained in one of those three. Draw the lattice of all subgroups of G, giving each subgroups in terms of at most two generators (cf. Exercise 2.5.12).

Solution. Put $x = a$ and $y^2 = b$, where $\langle a, b\rangle$ is Exercise 2.5.12. Then G contains a copy of $Z_2 \times Z_4$, with the necessary appending of $\langle y\rangle$ and $\langle xy\rangle$ to the lattice:



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Exercise 2.5.14

Let M be the **modular group** of order 16 with the following presentation:

$$\langle u, v \mid u^2 = v^8 = 1, vu = uv^5 \rangle.$$

It has three subgroups of order 8: $\langle u, v^2 \rangle$, $\langle v \rangle$, and $\langle uv \rangle$, and every proper subgroup is contained in one of those three. Prove that $\langle u, v^2 \rangle \cong Z_2 \times Z_4$, $\langle v \rangle \cong Z_8$, and $\langle uv \rangle \cong Z_8$. Show that the lattice of subgroups of M is the same as the lattice of subgroups of $Z_2 \times Z_8$ (cf. Exercise 2.5.13) but that these two groups are not isomorphic.

Solution. Since $|v| = |uv| = 8$, then $\langle v \rangle \cong \langle uv \rangle \cong Z_8$. One can apply the second relation to obtain $v^2 u = uv^2$, hence we deduce that $\langle u, v^2 \rangle$ is Abelian. We may write

$$\langle u, v^2 \rangle = \{1, v^2, v^4, v^6, u, uv^2, uv^4, uv^6\}$$

From the enumeration of $\langle u, v^2 \rangle$, we have the map $\varphi : Z_2 \times Z_4 \longrightarrow \langle u, v^2 \rangle$ defined by

$$\varphi(a) = u, \quad \varphi(b) = v^2.$$

This extends to a homomorphism. Moreover, $|\langle u, v^2 \rangle| = |Z_2 \times Z_4| = 8$. Since φ sends distinct elements of $Z_2 \times Z_4$ to distinct elements of $\langle u, v^2 \rangle$, it is injective. Being a homomorphism between two groups of the same finite order, it is also surjective, hence an isomorphism. We have the following lattice for M:

$$\begin{tikzcd}
 & G \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & & & \\
\langle v \rangle \arrow[dr, dash] & \langle uv \rangle \arrow[d, dash] & \langle u, v^2 \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & & \\
 & \langle v^2 \rangle \arrow[dr, dash] & \langle uv^2 \rangle \arrow[d, dash] & \langle u, v^4 \rangle \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
 & & \langle v^4 \rangle \arrow[dr, dash] & \langle u \rangle \arrow[d, dash] & \langle uv^4 \rangle \arrow[dl, dash] \\
 & & & 1 &
\end{tikzcd}$$

Since $vu = uv^5 \neq uv$, M is not Abelian, and therefore cannot be isomorphic to $Z_2 \times Z_8$. ∎

Exercise 2.5.15

Describe the isomorphism type of each of the three subgroups of D_{16} of order 8.

Solution. From $|r| = 8$, we deduce $\langle r \rangle \cong Z_8$. The remaining subgroups $\langle s, r^2 \rangle$ and $\langle sr, r^2 \rangle$ can be shown to be isomorphic to D_8 as follows:

- For the subgroup $\langle s, r^2 \rangle$, observe that $(r^2)^4 = s^2 = 1$, and $sr^2 = r^6 s = (r^2)^{-1} s$. Use r^2 in place of r and s in place of s to see that the relations match those of D_8.
- For the subgroup $\langle sr, r^2 \rangle$, observe that $(r^2)^4 = (sr)^2 = 1$, and

  $$(sr)r^2 = sr^3 = r^5 s = r^6 r^7 s = (r^2)^{-1}(sr).$$

  Use r^2 in place of r and sr in place of s to see that the relations match those of D_8. ∎

Exercise 2.5.16

Use the lattice of subgroups of the quasidihedral of order 16 to show that every element of order 2 is contained in the proper subgroup $\langle \tau, \sigma^2 \rangle$ (cf. Exercise 2.5.11).

Solution. Every element of order 2 generates a cyclic subgroup of order 2. Using the lattice, $\langle \tau, \sigma^2 \rangle$ properly contains all cyclic subgroups, except $\langle \tau\sigma \rangle$ and $\langle \tau\sigma^3 \rangle$, both of which are order 4. Then $\langle \tau, \sigma^2 \rangle$ contains all cyclic subgroups of order 2, hence contain all elements of order 2. ■

Exercise 2.5.17

Use the lattice of subgroups of the modular group M of order 16 to show that the set $\{x \in M \mid x^2 = 1\}$ is a subgroup of M isomorphic to the Klein 4-group (cf. Exercise 2.5.14).

Solution. Using the lattice in Exercise 14, we see that we have 3 candidates to be isomorphic to V_4, namely $\langle v^2 \rangle$, $\langle uv^2 \rangle$, and $\langle u, v^4 \rangle$. The first and second subgroups are cyclic, while $\langle u, v^4 \rangle = \{1, u, v^4, uv^4\}$. Since it is not generated by one element, each of these elements are of order 2, and $v^4 u = v^3 u v^5 = \cdots = uv^{20} = uv^4$ so that it is Abelian, then $\langle u, v^4 \rangle \cong V_4$. ■

Exercise 2.5.18

Use the lattice to help find the centralizer of every element of QD_{16} (cf. Exercise 2.5.11).

Solution. Note that $\sigma^4\tau = \sigma^3\tau\sigma^3 = \cdots = \tau\sigma^{12} = \tau\sigma^4$ so that $\sigma^4 \in Z(QD_{16})$ as σ^4 already commutes with powers of σ. Moreover, any power of σ does not commute with τ except for σ^4. The elements $\tau\sigma$ and $\tau\sigma^3$ do not commute with σ^2 as $(\tau\sigma)\sigma^2 = \tau\sigma^3 = \sigma\tau \neq \sigma^2(\tau\sigma) = \tau\sigma^2$, and $(\tau\sigma^3)\sigma^2 = \sigma^7\tau \neq \sigma^3\tau = \sigma^2(\tau\sigma^3)$. Next, $(\tau\sigma^2)\sigma^2 = \tau\sigma^4 \neq \tau = \sigma^2(\tau\sigma^2)$ so that σ^2 does not commute with $\tau\sigma^2$. Moreover, $(\tau\sigma^6)(\tau\sigma^2) = \tau\sigma^2\sigma^4\tau\sigma^2 = (\tau\sigma^2)(\tau\sigma^6)$ so that $\tau\sigma^2$ commutes with $\tau\sigma^6$, but $\sigma^2\tau\sigma^6 = \tau\sigma^4 \neq \tau = (\tau\sigma^6)\sigma^2$ so σ^2 does not commute with $\tau\sigma^6$. Lastly, $\sigma^2(\tau\sigma^4) = \tau\sigma^2 \neq (\tau\sigma^4)\sigma^2$, and $\sigma^2\tau = \tau\sigma^6 \neq \tau\sigma^2$ so that σ^2 does not commute with $\tau\sigma^4$ nor with τ. It follows that the centralizers of the elements of QD_{16} are

$$\begin{aligned}
C_{QD_{16}}(1) = C_{QD_{16}}(\sigma^4) &= QD_{16} \\
C_{QD_{16}}(\langle \sigma^k \rangle) &= \langle \sigma \rangle \text{ for } k = 1, 2, 3, 5, 6, 7 \\
C_{QD_{16}}(\langle \tau\sigma \rangle) = C_{QD_{16}}(\langle \tau\sigma^5 \rangle) &= \langle \tau\sigma \rangle \\
C_{QD_{16}}(\langle \tau\sigma^3 \rangle) = C_{QD_{16}}(\langle \tau\sigma^7 \rangle) &= \langle \tau\sigma^3 \rangle \\
C_{QD_{16}}(\langle \tau \rangle) = C_{QD_{16}}(\langle \tau\sigma^4 \rangle) &= \langle \sigma^4, \tau \rangle \\
C_{QD_{16}}(\langle \tau\sigma^2 \rangle) = C_{QD_{16}}(\langle \tau\sigma^6 \rangle) &= \langle \tau\sigma^2, \sigma^4 \rangle
\end{aligned}$$

■

Exercise 2.5.19

Use the lattice to help find $N_{D_{16}}(\langle s, r^4 \rangle)$.

Solution. Based on the placement of $\langle s, r^4 \rangle$, its normalizer may be itself, $\langle s, r^2 \rangle$, or D_{16}. Note that $\langle s, r^4 \rangle = \{1, s, r^4, sr^4\}$, and taking r^2 and $(r^2)^{-1} = r^6$, we have

$$r^2 \langle s, r^4 \rangle r^6 = \{1, sr^4, r^4, s\} = \langle s, r^4 \rangle$$

so that $r^2 \in N_{D_{16}}(\langle s, r^4 \rangle)$, and $\langle s, r^2 \rangle \leq N_{D_{16}}(\langle s, r^4 \rangle)$. Since $rsr^{-1} = r^2 s \neq r$, then $r \notin N_{D_{16}}(\langle s, r^4 \rangle)$ so that $N_{D_{16}}(\langle s, r^4 \rangle) = \langle s, r^2 \rangle$. ■

Exercise 2.5.20

Use the lattice of subgroups of QD_{16} (cf. Exercise 2.5.11) to help find the normalizers
(a) $N_{QD_{16}}(\langle\tau\sigma\rangle)$ (b) $N_{QD_{16}}(\langle\tau,\sigma^4\rangle)$.

Solution.

(a) Note that $\langle\tau\sigma\rangle = \{1, \tau\sigma, \sigma^4, \tau\sigma^5\}$. Moreover, $(\sigma^2)^{-1} = \sigma^6$ so that

$$\sigma^2\langle\tau\sigma\rangle\sigma^6 = \{1, \tau\sigma^4, \sigma^4, \tau\sigma\} = \langle\tau\sigma\rangle$$

and $\sigma(\tau\sigma)\sigma^7 = \tau\sigma^3 \neq \tau\sigma$ so that $\sigma \notin N_{QD_{16}}(\langle\tau\sigma\rangle)$. Then $N_{QD_{16}}(\langle\tau\sigma\rangle) = \langle\sigma^2, \tau\sigma\rangle$.

(b) $\langle\tau,\sigma^4\rangle = \{1, \tau, \sigma^4, \tau\sigma^4\}$, and

$$\sigma^2\langle\tau,\sigma^4\rangle\sigma^6 = \{1, \tau\sigma^4, \sigma^4, \tau\}$$

while $\sigma\tau\sigma^7 = \tau\sigma^2 \neq \tau$ so that $\sigma \notin N_{QD_{16}}(\langle\tau,\sigma^4\rangle)$. Then $N_{QD_{16}}(\langle\tau,\sigma^4\rangle) = \langle\sigma^2,\tau\rangle$. ∎